



**Web Service Interface –
Northbound Open RESTful
Integration Service (NORIS)
API Version 8.0**

Configuration



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Edition: 2024-10-31

Document ID: A6V10470112_en_b_80

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About This Document

Purpose

This document describes the interface between the outside world and the Management System through a Web Service Interface (Web API). It describes how the interface is to be accessed and what data in what format are exchanged.

Scope

The document provides a description of the public API which can be used to access data and the functionality provided by the system platform. This document applies to the system version 8.0.

Target Audience

- Testers of the public API
- Developers/Clients accessing the public API

Cyber security disclaimer

Siemens products and solutions provide security functions to ensure the secure operation of building comfort, fire safety, security management and physical security systems. The security functions on these products and solutions are important components of a comprehensive security concept.

It is, however, necessary to implement and maintain a comprehensive, state-of-the-art security concept that is customized to individual security needs. Such a security concept may result in additional site-specific preventive action to ensure that the building comfort, fire safety, security management or physical security system for your site are operated in a secure manner. These measures may include, but are not limited to, separating networks, physically protecting system components, user awareness programs, defense in depth, etc.

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<http://www.siemens.com/cert/en/cert-security-advisories.htm>

Document Conventions

The following table lists conventions to help you use this document in a quick and efficient manner.

Convention	Examples
Numbered Lists (1, 2, 3...) indicate a procedure with sequential steps.	1. Turn OFF power to the field panel. 2. Turn ON power to the field panel. 3. Open the panel.
One-step procedures are indicated by a bullet point.	● Expand the Event List.
Conditions that you must complete or must be met before beginning a procedure are designated with a ▷. Intermediate results (what will happen following the execution of a procedure step), are designated with an indented ⇒. Results, after completing a procedure, are designated with a ⇒.	▷ The report you want to print is open. 1. Click Print . ⇒ The Print dialog box displays. 2. Select the printer and click Print. ⇒ The print confirmation displays.
Bold font indicates something you should type or select, or when a dialog box or window is specified.	Type F for field panels. Click OK to save changes and close the dialog box. The Create a New Project dialog box displays.
Menu paths in procedures are indicated in bold .	Select File > Text, Copy > Group , which means from the File menu, select Text, Copy and then Group .
File paths containing placeholders display the placeholders in <i>italics</i> enclosed in square brackets.	[<i>installation drive:</i>]\[<i>installation folder</i>]\[<i>project</i>]\...
Error and system messages are displayed in Courier New font.	The message Report Definition successfully renamed displays in the status bar.
<i>Italics</i> are used to emphasize new or important terms.	The reaction processor continuously executes a user-defined set of instructions called the <i>control program</i> .
	This symbol signifies a Note. Notes provide additional information or helpful hints.
Cross references to other information in printed material are indicated with an arrow and the page number, enclosed in brackets: [→ 92]	For more information on creating flowcharts, see Flowcharts [→ 92].

Getting Help

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Safety Messages According ANSI Z535.6

The following examples show the ANSI standard safety messages used in this document to draw the reader's attention to important information.

ANSI distinguishes between *personal injury* safety messages and *property damage* warning messages.

The personal injury safety messages have safety alert symbols and the following alert level labels: DANGER!, WARNING!, CAUTION!

The label for property damage messages is: NOTICE.

Examples:

	NOTICE
	Property Damage Warning Message Equipment damage or loss of data may occur if you do not follow a procedure or instruction as specified.
	CAUTION
	Caution Safety Message Minor or moderate injury may occur if you do not follow a procedure or instruction as specified.
	WARNING
	Warning Safety Message Personal injury or property damage may occur if you do not follow a procedure as specified.
	DANGER
	Danger Safety Message Electric shock, death, or severe property damage may occur if you do not perform a procedure as specified.

Document Revision History

Document Identification

The document ID is structured as follows:

ID_Language(COUNTRY)_ModificationIndex_ProductVersionIndex

Example: A6Vnnnnnnnn_en_a_02

Document Revision History		
Modification Index	Edition Date	Brief Description
a	2019-04-11	Release Edition 4.0
b	2019-07-09	Edition 4.1
c	2019-07-30	Edition 4.2
d	2020-09-29	Edition 5.0
e	2021-06-30	Edition 5.1
f	2022-04-30	Edition 6.0
g	2023-04-21	Edition 7.0

1 Introduction

1.1 References

1.1.1 Links

Ref.	Abbreviation	Link Description
[1]	HAL	http://stateless.co/hal_specification.html
[2]	SignalR	https://github.com/SignalR/SignalR/wiki
[3]	Base64url	http://www.ietf.org/rfc/rfc4648.txt
[4]	OAuth	https://tools.ietf.org/html/rfc6749
[5]	Language-Tags	http://tools.ietf.org/html/rfc5646
[6]	JSON:	http://www.ietf.org/rfc/rfc4627.txt
[7]	REST	http://en.wikipedia.org/wiki/Representational_state_transfer
[8]	URI	https://tools.ietf.org/html/rfc3986
[9]	EBNF	http://en.wikipedia.org/wiki/Extended_Backus%E2%80%93Naur_Form
[10]	Media-Types	http://www.iana.org/assignments/media-types/media-types.xhtml
[11]	Swagger	http://swagger.io
[12]	Mutual TLS	https://datatracker.ietf.org/doc/html/rfc8705

1.1.2 Documents

Ref.	Name	Version
[20]	Regional Product Customization Guide	

1.1.3 Online Documentation

Web Service Interface also provides an interactive online documentation (web page). The documentation is based on Swagger[19] and is a runtime representation of the API. It requires a running Web Service Interface and can be accessed in two ways using the following URL:

i) Intranet :

[http\(s\)://\[hostname\]:\[port\]/swagger/ui/index](http(s)://[hostname]:[port]/swagger/ui/index)

Example

<https://myWebServiceHost:8443/swagger/ui/index>

ii) Internet (i.e via IIS)

Once Web Service Application is created in SMC, append `/swagger/ui/index` to WSI application URL shown.(in SMC)

Example

[https://\[hostname\]:\[websitePort\]/ApplicationName/swagger/ui/index](https://[hostname]:[websitePort]/ApplicationName/swagger/ui/index)

**NOTE:**

If you are using Internet Explorer make sure Compatibility View is turned off.

1.1.3.1 Exposure of Swagger via IIS is configurable

An advanced user can enable/disable the exposure of Swagger via IIS. To enable the swagger via IIS, user has to set the *enabled* flag to *true* for SwaggerRoutingRule under WSI web application web.config file. By default swagger via IIS is disabled. As shown below :

```
<rule name="SwaggerRoutingRule" enabled="false" stopProcessing="true">
  <match url="(/{^})(swagger/)(.*)" />
  <action type="Rewrite" url="http://localhost:8095/{R:2}{R:3}" />
</rule>
```

Path for WSI Web Application web.config :

Example :

<Drive:>\GMSProjects\WebSites\GmsWebsite\WsiWebApplicationName\

1.2 Definitions, Acronyms and Abbreviations, Conventions

1.2.1 Glossary

Terms Used in Web Service Interface	
Term	Description
HAL	Simple format that gives a consistent and easy way to hyperlink between resources in an API.
HTTP	Hypertext transfer protocol: http://en.wikipedia.org/wiki/Hypertext_Transfer_Protocol
JSON	JavaScript Object Notation: http://en.wikipedia.org/wiki/JSON
OAuth 2.0	Open standard to authorization. OAuth provides client applications a 'secure delegated access' to server resources on behalf of a resource owner.
Push Notification	In this document push notifications refer to real-time notifications (see real-time push below). Push notifications to mobile devices (see wake-up push below) are not covered in this document. However, a dedicated extension module is available to provide this functionality.
REST	Representational state transfer: http://en.wikipedia.org/wiki/Representational_state_transfer
Real-time push	Push notification from a server (our Web Service in this case) to a client; used for notification of changes of values as long as client application is running. Notifies the client about events in the system, including COV, as long as the client application is running.
Reverse Proxy	Retrieves resources on behalf of a client from a server. These resources are then returned to the client as though they originated from the server itself.
SignalR	Library for ASP.NET. SignalR allows bi-directional communication between server and client.
Wake-up push	Push notification to a mobile phone; notifies a user about an event in the system; takes advantage of third-party notification providers (Apple, Google, ...).

1.2.2 Conventions in this Document

The product is developed for different regions and known under the following brands:

- Desigo CC
- Desigo DMS
- Cerberus DMS

In this document, we therefore use the neutral term *System* for each of these brands.

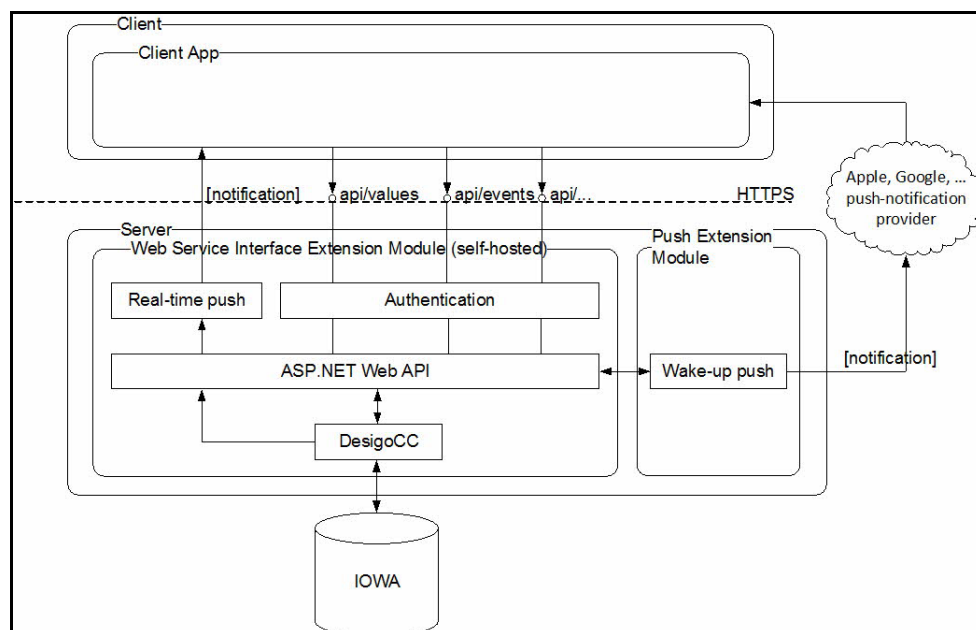
1.3 Open Issues in this Document Version

At the effective date of the current document version, the following open issues are identified:

- none

2 System Architecture

The following diagram shows the architecture for the Web Service Interface API.



2.1 System Limits

The following limits apply when using the Web Service Interface:

- Number of concurrent sessions: 100
- Delay for notifications: <1 second
- 5'000'000 value updates / day
- 20'000 value subscriptions
- 250'000 event notifications / day
- 250'000 event-counter notifications / day
- 1'000'000 trends / min



NOTICE

IIS limits imposed by Operating System version

The Client versions of Windows (e.g. Windows 7, Windows 8) have intentionally feature-limited versions of IIS. One limitation is the number of concurrent connections which can be as low as 10.

This especially affects the usage of push notifications with SignalR (see [2] [→ 19]).

3 Basic System Concepts

3.1 Glossary of the System

Aggregator

A parent object summarizing child objects, such as data points, displayed in the System Browser.

Alarm

See Event.

Attribute

The **Main** attributes belong to all existing objects visible in the System Browser. The **Details** attributes belong to each of the properties within an object defined in an Object Model. There are two kinds of attributes:

- Simple attribute, for example, unit or text group.
- Complex attribute, for example, commands or alarms.

COV

Change of Value. An updated value that represents the extent of change in a measured reading when change criteria have been met.

Data point

An input/output function consisting of all assigned information describing fully the data point's meaning.

Discipline

An area of building control, such as HVAC, lighting, blind control, fire or video. Building control disciplines can also contain subdisciplines.

Event

A change of state or value detected for processing and/or reporting.

Function

Is the semantic description for one or more data point types, for example, fire detector.

Object

A specific instance of an object type. While an object type is identified by a unique object type property, an object is identified by its object identifier property.

Project data points imported in the system database. Instances are customized with attributes, such as discipline or text group, during data import.

Object Model

A data point type and its attributes specific to a subsystem and protocol. Native presentation of subsystem objects in the system, for example, analog input.

Property

Each object has a different number of properties, depending on the Object Model or the defined Function. Properties show online values (COV or polling) and can have an operation function assigned to them, for example, Acknowledge.

Properties collect data from the field system and need more system resources than a static attribute in the database.

Propagation

A mechanism for relaying Common Status information up the building control system's hierarchical tree. Propagation is based on the parent-child-grandchild model. Each object in the system can have one or more parent objects and one or more child objects. Each child in the system can simultaneously propagate multiple active Common Status properties to a parent—for example, fire alarm, security fault, or technical excluded.

System Manager

A system application that allows users to select and act on objects for monitoring and controlling a facility. The System Manager contains the following panes: System Browser (selection of data points), Primary, Secondary, and Contextual.

System Browser

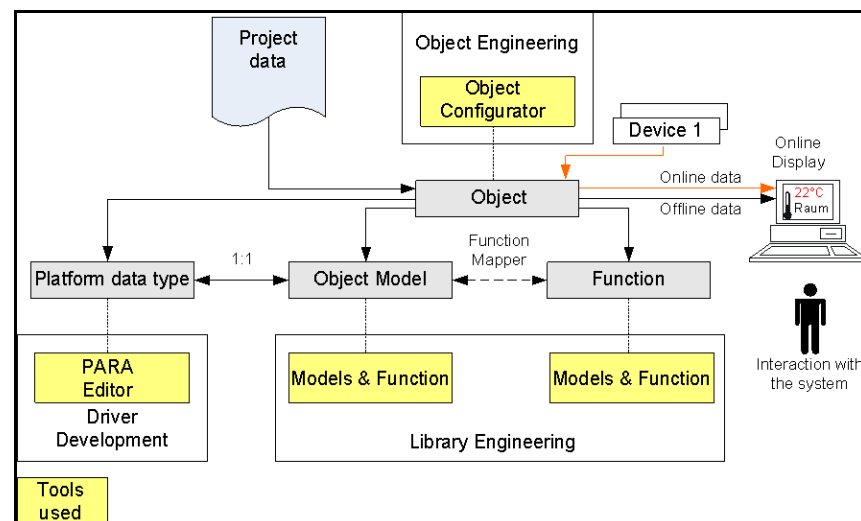
Shows the list of hierarchical objects, such as data points, in one or more views. Compiles objects into an aggregator object.

View

In System Browser, there are two basic views: Management View and Application View. Additional project views can be defined as needed. The objects are displayed hierarchically, depending on the selected view. An object can be assigned to more than one view.

3.2 Basic Data Concept in the System

Instantiated objects for individual subsystems are always related to an Object Model and a Function. This concept ensures that the user experiences the same operating and monitoring philosophy for the same or similar Functions.



The system concept impacts the following main elements:

Platform data type	It forms the basis for the Object Model set up on the system. Platform tools are required in individual cases for driver development of additional subsystems.
Object Model	All system basic data types are designated Object Models. The corresponding default value is defined per subsystem for each Object Model.
Function	Function includes the semantic description (for example, fire detector or temperature sensor) for one or more specific data points.
Object	Imported project data can be modified specifically to the object using the Object Configurator.
Project data	The project data contains the specific information on a data point or data point amount by the system and is responsible for system response. Collecting data using auto-discovery is possible if supported by the subsystem. The Function or Object Model supplements missing information on a project data point to ensure uniform functioning of system.

The tree structure ensures a consistent operating, commanding and alarming philosophy for the various integrated subsystems. The benefits of this concept include:

- Discipline-specific project data from fire, HVAC, intrusion, and so on can be configured using the same mechanism.
- Additional or new systems can easily be added.

3.3 Alarm Concepts

The system automatically controls and monitors building services plants. Events occur (such as a fault) that are important for the operator, requiring user intervention. The alarm system processes such events and reports them accordingly.

The system reacts automatically to a fault with a system reaction (for example, automatically locking a ventilation plant during a fire alarm).

For control deviations such as a dirty filter, a simple alarm is triggered and only an alarm message is sent.

The system switches to an alarm state for both cases and a corresponding alarm is issued. After trouble-shooting the cause of the alarm, the user resets the alarm state and the system returns to normal status.

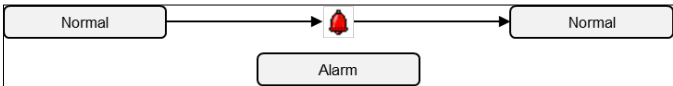
Alarm Types

- Field system alarms (alarm behavior configured in the field system)
- Management platform alarms
 - Discrete alarm type (for example, alarm range between 10 and 20 or between 30 and 40).
 - Continuous alarm type (for example, alarm range ≤ 30 or ≥ 70).

Alarm Handling		
Alarm Type	Name	Alarm Behavior
Alarm_NoAckNoReset	Simple	No user action necessary
Alarm_NoReset	Basic	Acknowledge the alarm
Alarm	Extended	Acknowledge and reset the alarm
Alarm_Dynamic	User defined	Acknowledge and reset the alarm executing commands defined on the alarm source

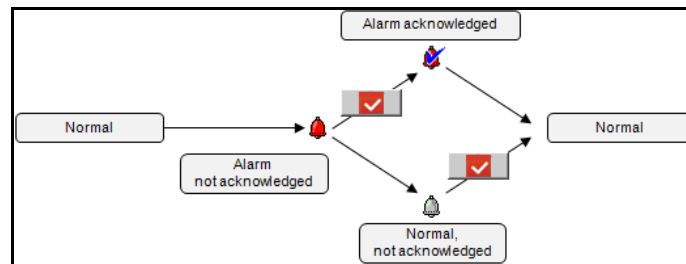
Simple Alarm

Simple alarms do not require you to acknowledge or reset, they simply return to the normal state when the monitored condition (for example, a dirty filter) returns to normal.



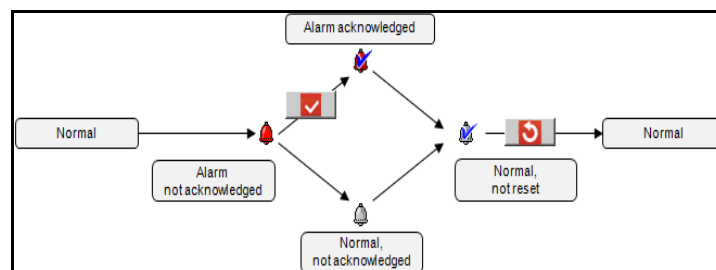
Basic Alarm

More important alarms (for example, lack of water) must be acknowledged. After the alarm state is eliminated, the plant is once again released after acknowledgement.



Extended Alarm

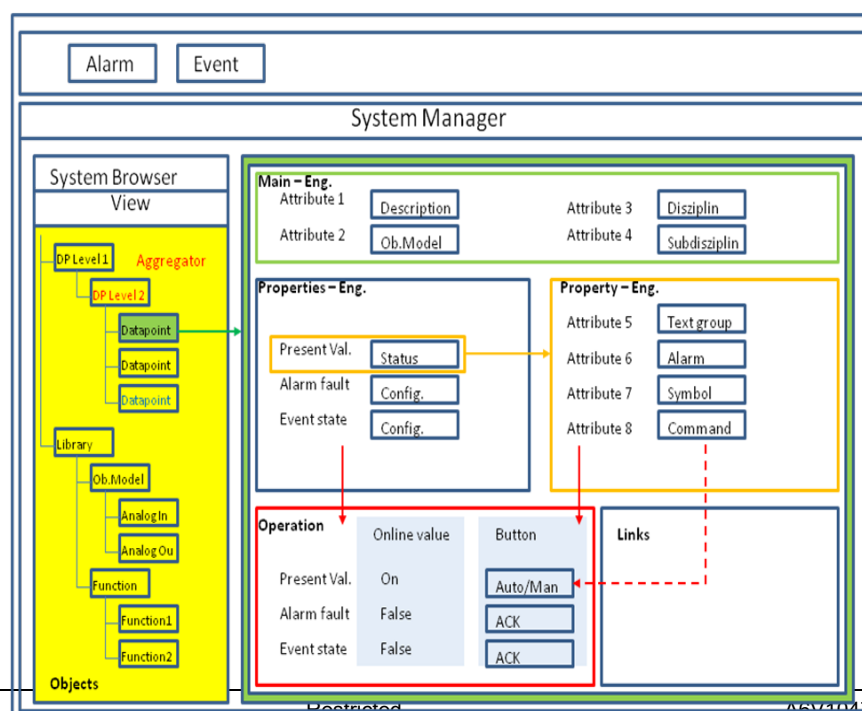
For an extended alarm (for example, velocity monitor), the reset returns the plant to normal. Provided the monitored state returns to the normal range. The idea is that important alarms are not overlooked when the alarm condition clears (Normal, not acknowledged situation). The plant must be reset to re-start operation.



3.4 User interface Concepts

The user interface contains the information for:

- Displaying Alarms and Events in a list
- Object navigation in the System Browser tree
- Engineering dialogues to customize the objects and properties
- Operating the system
- Links to further information on an object



Displaying Alarms and Events

Alarms or events are displayed in the Event Summary bar. From there, Alarms or events can – depending on their type - be acknowledged or reset.

System Browser Navigation

If an object (green) is selected in the System Browser tree, all associated objects and properties (green frame) are displayed as part of the Engineering dialogue.

Engineering Dialogue

Each object in System Browser has the same number of standard attributes, that are displayed in the **Main** section (light green frame). The number of properties can vary depending on the type of the object. When selecting a property, its attributes (blue frame) are displayed.

Operating the System

Only properties that are essential to the operation of the system are displayed in the **Operation** tab (red frame). The value of a property is read from the system database or the field network (online value).

If a property has a command attribute, an interactive element is displayed (text field, list box or button). To change a value, you need to perform two steps:

- Set the value
- Send the value

3.5 Search Mechanism Behavior

The result of a search is different depend which commanding you are using:

- /properties/object.property
- /properties/object

/properties/object.property

The result shows list all data points with the requested property information.

Example:
XXX.YYY.Present_Value
XXX.YYY.Alarm.Fault
XXX.YYY.Event_State

readAllProperties	false ▾	If true all the properties for the requested Object OR Property Id will be returned in the response.	query	boolean
-------------------	---------	--	-------	---------

/properties/object

The result shows list all data points with the default property information.

NOTE: The default property is defined for each Object Model as a main attribute.

Example:
XXX.YYY.Present_Value
XXX.ZZZ.Fault
YYY.ZZZ. Present_Value

4 Using the Web Service Interface

4.1 Communication

The API can be accessed over HTTP or HTTPS. In a production environment it's highly recommended to use HTTPS because otherwise primary credentials are transmitted in plaintext.

4.1.1 Authentication

For authentication we use OAuth 2.0 Resource Owner Password Credentials Grant [4] [→ 19]. In order to access a resource of the Web Service Interface the Client needs a valid access token and a valid session within the system. In case no token is presented or the token is invalid (e.g. expired) or the system session is not available the user will get a status code 401 (*Unauthorized*) and needs to request an access token. The access token can be requested through a dedicated resource (/token) by presenting the primary credentials. In case the credentials are provided, the system creates a new session and returns an access token.

1. Client tries to access a resource.
2. Server responses with status code 401 (*Unauthorized*).
⇒ Client asks the end-user for credentials.
3. Client sends credentials to a dedicated resource.
4. Server returns an access token, the system creates an internal session.
5. Client again tries to access a resource and includes the access token in an authorization header.
6. Server returns content with status code 200 (*Succeeded*).



NOTICE

Session alive

The Client either needs to subscribe for notifications (e.g. change of values) or needs to access the API at least once every 10 minutes otherwise its system session expires. The API provides a dedicated resource (/heartbeat) just for the sake of keeping a session alive.

4.1.2 Client Certificate Authentication

Optionally, a client can authenticate to the server by providing a client certificate. The Web Service Interface expects the client certificate to be provided in a header called *X-ARR-ClientCert*. In case the interface is accessed through a reverse proxy (almost always the case in a production environment), the reverse proxy asks the client for an optional client certificate. In this case the reverse proxy passes the client certificate to the backend server as HTTP header with the default header configured as *X-ARR-ClientCert*.

4.1.3 Subscriptions (Push Notifications with SignalR)

Optionally, for specified services the API provides not only data on request (pull) but also in case of an event determined by the respective service (push).

Any Client subscribing for any push notifications needs to support SignalR (see [2] [→ 19]). The Client first needs to connect to a dedicated SignalR hub and can then subscribe for notifications by providing a connection ID which it gets after a connection between Client and server is established. The Client then needs to implement a function (see signature in respective service chapters) which will be called from the server in case a notification is due.

Noris hosts signalr url at `/signalr` route.

By default Javascript & C# signalr client tries to connect to `/signalr` route if no route is specified. Please verify with signalr client package's documentation for setting up signalr url.

Signalr also generates hub proxy file for javascript clients, this can be accessed at `/signalr/hubs` route.

i) Intranet :

`http(s)://[hostname]:[port]/signalr`

`http(s)://[hostname]:[port]/signalr/hubs`

ii) Internet (i.e via IIS)

`https://[hostname]:[websitePort]/ApplicationName/signalr`

`https://[hostname]:[websitePort]/ApplicationName/signalr/hubs`

Example

```
//norisHub is name of our hub on the server side
var hub = $.connection.norisHub;
//register notifyEvents as a method on the client side for events
hub.client.notifyEvents = function (events) {
    //do something in case a notification arrives
}
//register notifySystems as a method on the client side for systems
hub.client.notifySystems = function (systems) {
    //do something in case a notification arrives
}
// start hub
$.connection.hub.start().done(function() {
    //do other stuff
});
```

4.1.4 Latency Push Notification to Request-Response

In the workflow of subscription and notification it may happen that push notification from server is received before subscription success response. This latency will be experienced because of network communication delay of the messages on two independent channels of SignalR hub connection for push notification and response of subscription request.

In this kind of scenario Clients of WSI should consider received notification as a valid notification and design their solution accordingly.

4.1.5 SignalR Hubs

For signalR notification client has to setup hub connection with server.

Legacy implementation of Noris provides specific hub for almost all the subscription services respectively.

Noris also introduces singleHub solutions as NorisHub. Now client should create only one NorisHub connection and use it for all the services for subscription-notificaitons. It is the recommended singlanR connection approach.

Service specific list of Hubs are as below:

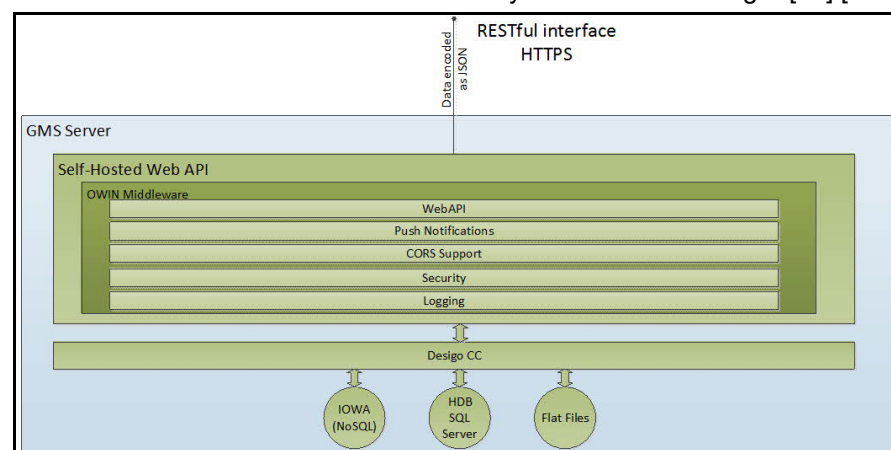
Subscription Service	Legacy Hub	Recommended
EventsSubscriptions	eventsHub	norisHub
EventcountersSubscriptions	eventCountersHub	norisHub
SystemBrowserSubscriptions	systemBrowserHub	norisHub
ValuesSubscriptions	valuesHub	norisHub
CommandsSubscriptions	commandsHub	norisHub
EventCategorySoundsSubscriptions	eventSoundsHub	norisHub
SystemsSubscription		norisHub

4.2 Supported Client Environments

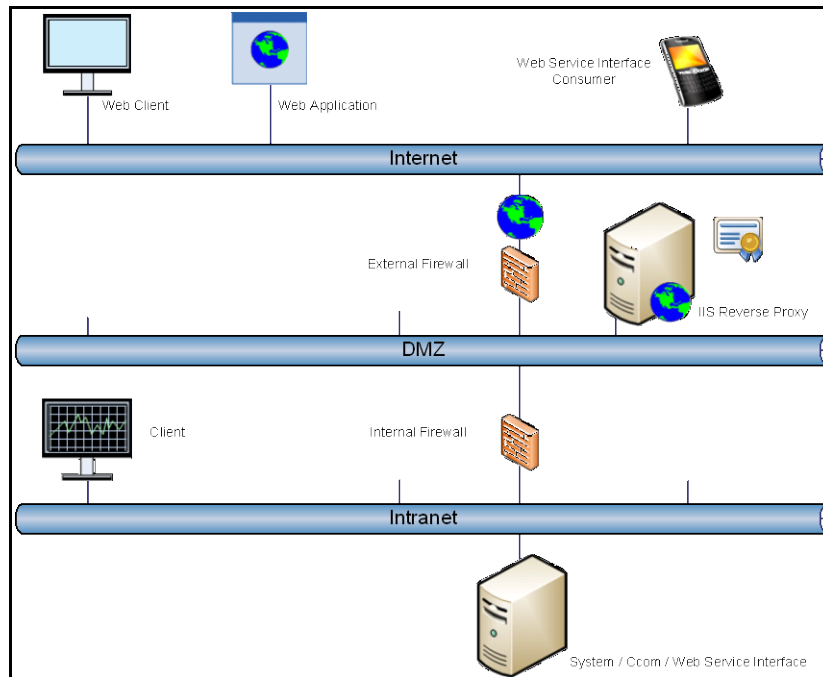
There are no Client dependencies except SignalR version 2.2.0 Client library (see [2] [→ 19] for available Client libraries) for Clients interested in push notifications.

4.3 Deployment

The Web Service Interface will be deployed as a self-hosted component. It's an executable and can be commanded as any other IOWA manager [12] [→ 19].



In a production environment the Web Service Interface (and System) will most likely only be accessible through a reverse proxy.



Network Topology

4.4 Configuring Web Service Interface in a Project



NOTE:

A user has the same user rights as defined in the system. Therefore, you can check the user rights in advanced in the system bevroe using the Web Service Interface.

4.4.1 Stopping a Project

- ▷ At least one project is available under the **Projects** node and is started.
- 1. In the System Management Console tree, select the **Projects** node.
- 2. In the **Projects** tree, select the project that you want to stop.
- 3. Click the **Project Settings** tab, if not yet selected.
 - ⇒ **Project Settings** displays.
- 4. In the **Project Settings** toolbar, click **Stop** .
 - ⇒ A confirmation message displays.
- 5. Click **OK** to stop the started project.

4.4.2 Showing Settings

- ▷ The Extension module **Connectivity > Web Service Interface** is installed.
- ▷ At least one project is available under the **Projects** node and is started.
- 1. In the System Management Console tree, select the **Projects** node.
- 2. In the **Projects** tree, select the project.
- 3. Click the **WSI Settings** tab.
 - ⇒ **WSI Settings** displays and shows the used protocol type and port number.



NOTE:

If you want to change the settings, you must stop the project first.

4.4.3 Defining Protocol Type and Port

- ▷ The *Web Service Interface* extension module installed for the selected project.
- ▷ The project is stopped.
- ▷ IIS is installed according to the settings in the Installation Manual chapter 4.
- 1. Click **Edit** .
- 2. Click **Next** .
 - ⇒ The **Web Service Interface Settings** expander displays.
- 3. Select the protocol:
 - **Unsecured**
 - Under **Port**, enter the port number (Default: 8080).
 - **Secured**
 - Under **Port**, enter the port number (Default: 8443).
 - Under **Host Certificate**, click **Browse**.
 - Select the appropriate certificate and click **OK**.
- NOTE:** Additional information on certificates is available in the SMC Help.
- 4. Click **Save Project** .
 - ⇒ The protocol settings are saved.
- 5. Click **Start Project** .



NOTE:

HTTP is insecure and is vulnerable to man-in-the-middle and eavesdropping attacks, which can let attackers gain access to sensitive information.

Many organizations require certificates from third-party authorities instead of self-signed certificates to remain compliant with current regulations.

4.4.4 Creating a WSI on Server

- ▷ At least one Web site is created and available under the **Websites** root node in the System Management Console tree.
- 1. In the System Management Console tree, select the **Websites** node.
- 2. In the **Websites** tree, select the **Website**.
- 3. Click the **Management** tab.

- ⇒ The **Details** expander displays.
4. In the **Website** toolbar, click **Web Service Interface** .
⇒ The **Web Application Details** expander displays.
 5. In the **Web Application Details** expander, provide the Web application details as follows:
 - From the **Project Name** drop-down list, select a project that you want to link to the Web application.
 - In the **Name** field, type a unique name of the Web application.
 - Browse for the **Path** to save the Web application on the disk. The default path is `[installation drive:]\[installation folder]\[WebSites]\[Web site Name]`.
 - Browse for a web application **User** using the Select User dialog box. You can select a different user than the one you selected during Web site creation.
NOTE: The Web application must be a member of the IIS_IUSRS group. If you select a user that is not a member of the IIS_IUSRS group, SMC prompts you to add it.
 - Enter the valid **Password** for the Web application user.
 6. Click **Save** .
⇒ A confirm message displays.
⇒ Click **OK** to initiate the Web application creation (or **Cancel** to abort).
- ⇒ If confirmed, the data is validated on successful creation.
- ⇒ The **Web Service Interface Settings** expanders displays the defined settings.

4.4.5 Starting a Project

- ▷ The project is stopped.
1. In the System Management Console tree, select the **Projects** node.
 2. In the **Projects** tree, select the project that you want to start.
 3. Click the **Project Settings** tab, if not yet selected.
⇒ **Project Settings** displays.
 4. In the **Project Settings** toolbar, click **Start Project** .
⇒ A confirmation message displays.
 5. Click **OK** to start the stopped project

4.5 Service Category

Based on frequency of change in design, all of the services and APIs are categorized as below three types:

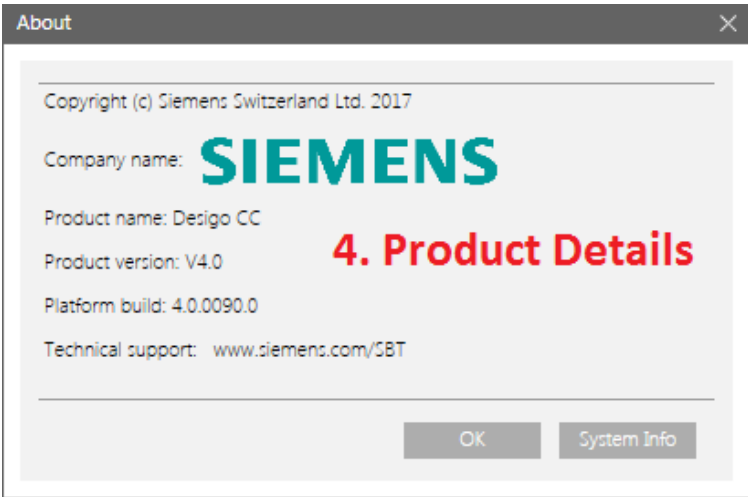
Category	Scope	Description
Public	Service and API	Any change in APIs will not be breaking. Backward compatibility will always be supported. Code: Green
Partial Internal	Service only	Service contains both internal and Public APIs. Code: Yellow
Internal	Service and API	Service is developed strictly only for internal development. Interface might change with newer version. Backward compatibility is not guaranteed. Code: Red

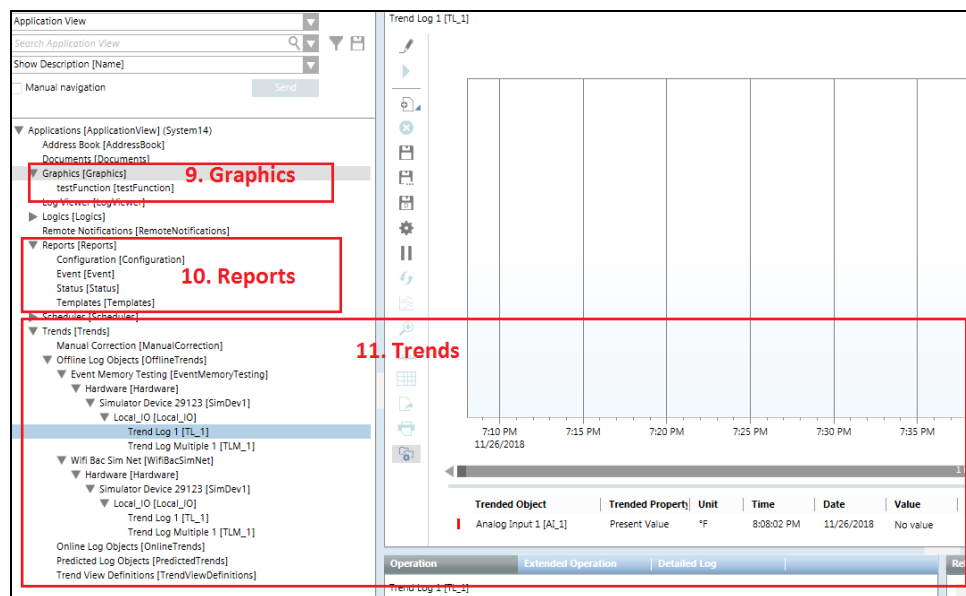
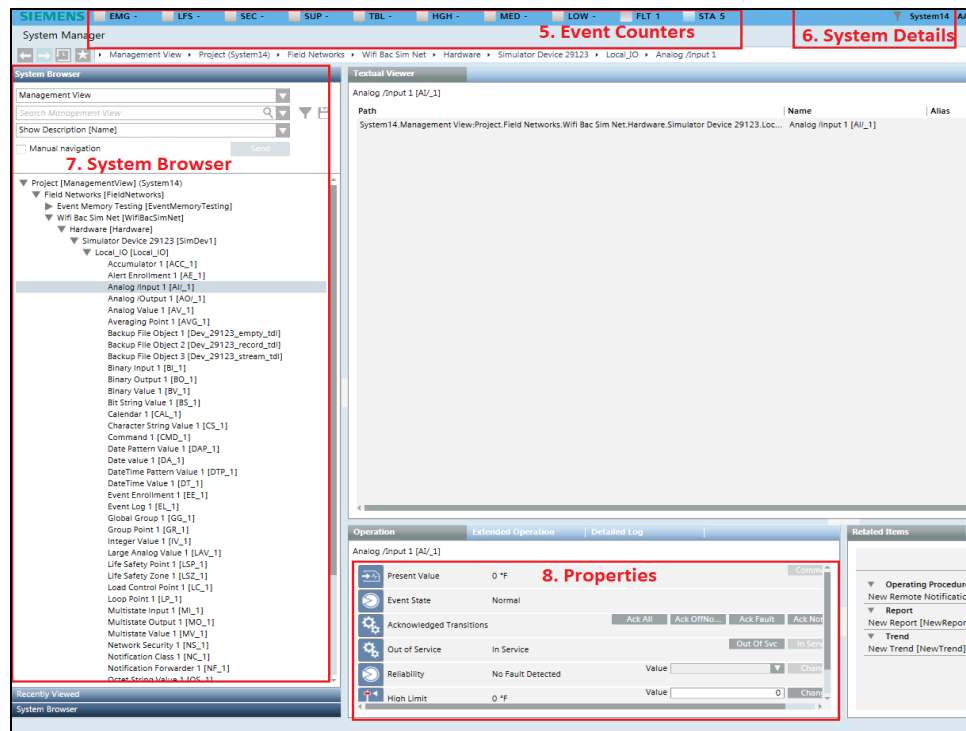
4.5.1 Summary

For quick reference of service category refer below table

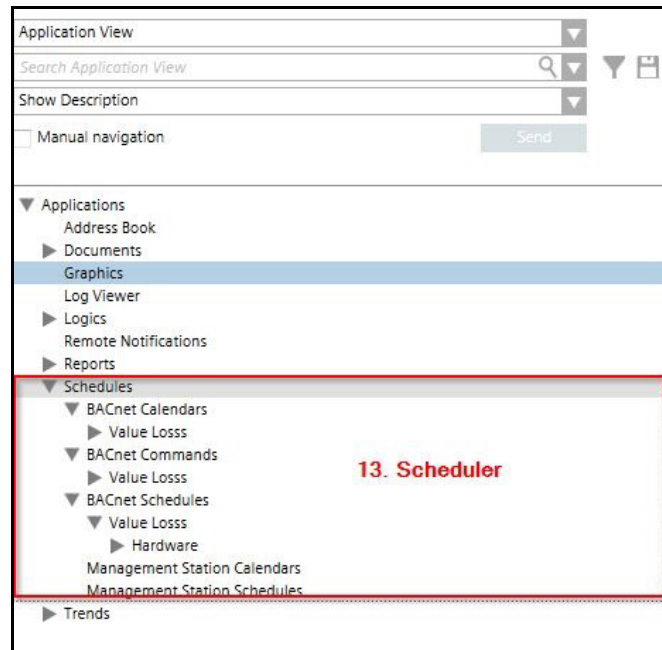
S.No.	Service	Code	Service Category
1	Token		Public
2	HeartBeat		Public
3	Command		Public
4	CommandSubscriptions		Public
5	Diagnostics		Public
6	EventCategorySounds		Public
7	EventCategorySoundsSubscriptions		Public
8	EventCounters		Public
9	EventCountersSubscriptions		Public
10	EventsCommands		Public
11	Events		Public
12	EventsSubscriptions		Public
13	Files		Partially Internal
14	HistoryLog		Public
15	Graphics		Internal
16	Images		Internal
17	Language		Public
18	Products		Public
19	Properties		Public
20	PropertyValues		Public
21	Settings		Internal
22	SystemBrowser		Public
23	SystemBrowserSubscriptions		Public
24	Systems		Public
25	Tables		Partially Internal
26	TrendSeries		Public
27	TrendSeriesInfo		Public
28	Users		Internal
29	Values		Public
30	ValuesSubscriptions		Public
31	Systems		Public
32	SystemsSubscriptions		Public
33	Scheduler		Internal
34	Related Items		Public
35	AccessRights		Internal
36	AccessRightsSubscriptions		Internal

4.6 Service mapping with Desigo CC(Basic reference)





12. Event List		
	Cause	Path
	At least one service not running properly	Project.Management System.Servers
	BACnet Driver Not Communicating (Configuration Error)	Project.Management System.Servers.Main Se rver.Drivers
	BACnet Driver Not Communicating (Configuration Error)	Project.Management System.Servers.Main Se rver.Drivers
	History Database (Not Connected)	Project.Management System.Servers.Main Se rver.History Database
	History Database (Not Connected)	Project.Management System.Servers.Main Se rver.History Database
	Transaction Mode Simple	Project.Management System.Servers



S.No.	Workflow	Services [45]
1	Image Download	Files[149]
2	Login Support	Token[45]
3	Change Password	Users[158]
4	Product Details	Products[146]
5	Event Counters	EventCounters[56]
6	Systeme Details	Systems[142]
7	System Browser	SystemBrowser[62]
8	Properties	1. Value[73] 2. Properties[86] 3. PropertyValue[83]
9	Graphics	Graphics[150]
10	Reports	Histrory Logs[118]
11	Trends	Trend[109]
12	Events	1. Event[49] 2. EventsCommands[54]
13	Scheduler	3. Scheduler[230]

4.7 Access rights for application services

Certain services in Web service interface can be related to applications in DesigoCC client. Application rights for such applications are also enforced service endpoints as well. Access to operation would be denied if application rights are not available with user.

Service	Endpoints	Application [ID- Name]	Operation [ID- Name]
Trend Service	Delete TrendViewDefinition	54 - Trends	1730 - Configure
	Delete TrendLogOnline	54 - Trends	1730 - Configure
	Create/Update TrendViewDefinition	54 - Trends	1730 - Configure
Schedule Service	Save Schedule	46 - Schedules	1474 - Configure
	Save Exceptions	46 - Schedules	1474 - Configure
	Save Schedule Options	46 - Schedules	1474 - Configure
	Save Calendar	46 - Schedules	1474 - Configure

4.8 WSI Configuration keys

Configuration keys in project config can be modified to change default behavior of NORIS. It follows Hierarchical configuration i.e. configuration defined at instance level can be used to override configuration at base configuration.

By default configuration at instance level is not defined and it will inherit config from base e.g. *MaxVirtualMemorySizeForProcess* is defined at base config only which can be overridden at instance level if certain instances require more memory for operation.

Configuration Key	Data Type	Default Value	Description
WhiteListDomain	String		Update CORS policy to allow only whitelisted domains in access-control response headers. Multiple domain names need to be separated by semi colon (;) . Key is optional and default configuration allows all domains. e.g. WhiteListDomain = http://xyz.com ; https://xyz.com ; https://abc.in
CachingDefaultMaxAge	Unit		Time duration in minutes to set Cache control max-age value in WSI response. Key is optional and default configuration sets max-age to 30 mins. Setting value as 0 will disable caching headers in response. e.g. CachingDefaultMaxAge = 60
CachingDefaultVaryHeaders	String		Header name to revalidate/refresh cache on browser side. This will set the Vary header in response if caching is enabled. Multiple header names need to be separated by semi colon (;) . Key is optional and default configuration sets vary to 'Authorization'. e.g. CachingDefaultVaryHeaders = Header1;Header2;Header3
MaxSnapshotsForSingleSession	String	5	Maximum Snapshots time for single session
MaxSnapshotInactiveTime	String	10	Maximum Inactive Time for Snapshot
MaxVirtualMemorySizeForProcess	String	4294967296	Maximum Virtual Memory Size For Process
maxBcmBufferSize	String	100000	Maximum Bcm Buffer Size
DirectoriesAccess	String	libraries	Directories Access

SignalrMessageBuffer	String	3000	Signalr Message Buffer size
SignalrKeepAlive	String	6	Signalr Keep Alive
ExpiresUtcTicketForCodeGrant	String	5	UtcTicket Expires time For Code Grant
LicensedClient	String	40	Maximum number of clients of type ,LicensedClient' allowed to login.
NonLicensedClient	String	10	Maximum number of clients of type , NonLicensedClient ' allowed to login.
WebApiThrottlingConfiguration	String	<pre>[{"ApiRoute": "/api/token", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/diagnostics", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/products", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/systems", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/users", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/licenses", "MaxRequestLimitPerSecond": 25}]</pre>	WSI endpoints web api throttling configuration used for limiting maximum allowed request rate. For more details please refer section 4.8.1.
EncryptionKeyExpiryDuration	String	4	Expiry duration (in minutes) for the asymmetric key pair generated on WSI server for secured credentials transaction from client to server.

4.8.1 Web Api Throttling Configuration

Web api throttling configuration can be given in quoted json format as shown in WSI Config keys table in section 4.8. This configuration supports three key-value pairs for each entry, details about which can be found in following table.

Key	Value
ApiRoute	Route of the WSI endpoint for which throttling is to be configured.
MaxRequestLimitPerSecond	Maximum number of allowed calls per second for endpoint mentioned in ApiRoute.
MaxRequestLimitPerMinute	Maximum number of allowed calls per minute for endpoint mentioned in ApiRoute.

Default configuration for web api throttling is added to the project while creating new project and upgrading the project which doesn't have this config-

uration present in WSI Config section of the Project config file. If this configuration is already present in the config file, then it will not be altered while upgrading the project. When this configuration is present in config file, the mentioned endpoints will be monitored, and the maximum number of calls will be limited to the configured number, beyond which the IP will be denied for more calls in that time interval. E.g., if for ApiRoute : "api/token", MaxRequestLimitPerSecond is set to 25, then for every second only 25 calls are processed for any IP and the requests more than 25 are denied to process per second. Both MaxRequestLimitPerSecond and MaxRequestLimitPerMinute can be set for any ApiRoute, in this case both the limits will be checked before processing requests. E.g., if for ApiRoute : "api/token", MaxRequestLimitPerSecond is set to 25 and MaxRequestLimitPerMinute is set to 200 then in this case the maximum number of allowed calls per second will limit to 25 and allowed calls per minute will be limited to 200, that means after processing 25 requests/second for 8 second, it will deny processing new requests till completion of one minute.

When requests to the api endpoints are denied due to frequent requests more than allowed limit, one will get message "Exceeded maximum number of allowed calls." and 403 (Forbidden) as a response code.

4.9 Time sync

It could be scenario where NORIS server and Client application has some time difference. This time difference can result in inconsistent data – Event time represented incorrectly, during commanding incorrect data being sent to server.

To sync the time between server and client , NTP time sync could be implemented - [Clock_synchronization_algorithm](#).

NORIS provides server's UTC datetime information as well as time difference between server time and client time if client pass client information to the server.

The information provided by Diagnostics Service [→129] can be used calculate time difference at client side.

5 Conventions

5.1 Case Sensitivity

Unless otherwise specified the system is case sensitive. This affects the API in terms of filters and search results.

5.2 Number Format

The API uses an invariant culture for the number formatting regardless of the logged in user. The decimal mark is a period.

5.3 Date/Time Format

Date and time representations under ISO-8601 acceptable formats are :

```
yyyy-MM-ddTHH:mm:sZ  
yyyy-MM-ddTHH:mm:ssZ  
yyyy-MM-ddTHH:mm:ss.FFFZ
```

**NOTE:**

Unless otherwise noted, the Zulu time zone is expected.

Examples:

```
2016-03-06T12:35:1Z  
2016-03-06T12:35:01Z  
2016-03-06T12:35:01.711Z
```

5.4 Terminology

Terminology	
Term	Description
Name ¹	Literal; supposedly unreadable; usually not displayed in a UI.
Descriptor ¹	Localized string in a readable form; supposed to be displayed to an end- user.
Designation ¹	Unique address of a node; supposedly unreadable; usually not displayed in a UI.
Location ¹	Localized, fully expanded path of a node; supposed to be displayed to an end-user.
ObjectId	Identifies an object in the system.
PropertyId	Identifies a property and its object in the system.
AttributeId	Identifies an attribute, its property and its object in the system.
ObjectOrPropertyId	An object ID or a property ID is expected.
PropertyName	Name of a property.

¹ see also Naming [→ 245]

5.5 Maximum page size

Maximum page size which is supported in case of paging is 100000 records per page.

In case if page size is not specified then above mentioned size will be considered as page size. And if valid page size is specified then minimum from the both will be considered as page size.

5.6 Request per system

In a distributed environment (multiple systems) a request may only contain data points to a single system. If one needs to process data points of multiple systems multiple requests have to be invoked.



NOTE:

Unless explicitly stated, a service request may only affect a single system.

6 General API Specification

The REST API provides programmatic access to read, write and command the system data and functionality. The API identifies users using OAuth [4] [→ 19]; responses are available in JSON [6] [→ 19].

6.1 HTTP Methods

Following REST principles, one can call the following HTTP methods on the API resources: GET, POST, PUT, and DELETE.

Method (HTTP Verb)	Description
GET	Retrieves a list of resources, or a specific resource.
POST	Creates a resource, passes in a parameter, or performs an action on a resource.
PUT	Updates an existing resource.
DELETE	Deletes an existing resource or subscription.

6.2 Resources

The Web Service Interface API follows REST principles and therefore exposes data and functionality as resources [7] [→ 19].

Every resource is represented as an URI (see Appendix A in [8] [→ 19]). As a summary a resource looks like the following:

`scheme://domain:port/path?query`

Part	Description
scheme	Usually HTTPS; HTTP should only be used for testing
domain	Hostname of host where manager is running
port	Usually 8080 for HTTP and 8443 for HTTPS
path	Starts with api followed by the service-identifier and optional sub paths (e.g. api/languages/en-us); see sections below
query	Optional; depends on resource (see sections below)

Example

`https://ch1w80012:8443/api/events?sorting=1`



NOTE:

In a production environment, the API runs most likely behind a reverse proxy. In this case, the URI might look differently and might hide some information from the Client.

Example

`https://www.example.com/example/api/events?sorting=1`

Therefore the general format of a URL is:

`{entry-point}/api{resource-address}`

Where **entry-point** is `http://[hostname]:[port]` by default or dependent on the deployment, and **resource-address** in EBNF is:

`resource-address = path [ "?" query ]`

`path = /root-resource | path "/" sub-resource`

`query = parameter { "&" parameter }`

`parameter = key [ "=" value ]`

6.3 Status Codes

6.3.1 HTTP Status Codes

Status Code	Meaning	Description
200	OK	Request processed. NOTE: Although the request has been processed as a whole, some of the sub results could still be invalid. See Error Code [→ 44] to check sub-results.
204	No Content	The server successfully processed the request and is not returning any content.
304	Not Modified	Resources are not modified from last request. NOTE: This status code is returned only if Etag header is present in earlier response and request contains If-None-Match header with Etag value and resource is not modified. See caching support [→248] .
400	Bad Request	Request rejected by application.
401	Unauthorized	Invalid or expired token presented. User needs to login.
403	Forbidden	The server understood the request but refuses to authorize it.
404	Not Found	Resource couldn't be found for the provided URL and query parameters.

6.3.2 Error Code

Code	Description
0	No error occurred.

6.4 Encoding

Data is encoded in UTF-8.

6.5 Array in Query Parameter

In case a query parameter represents an array, the array is passed as a JSON array [6] [→ 19].

Example

```
someurl?Ids=[1,2,3]
```

6.6 HTTP caching

Browser can create local cache of response data based on certain headers in HTTP headers, thus avoiding repetitive calls to server for same set of data .For more details on HTTP caching refer : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Caching>

Details of implemnetation for HTTP caching in NORIS is mentioned in Support for Browser Caching[→9.4]

7 Services

7.1 Token Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/token	Public	x (Logout only)	

Note : For clients of type **LicensedClient**, the maximum and default number of logins allowed is 40. For clients of other types, the maximum and default number of logins allowed is 10. The numbers can be configured in the project config file > 4.8. But maximum values are limited to 40 and 10 for LicensedClient and NonLicensedClient respectively.

7.1.1 Logging with Username and Password

Resource Path	/token		
Category	Public		
HTTP Method	POST		
Content Type	application/x-www-form-urlencoded		
Body	Property	Content	Description
	grant_type	password	grant_type=password
	username	[username]	Username provided by user.
	password	[password]	Password provided by user.
	client_id	[client identifier]	Optional : LicensedClient Client identifier of application , certain services may require use of license that are available with license server .
Response	HTTP 200 OK A Login object (see Token Service Login [203]) HTTP 403 Forbidden User password is expired and need to be changed first		
Examples	Request: POST /WSI/api/token Post Data: grant_type=password&username=DefaultAdmin&password=Siemens@12345 Response: { "access_token": "AQAAANCMnd8BFdERjHoAwE_Cl-sBAAAAJbLmifXxrkmVwrHtNdOoBQAAAAACAAAAAAQZgAAAAEAACAAAAAD1DER3xtPANmCUT4lAg_C0ms9Y6UL6NQxW5WZkUNyMAAAAAAOGAAAAIAACAAAAA0bZ568TgfMVBn6qRpR-wtLTaxWHhjGF8dPPuv3AXH7ZABAABx6NBN9NVK-DxS0muRa7LnUlKUz8gxOaq3Nvd9jZptozEzQxSsdNs0JV5mXfuSDn2qIqjihwaxv4i4ge7qyxQ_xIPuMW4cwWs3qSnqCDBJBku2WwqTt12zrZhVfVR4uKnXnPHdXV-_lqqpPgdtay9CPWR_fns4WyaY_Rj_CYmaWidZlUmVuJUZNkn96LeTD5nN2Df1EKKJ8g2L7DDg8ZLpCERCHoCist6CW47w7hlhMElZySDUIUbZ5GotMIUMSaJrMk9vQLPSn_HMVgyPYog79HWQC0xZelSeUYCAD0Pj5fzw6luzjYz8fQ20BjIw4hH22Ef9xQocYnjGI4SlhxsAt1Bx6w_T9FPczSjJ8OCHb51a9hmWE8mbIw7mGsJbn2P1Y_aDvEc85qBwt_XEMQ3vMPWcfMoeQgTGO38NB04YsZPoeyR_Md_4p7jdz07qUHWqDR2tdzI5iM5tEVyMGIysuXr6pKaxj_9IT67g9FMFzePvUCb0qieCT2ksQVwrVC5F7ZtY1uQb-a0jNefC76cXQAAAACFzePjZimp2xZRTFV-rwxk_cclpFGlU44GHRyt76ZrgWsIMgcQZKE8M5OySwVih10d5ukHFgcXa7		

	<pre>BQJ4M37ffI", "token_type":"bearer","expires_in":2591999, "user_name":"DefaultAdmin", "user_descriptor":"Default Administrator", "user_profile":"DEFAULT.ldl", "flex_user_profile":"DEFAULT", "user_inactivity_timeout":"0", "password_reminder_counter":"0", "personally_used":"false" }</pre>
	Example: User password is expired Request: POST /WSI/TestClient/api/token Post Data: grant_type=password&username=testuser&password=123 Response: HTTP 403 Forbidden { "Id":2400003,"Error":"password_expired","Details":"Logon rejected. Password of this account is expired and need to get changed first." }
	Example: User account locked Request: POST /WSI/TestClient/api/token Post Data: grant_type=password&username=testuser&password=123 Response: HTTP 403 Forbidden { "Id": 2400005,"Error":"exceeded_bad_attempts_limit", "Details":" Entered wrong credentials too many times." }

7.1.2 SSO Log in

It is only available for internal gms application developers. Get Single sign on (SSO) token directly from Logon orchestrator and provide it to WSI using below interface.

Resource Path	/token		
Category	Public		
HTTP Method	POST		
Content Type	application/x-www-form-urlencoded		
Body	Property	Content	Description
	grant_type	client_credentials	grant_type= client_credentials To login provide SSO token
	Client_Id	SSO token	Retrived from Logon Orch.
Response	HTTP 200 OK A Login object (see Token Service Login [→87])		
Examples	Request: POST /WSI/TestClient/api/token Post Data: grant_type=client_credentials&Client_Id=811d8bb9856598ff60c05ad525746bef40517ab51020faae303c28afe932a323d45469eb8b9e2ff85ef1c00dc7b4cfcf3e1bc33e6cd878e5ed6b63d2271b9d6ef98248544bd5ea16d6fdb72aacd960ad8611c98f01d7f85e8e924829edb7dc689fb248ba9fd5093faf6cb942587e9516a17bf0c13af994c0814e53dce98923a67a0fa02015f44580e0016e862aa600a8ecdd2a2597a3074bd1c5d8af3e2db5a2868de2176a1fa533e4dbba308279		

	<pre>8dde4938cdc95c086055bedc6b4d923ebec2ce16d8eab3405955ae6c88 1d6d4f274796b166cc541e18a16039282de247b3b0083508e60674088a 3f7d6fcff12c0d2144414faa09cdc376581c0e2ff1711d34</pre> Response: <pre>{ "access_token": "AQAAANCMnd8BFdERjHoAwE_Cl- sBAAAAJbLmifXxrkmVwrHtNdOoBQAAAAACAAAAAAQZgAAAAEAACAAAAAC Nj5CbeMxGu7jEh6ZgK_sEnNOSLtVZfaE2SriUxy8FwAAAAAOGAAAAIAAC AAAAAILv8nDmujV57WEjOGUGyCH6ofwAI6RQyt7sgu6GFM1ZABAABEXjxX AqZnkyYVctuZdpqrEUHVqh5uc16PT4qC3EjVSK9cnq5qD2rtDLFaLJQoO- Nklg7fpSmQBF84Ts7FpkGmu__EYhq4lMgzJWWOKiXbWTJhG1Lj12W3wmyx 3zfENU8chB_uoKM4_nB57Kule77cYocsk0zIwaayqLTvLQ1sXbIIESoNGY ExyHoEu-Cetv2zAw-00hAj6p9UG8rIQXfyUxYIDpkBYLdcrvqEt18GQ8- pjAWIQac46WjA_W7nYm7MZGJYmHgtiet5SXtQz5cm88y1JBMxrmgCMhDx ub7QZ- Xq6V6V11iPhMEDnoPnd5KIrlmdTDPGuD7XxNFbKOFdsxa9GQxN6KLxM6og WlWMP3R3edlTFP2KvF4C6V8o6R2fKFU71HaDatJdqLjyc5MdUpIs19daZz G-eW7XNM6_haiHEtN9F-6Sxk2q-5O9wNbeYGuPHv_PbRxxEYF- bRauayKQnJ3Xp0321zeppaXjyep7KtlezrtBaZ1E- 3WIf4QKfKjPnaxT84XMWYdXe- PQAAAAp5JIPc410Sk4D_ATLk7FQX000sKZINkr6JzQQCVQXMuK9qC9hQX WJSrOCk3wVhUkSMgKz1xSvi2xRQ_V9T40k", "token_type": "bearer", "expires_in": 2591999, "user_name": "DefaultAdmin", "user_descriptor": "Default Administrator", "user_profile": "DEFAULT.ldl", "flex_user_profile": "DEFAULT", "user_inactivity_timeout": "0", "password_reminder_counter": "0", "personally_used": "false" }</pre>
--	---

7.1.3 Default Logon (Mutual TLS)

This method enables the client to login with a client certificate, on page load.

GMS certificate should be configured beforehand. Login is successful if the certificate is valid.

Resource Path	/token		
Category	Public		
HTTP Method	POST		
Content Type	application/x-www-form-urlencoded		
Body	Property	Content	Description
	grant_type	client_credentials	grant_type= client_credentials To login, provide client certificate
	token_endpoint_auth_method	tls_mutual_auth	Fixed parameter to be passed for mutual tls. [12]
	Client_Id	DefaultLogon	Fixed string value.
Response	HTTP 200 OK A Login object (see Token Service Login [→87])		
Examples	Request: POST /WSI/api/token Post Data: grant_type=client_credentials&Client_Id=LicensedClient&token_endpoint_auth_method=tls_mutual_auth Response: <pre>{ "access_token": "AQAAANCMnd8BFdERjHoAwE_Cl- sBAAAAJbLmifXxrkmVwrHtNdOoBQAAAAACAAAAAAQZgAAAAE- AACAAAAABpcv7QPhIBZ6ScppqXVufFpHYKWCDEompqv9waXbL- BAAAAAOGAAAAIAACAAAABrUNEhn- 4dsVwkv1LL6EMU5_Z6bE_5aVLn-TAszFNOy5ABAAB_ZYoK6ka-</pre>		

	<pre> 3EVBzGM79LcR__Cq3fMUlwXlXki_hloj3vEzb9pODrwwbWwF3NHIS4 NAm5Ugf-S3oVm5y8- 4n4tH86ydUzoVVl_6wFYmuiBmDRI8nyEFJrz1U2O26sr0PeUDjTYx6 kxTzsZL6w6jQT95z5PTQh8SDzUnJJQO0i5zFtNH-b- 0I6skazkjVzuEwVBKqSsqkGA0DG64RNqbCUEhZMWDxnjtEdQi7D5_w S2TEhdfemBZ25XcgYu4xbMo8- uzBLKPO0EB82YplGJvUKgzMbeouqle2- AUMP3mv01J8VnlRXls8IQprQP9NpHmK23sVz9zEgznYEkatWU_3hc6 _5a5pQA2Y4_DTMS7mWUNN6eUdlQT_2WbvbEzAldTIJ430Zo_JraBnC jgaHm3FDMpRcHtji2AGlCm4cmMLPEsnk- PBZemSvUp4HR62VzK4nlyH0CXZBPAO6YR07t7mHdq_GNIWksBQH_QL eSws0EHstfGsgmPgqm0FpJibBW- lvG41ldW7RzAVKkHZ6sLc03rQAAAAJ_25yeNdodB63HNqT_0KWF6zb VVtX9H7tCtzzS1MPhxSSEGKw8tEpMmQn65yb8n5SAzBBwDJmO8OEg7 rutH3L8", "token_type":"bearer","expires_in":2591999, "user_name":"SWAccForDefaultLogon", "user_descriptor":"Software account for default lo- gon", "user_profile":"DEFAULT.ldl", "flex_user_profile":"DEFAULT", "user_inactivity_timeout":"0", "password_reminder_counter":"0", "personally_used":"false" } </pre>
--	--

7.1.4 Logging out

Logging out a User	
Resource Path	/token
Category	Public
HTTP Method	DELETE
Response	HTTP 200 OK
Examples	Request: DELETE /NBWSITest/NorthboundWebApiClient/api/token Response: HTTP 200 OK

7.2 Heartbeat Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/heartbeat	Public	x	

7.2.1 Extending the Lifetime of a Session and its Bearer Token

If a Client logs in, a session is created and bound to the bearer token that the Client receives as part of a successful login request. The session and its bearer token expire after some time of inactivity (default: 10 minutes). Inactivity is detected if no subscription is alive and no protected (authenticated) call is made within this timeout.

The **/heartbeat** resource can be called to artificially signal activity and to reset the timeout.

Lifetime of a Session.	
Resource Path	/heartbeat
Category	Public
HTTP Method	POST
Response	HTTP 200 OK
Examples	Request: POST /NBWSITest/NorthboundWebApiClient/api/heartbeat Response: HTTP 200 OK

7.3 Event Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/events	Public	x	
/eventssubscriptions	Public	x	x

Events depend on the event type (see [20]). The current event type can be read with the following object ID: ManagementView_System_Settings.EventSchema.

7.3.1 Retrieving a List of Events

**NOTE:**

1. In order to retrieve grouped events right combination or unique identifier have to be provided in parameters
2. Wild card characters are supported in eventfilter search wherever applicable.
3. Events extension commands are also supported in response.(e.g. StartReno etc)

Resource Path	/events		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	sorting	int	See EventSorting [→ 206]
	page	int	
	size	int	
	ids	Array of string	
	caseSensitive	bool	
	categories	Array of int	

	<table><tr><td>cause</td><td>string</td><td></td></tr><tr><td>creationTime</td><td>int</td><td>See EventTimeFilter [→ 51]</td></tr><tr><td>creationTimeFrom</td><td>DateTime</td><td></td></tr><tr><td>creationTimeTo</td><td>DateTime</td><td></td></tr><tr><td>disciplines</td><td>Array of int</td><td></td></tr><tr><td>Systemid</td><td>Uint</td><td>System Identifier</td></tr><tr><td>states</td><td>Array of string</td><td>See EventState [→ 205]</td></tr><tr><td>designations</td><td>Array of string</td><td></td></tr><tr><td>sourceName</td><td>string</td><td></td></tr><tr><td>sourceDescription</td><td>string</td><td></td></tr><tr><td>locationDescription</td><td>string</td><td></td></tr><tr><td>locationName</td><td>string</td><td></td></tr><tr><td>objectState</td><td>Array of string</td><td>See ObjectState [→ 205]</td></tr><tr><td>maintenance</td><td>string</td><td>See MaintenanceState [→206]</td></tr></table>	cause	string		creationTime	int	See EventTimeFilter [→ 51]	creationTimeFrom	DateTime		creationTimeTo	DateTime		disciplines	Array of int		Systemid	Uint	System Identifier	states	Array of string	See EventState [→ 205]	designations	Array of string		sourceName	string		sourceDescription	string		locationDescription	string		locationName	string		objectState	Array of string	See ObjectState [→ 205]	maintenance	string	See MaintenanceState [→206]
cause	string																																										
creationTime	int	See EventTimeFilter [→ 51]																																									
creationTimeFrom	DateTime																																										
creationTimeTo	DateTime																																										
disciplines	Array of int																																										
Systemid	Uint	System Identifier																																									
states	Array of string	See EventState [→ 205]																																									
designations	Array of string																																										
sourceName	string																																										
sourceDescription	string																																										
locationDescription	string																																										
locationName	string																																										
objectState	Array of string	See ObjectState [→ 205]																																									
maintenance	string	See MaintenanceState [→206]																																									
Response	HTTP 200 OK A Page<Events, Event>-object (see Page<Name, Type>) [→ 199]																																										
Examples	Request: GET /api/events?sorting=0 Response: { "Events": [{ "Id": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0", "EventId": 86, "CategoryId": 9, "CategoryDescriptor": "Fault", "State": "Unprocessed", "Cause": "Services running properly", "SrcPropertyId": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value", "SrcObservedPropertyId": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value", "SrcState": "Quiet", "SrcSystemId": 1, "SrcViewName": "ManagementView", "SrcViewDescriptor": "Management View", "SrcDesignation": "ManagementView.ManagementSystem.Servers.Server", "SrcLocation": "Project.Management System.Servers.Main Server", "SrcName": "Server", "SrcDescriptor": "Main Server", "SrcDisciplineId": 0, "SrcDisciplineDescriptor": "Management System", "CreationTime": "2015-08-04T09:51:57.162Z", "Direction": "None", "SuggestedAction":WaitforCondition, "NextCommand":NoActionPossible, "InfoDescriptor": "At least one service not running properly", "Commands": [{ 																																										

	<pre> "Id": "Suspend", "EventId": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0", "Configuration"=1 "_links": [{ "Rel": "command", "Href": "api/eventscommands/System1%3aManagementView_ManagementSystem_Servers_Server.SoftwareRunning%3a_alert_hdl.2._value~635742985171620000~0/Suspend", "IsTemplated": false }] }, { "Id": "Ack", "EventId": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0", "Configuration"=1 "_links": [{ "Rel": "command", "Href": "api/eventscommands/System1%3aManagementView_ManagementSystem_Servers_Server.SoftwareRunning%3a_alert_hdl.2._value~635742985171620000~0/Ack", "IsTemplated": false }] }], "_links": [{ "Rel": "category", "Href": "api/tables/categories/9", "IsTemplated": false }, { "Rel": "discipline", "Href": "api/tables/disciplines/0", "IsTemplated": false }] }], "Total": 1, "Page": 1, "Size": 100000, "_links": [{ "Rel": "page", "Href": "api/events?page={page}", "IsTemplated": true }] } </pre>
--	---

7.3.2 Creating a Subscription for Events (Obsolete)

Obsolete: Use channelized subscription instead.

Resource Path	/sr/eventssubscriptions/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	connectionId	string	Identifier of subscription Required
	sorting	int	See Event Sorting [→ 206]
Body	EventFilter[→ 206]		
Response	HTTP 200 OK		
Examples	Request: POST /api/sr/eventssubscriptions/983a0ade-7e7a-4812-bdf3-ea9ea59a7e9e?sorting=0 Response: HTTP/1.1 200 OK		

If subscription Post request submitted again then latest request will be considered and notifications will be pushed with the filter criteria of latest submitted request.

7.3.2.1 Push Notification

After subscribing for events, a list of current events will be sent in an initial notification. Afterwards, the Client will get notified about events according to the filter set (if any).

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. eventsHub : legacy		
Function	notifyEvents (data)		
	Property	Type	Description
	data	Array of Event	See Event [→ 204] Check the flag Deleted for checking whether the event got removed.

7.3.3 Creating Channalized Events Subscription

If subscription status will be pused through notifySubscriptionStatus function available at client side. Events updates will be notified over notifyEvents [→5259].

Resource Path	/sr/eventssubscriptions/channelize/:requestId/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required

	connectionId	String	Identifier of subscription hub connection Required
	sorting	Int	See Event Sorting [→ 206]
Body	EventFilter[→ 206]		
Response	HTTP 200 OK		
Examples	Request: POST /api/sr/eventssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/983a0ade-7e7a-4812-bdf3-ea9ea59a7e9e?sorting=0 Response: HTTP/1.1 200 OK notifySubscriptionStatus: <pre>{ "ErrorCode": 0, "SortingOrder": 0, "EventFilter": { "Ids": [], "Categories": [1, 2, 3, 10], "CreationTime": 6, "Disciplines": [0, 20, 100, 200], "States": ["Unprocessed", "ReadyToBeReset", "ReadyToBeClosed", "WaitingOPCompletion", "Acked"], "systemID": 1 }, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5" }</pre>		

If subscription Post request submitted again then latest request will be considered and notifications will be pushed with the filter criteria of latest submitted request.

7.3.3.1 Subscription Status Notification

After subscribing for event counters, the subscription status will be sent only once.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. eventsHub : legacy		
Function	notifySubscriptionStatus(status)		
	Property	Type	Description
	Status	EventSubscription	See

		Status object	EventSubscriptionStatus[→206]
Example	<pre>{ "ErrorCode": 0, "SortingOrder": 0, "EventFilter": { "Ids": [], "Categories": [1, 2, 3, 10], "CreationTime": 6, "Disciplines": [0, 20, 100, 200], "States": ["Unprocessed", "ReadyToBeReset", "ReadyToBeClosed", "WaitingOPCompletion", "Aked"], "systemID": 1 }, "RequestID": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor": "notifyEvents" }</pre>		

7.3.4 Modifying a Subscription for Events

Change an existing subscription for events: the same procedure as Creating a Subscription for Events [→ 52] but use PUT as HTTP method.

7.3.5 Deleting a Subscription for Events

Delete an existing subscription for events: the same procedure as Creating a Subscription for Events [→ 52] but use DELETE as HTTP method.

7.4 EventsCommands Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/eventscommands	Public	x	

7.4.1 Executing a Command on an Event

Resource Path	/eventscommands/:eventId:commandId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description

	eventId	string	Required
	commandId	string	See EventCommandId [→ 51] Required
	commentPasswordData	ValidationData See[→ 208]	Event-command parameter details (comment ,password and supervisor details) sent from client if required.
	treatmenttype	string	Optional query parameter to specify type of treatment being performed. This parameter is applicable only id command is Select.
Response	HTTP 200 OK		
Examples	Request: POST /api/eventscommands/System1%3aManagementView_ManagementSystem_Servers_Server.SoftwareRunning%3a_alert_hdl.2._value~635742985171620000~0/Ack Request Body: {"MultiLangComments":["comment1","comment2"], "Password":"password","SuperName":"supervisorUserName", "SuperPassword":"supervisorPassword"} Response: HTTP/1.1 200 OK		

7.4.1.1 EventCommandId

:commandId must have one of the following values.

Note :Newly added commands:Events extension commands are also supported in response.(e.g. StartReno etc)

:commandId	Description
select	Signals operator is treating an event
suspend	Signals operator stopped treating an event
ack	Acknowledge event
reset	Reset event
silence	Silence event
unsilenced	Unsilence event
close	Close event

7.4.2 Bulk Commanding on Events

Resource Path	/eventscommands/:commandId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	commandId	string	See EventCommandId [→ 51] Required
	BatchEventsCommandExecution	BatchEventsCommandExecution See[→209]	Event-command parameter details (eventId,comment ,password and supervisor details) sent from client.

			Required
	treatmenttype	string	Optional query parameter to specify type of treatment being performed. This parameter is applicable only if command is Select.
Response	HTTP 200 OK		
Examples	Request: POST /api/eventscommands/Ack Request Body: <pre>{ "ValidationInput": { "MultiLangComments": ["Comment en-US", "Comment de-DE"], "Password": "currentUserPassword", "SuperName": "Supervisor1", "SuperPassword": "SupervisorPassword" }, "EventIds": ["System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0"] }</pre> Response: <pre>[{ "EventId": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0" }]</pre> HTTP/1.1 200 OK		

7.5 EventCounter Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/eventcounters	Public	x	
/eventcounterssubscriptions	Public	x	x

7.5.1 Retrieving Event Counters

When retrieving Event counters two options are available with parameter categoryId which is optional. If categoryId is provided then Event counter for that particular categoryId will be returned as EventCounter object. And if categoryId is not provided then Event counters for all the categories in the system will be provided as an EventCounterList object. In both the cases response will be different depending on the request made. For more details see the examples below.

Note : Link has also made available to provide event category colors information. See [→210]

Resource Path	/eventcounters/:categoryId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	categoryId	int	Identifier of category
Response	HTTP 200 OK 1. When categoryId is provided: EventCounter object see [→ 52] 2. When categoryId is not provided: EventCounterList object see [→ 52]		
Examples	Request1: categoryId is provided GET api/eventcounters/9 Response1: <pre>{ "CategoryId": 9, "CategoryDescriptor": "Fault", "TotalCount": 1, "UnprocessedCount": 1, "TotalSubsequentGrouping": 0, "UnprocessedSubsequentGrouping": 0, "_links": [{ "Rel": "eventcounter", "Href": "api/eventcounters/9", "IsTemplated": false }, { "Rel": "categorycolor", "Href": "api/tables/categories/9/subtables/colors", "IsTemplated": false }] }</pre> Request2: categoryId is NOT provided GET api/eventcounters Response2: <pre>{ "TotalCounters": 1, "TotalUnprocessedCounters": 1, "EventCategoryCounters": [{ "CategoryId": 1,</pre>		

	<pre> "CategoryDescriptor": "Emergency", "TotalCount": 0, "UnprocessedCount": 0, "TotalSubsequentGrouping":0, "UnprocessedSubsequentGrouping"=52, "_links": [{ "Rel": "eventcounter", "Href": "api/eventcounters/1", "IsTemplated": false }, { "Rel": "categorycolor", "Href": "api/tables/categories/1/subtables/colors", "IsTemplated": false }] }, { "CategoryId": 2, "CategoryDescriptor": "Life Safety", "TotalCount": 0, "UnprocessedCount": 0, "TotalSubsequentGrouping":0, "UnprocessedSubsequentGrouping"=52, "_links": [{ "Rel": "eventcounter", "Href": "api/eventcounters/2", "IsTemplated": false }, { "Rel": "categorycolor", "Href": "api/tables/categories/2/subtables/colors", "IsTemplated": false }] }, { "CategoryId": 3, "CategoryDescriptor": "Security", "TotalCount": 0, "UnprocessedCount": 0, "TotalSubsequentGrouping":20, "UnprocessedSubsequentGrouping"=16, "_links": [{ "Rel": "eventcounter", "Href": "api/eventcounters/3", "IsTemplated": false }, { "Rel": "categorycolor", "Href": "api/tables/categories/3/subtables/colors", "IsTemplated": false }] }] </pre>
--	---

	}] }
--	-------------

7.5.2 Creating a Subscription for Event Counters (Obsolete)

Obsolete: Use channelized subscription instead.

Resource Path	/sr/eventcounterssubscriptions/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	connectionId	string	Identifier of subscription Required
Response	HTTP 200 OK		
Examples	Request: POST api/sr/eventcounterssubscriptions/448b3117-44ef-4d8f-baa2-7274aa3050af Response: HTTP/1.1 200 OK		

7.5.2.1 Push Notification

After subscribing for event counters, the current event counters will be sent in an initial notification. Afterwards, any change will be notified.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. eventCountersHub : legacy		
Function	notifyEventCounters(data)		
	Property	Type	Description
	data	EventCounterList	See EventCounterList [→ 52]

7.5.3 Creating Channelized Event Counters Subscription

If subscription status will be pushed through notifySubscriptionStatus function available at client side. EventCounter changes will be notified over notifyEventCounters [→59].

Resource Path	/sr/eventcounterssubscriptions/channelize/:requestId/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request.

			Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub connection Required
Response	HTTP 200 OK		
Examples	Request: POST api/sr/eventcounterssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/448b3117-44ef-4d8f-baa2-7274aa3050af Response: HTTP/1.1 200 OK notifySubscriptionStatus: { "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifyEventCounters" } 		

7.5.3.1 Subscription Status Notification

After subscribing for event counters, the subscription status will be sent only once.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. eventCountersHub : legacy		
Function	notifySubscriptionStatus(status)		
	Property	Type	Description
	Status	Subscription Response	See Subscription Status [210]
Example	{ "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifyEventCounters" } 		

7.5.4 Deleting a Subscription for Event Counters

Delete an existing subscription for event counters: the same procedure as creating event counters subscription [→ 32] but use DELETE as HTTP method.

7.5.5 Retrieving Event Suppressed objects count

Event Suppressed object returns the current count of all the suppressed object. For more details see the examples below.

Resource Path	/sr/eventcounters/suppressedobjects
Category	Public
HTTP Method	GET
Response	HTTP 200 OK
Examples	Request: GET

	<code>api/eventcounters/suppressedobjects?api_key=defaultadmin%3Aa</code> Response: { "SuppressedCount":3 }
--	--

7.5.6 Creating Channelized Event Suppressed Objects Subscription

It will subscribe the Suppressed Objects. Subscription status will be available at client side through `notifySuppressedObjects` function.

Resource Path	/sr/eventcounterssubscriptions/channelize/suppressedobjects/{requestId}/{connectionId}		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for each request. Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub connection Required
Response	HTTP 200 OK		
Examples	Request: POST <code>api/sr/eventcounterssubscriptions/channelize/suppressedobjects/0db77356-bc3f-0275-7262-c28ff19cfb73/8468d443-17b8-41c0-9e41-b12f45a07d22</code> Response: HTTP/1.1 200 OK notifySuppressedObjects: { "ErrorCode":0, "RequestId":"c8ead472-10b8-0f85-b20e-eabd58bc7054", "RequestFor":"notifySuppressedObjects" }		

7.5.6.1 Suppressed Objects Subscription Status Notification

After subscribing for Suppressed object, the subscription status will be sent only once.

Type	Name (case sensitive)		
Hub	norisHub : recommended		
Function	notifySuppressedObjectsSubscription (status)		
	Property	Type	Description
	Status	Subscription	See Subscription Status [210]

	Response
Example	<pre>{ "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifyEventCounters" }</pre>

7.5.7 Deleting a Subscription for Suppressed Objects

Deleting Suppressed objects subscriptions.

Resource Path	/sr/eventcounterssubscriptions/suppressedobjects/{connectionId}		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	connectionId	string	Identifier of subscription Required
Response	HTTP 200 OK		
Examples	Request: DELETE api/sr/eventcounterssubscriptions/suppressedobjects/7cb4f8c2-d524-4532-9305-8fac31db175e Response: HTTP/1.1 200 OK		

7.6 System Browser Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/systembrowser	Public	x	
/systembrowsersubscriptions	Public	x	x

7.6.1 Retrieving a List of Views of a System

Retrieves list of Views in the current system. This interface have two optional query parameters systemId and viewId. If systemId is received as zero or if it is not provided then current systems id will be taken else received systemId will be taken for further operations. If viewId is provided then only the view, which have viewId same as that of in the request will be returned in the response. If viewId is not provided then all the views in the system will be returned in the response.

Resource Path	/systembrowser		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	ID of a system 0 or Not Provided: All system will be

			considered >= 1: Specific System will be considered
	viewId	uint	ID of specific View to be retrieved Not Provided : All views will be returned > 0: Views will be returned for provided ViewId
Response	HTTP 200 OK An array of View objects see [→ 55]		
Examples	Request: GET api/systembrowser?systemId=1&viewId=10 Response: <pre>[{ "SystemId": 1, "ViewId": 10, "Descriptor": "Application View", "Designation": "System1.ApplicationView", "Name": "ApplicationView", "SystemName": "System1", "ViewType": 0, "_links": [{ "Rel": "systembrowser", "Href": "api/systembrowser/1/10/System1.ApplicationView", "IsTemplated": false }] }]</pre>		

7.6.2 Retrieving a List of Browser Objects

Retrieve a list of browser objects for a specific view. To retrieve child objects of an object pass Designation of the parent object as node parameter.

Resource Path	/systembrowser/:systemId/:viewId/:node		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	viewId	uint	ID of a view Required
	node	string	Designation of a parent browser object (Node or View) Required
Response	HTTP 200 OK An array of BrowserObject objects see [→ 55]		
Examples	Request: GET api/systembrowser/1/9/System1.ManagementView Response: <pre>[</pre>		

```

{
  "HasChild": true,
  "SystemId": 1,
  "ViewId": 9,
  "Name": "ManagementView",
  "Descriptor": "Project",
  "Designation":
"System1.ManagementView:ManagementView",
  "ObjectId": "System1:GMS_Aggregator_5",
  "Attributes": {
    "Alias": "Project Alias",
    "DefaultProperty":
"StatusPropagation.AggregatedSummaryStatus",
    "ObjectId": "GMS_Aggregator_5",
    "DisciplineDescriptor": "Management System",
    "DisciplineId": 0,
    "SubDisciplineDescriptor": "System Settings",
    "SubDisciplineId": 4,
    "TypeDescriptor": "View Element",
    "TypeId": 8000,
    "SubTypeDescriptor": "System Folder",
    "SubTypeId": 8014,
    "ManagedType": 56
  },
  "Location": "System1.Management View:Project",
  "_links": [
    {
      "Rel": "systembrowser",
      "Href":
"api/systembrowser/1/9/System1.ManagementView%3aManagement
View",
      "IsTemplated": false
    },
    {
      "Rel": "properties",
      "Href":
"api/properties/System1%3aGMS_Aggregator_5?requestType=0",
      "IsTemplated": false
    }
  ]
}
]

```

7.6.3 Searching for Browser Objects

Search for browser objects in a provided system.

Resource Path	/systembrowser/:systemId/:viewId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	viewId	uint	ID of the view in which search needs to be performed
	searchString	string	String to be searched for

		Required
searchOption	int	See SearchOption [→ 37]
caseSensitive	bool	Only valid for search options 0 or 1
groupByParent	bool	If true, group search results by parent node
size	int	Number of browser objects per page
page	int	Current page number (default: 1)
disciplineFilter	string	Discipline and subdiscipline filters See Discipline- and Objecttype Filters [→ 677]
objecttypeFilter	string	Type and subtype filters See Discipline- and Objecttype Filters [→ 677]
aliasFilter	string	Case-sensitive alias filter
alarmSuppression	Boolean	When a Boolean is provided, will search for objects where the alarm suppression is set to the Boolean value (true or false).
outOfScan	Boolean	When a Boolean is provided, will search for objects where the out of scan flag is set to the Boolean value (true or false).
Response	HTTP 200 OK A Page<Nodes, BrowserObject>-object (see Page<Name, Type [→ 50]>)	
Examples	Request:: GET /api/systembrowser/1?searchstring=*view*&searchOption=1&caseSensitive=false&groupByParent=false Response: <pre>{ "Total": 1, "Page": 1, "Size": 100000, "Nodes": [{ "SystemId": 1, "ViewId": 9, "Name": "TxG_BA_BACnetAkedTransitionsEnum", "Descriptor": "(0000) BACnet Aked Transitions Enum list", "Designation": "System1.ManagementView:ManagementView.SystemSettings.Libr aries.Headquarter (HQ).BA.Device.BA_Device_PX_HQ_2.TextGrou ps.TxG_BA_BACnetAkedTransitionsEnum", "ObjectId": "System1:TxG_BA_BACnetAkedTransitionsEnum", "Attributes": { "DefaultProperty": "Read_Only", "ObjectId": "TxG_BA_BACnetAkedTransitionsEnum", "DisciplineDescriptor": "Management System", "DisciplineId": 0, "SubDisciplineDescriptor": "Unassigned", "SubDisciplineId": 0, "TypeDescriptor": "Library Element", "TypeId": 4600, "SubTypeDescriptor": "Text Group", "SubTypeId": 4609, </pre>	

	<pre> "ManagedType": 20 }, "Location": "System1.Management View:Project.System Settings.Libraries.Headquarter.BA.Device.Desigo PX.Text Groups.(0000) BACnet Acked Transitions Enum list", "_links": [] }], "_links": [{ "Rel": "page", "Href": "api/systembrowser/1?searchString=*view*&searchOption=1&ca seSensitive=False&groupByParent=False&size=100000&page={pa ge}&disciplineFilter=null&objectTypeFilter=null&aliasFilt er=", "IsTemplated": true }] } </pre>
--	---

7.6.3.1 Searching with Wildcards

Two wildcard characters are supported in System Browser—the asterisk (*) and the question mark (?). Each functions differently. The asterisk wildcard serves as a placeholder for zero or more characters. The question mark wildcard serves as a placeholder for exactly one character only. Therefore, each wildcard serves different purposes.

: Allows you to add zero or more characters to your search criteria. For example, "a" matches and displays, "a", "ab", "abc", and "abcd".

?: Allows you to add one character to your search. For example, "ab?" matches and returns "abc" but does not match or return "ab" and "abcd".

7.6.3.2 SearchOption

searchOption must have one of the following values.

searchOption	Description
0	Search in full name (Designation); default
1	Search in full description (Location)
2	Search in ObjectID
3	Search in Alias

7.6.3.3 Discipline and Objecttype Filters

This JSON string has two level filtering options—, group level (e.g. disciplines/types) and subgroup level (e.g. subdisciplines/subtypes). In case if SubGroupID/SubObjectTypeID is not provided along with GroupID/ObjectTypeID respectively then all the SubGroups/SubObjectTypes from the metioned Group/Object Type will be taken for filtering.

- **Example**

```
{ "<GroupID>": [ <SubGroupID>, . . . , <SubGroupID> ], "<GroupID>": [ <SubGroupID>, . . . , <SubGroupID> ] }
```

- **Concrete example**

```
{ "0": [ 0, 1, 2, 4 ], "20": [ 0, 21, 22 ] }
```

7.6.4 Search multiple Object Ids

Along with normal searches available, This API also provides bulk search option for searching multiple Object Ids.

Note : List of objectOrPropertyIds can be same or different SystemID.(**Distributed System is supported.**)

Resource Path	/systembrowser/:systemId/:viewId		
Category	Public		
HTTP Method	POST		
Parameters	systemId	uint	ID of a system Required
	viewId	uint	ID of the view in which search needs to be performed
	groupByParent	bool	If true, group search results by parent node
Body	Array of ObjectID	String[]	Array of object ids to be searched
Response	HTTP 200 OK Array of Multiple Object Ids search Browser Objects see [→ 55] An array of searched browser objects		
Examples	Request: GET api/systembrowser/1/9?searchOption=2&caseSensitive=false&groupByParent=false Post Data: [		

	<pre> "System1:ManagementView_System_Settings", "System1:JournalingRootFolder"] Response: [{ "ErrorCode": 0, "ObjectId": "System1:ManagementView_System_Settings", "Nodes": [{ "SystemId": 1, "ViewId": 9, "Name": "SystemSettings", "Descriptor": "System Settings", "Designation": "System1.ManagementView:ManagementView.SystemSettings", "ObjectId": "System1:ManagementView_System_Settings", "Attributes": { "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus", "ObjectId": "ManagementView_System_Settings", "DisciplineDescriptor": "Management System", "DisciplineId": 0, "SubDisciplineDescriptor": "System Settings", "SubDisciplineId": 4, "TypeDescriptor": "View Element", "TypeId": 8000, "SubTypeDescriptor": "System Folder", "SubTypeId": 8014, "ManagedType": 153 }, "Location": "System1.Management View:Project.System Settings", "_links": [] }], { "ErrorCode": 0, "ObjectId": "System1:JournalingRootFolder", "Nodes": [{ "SystemId": 1, "ViewId": 9, "Name": "JournalingRootFolder", "Descriptor": "Journaling", "Designation": "System1.ManagementView:ManagementView.SystemSettings.Jour nalingRootFolder", "ObjectId": "System1:JournalingRootFolder", "Attributes": { "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus", "ObjectId": "JournalingRootFolder", "DisciplineDescriptor": "Management System", "DisciplineId": 0, "SubDisciplineDescriptor": "System Settings", </pre>
--	---

	<pre> "SubDisciplineId": 4, "TypeDescriptor": "Journaling", "TypeId": 4500, "SubTypeDescriptor": "Journaling Printer", "SubTypeId": 4501, "ManagedType": 63 }, "Location": "System1.Management View:Project.System Settings.Journaling", "_links": [] }] }]</pre>
--	---

7.6.5 Creating a Subscription for Changes in the System Browser views (Obsolete)

Obsolete: Use channelized subscription instead.

In System browser views subscription, client can provide one or many designations for which client is interested for changes. This basically provides the feature of subtree subscription. All the nodes matching (or underneath) the provided designation will be subscribed for the changes. And any change happening under the provided designation will be notified to the subscribed clients.

Resource Path	/sr/SystemBrowserSubscriptions/:connectionId/		
Category	Public		
HTTP Method	POST		
Parameters	connectionId	string	Identifier of subscription Required
Body	designationsList	String[]	JSON Array of designation(s) for which client want to subscribe
Response	SystemBrowserSubscriptionKeyRepresentation object See [→ 78]		
Examples	Request: <pre>POST /api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904</pre> Post Data: <pre>["System1.ApplicationView", "System1.ManagementView"]</pre> Response: <pre>{ "Designations": ["System1.ApplicationView", "System1.ManagementView"], "Key": 2, "ErrorCode": 0, "_links": [{ "Rel": "unsubscribe", "Href": "api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904?subscriptionKey=%5b2%5d", "IsTemplated": false }] }</pre>		

	}
--	---

7.6.5.1 Push Notification

After subscribing for changes in the System Browser views, for any changes in the System Browser views a notification will be sent.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. systemBrowserHub : legacy		
Function	notifySystemBrowserChanges(data)		
	Property	Type	Description
	data	SystemBrowserSubscription	See System Browser subscription object [→ 78]

7.6.5.2 Distributed System support for subscription

- For distributed systems subscription support, A filter "*" is explicitly supported for the System part. That means client can specify paths 'for all systems' using * as shown below.
 "*.ApplicationView" – this will subscribe Application View of all the systems added in the distributed system.
- And if value of designation array is not provided then, no such filtering as described above will be applied. In case of NULL, client will receive notifications of all the changes in the all of the systems.

7.6.6 Creating Channelized Systembrowser Subscription

If subscription status will be pused through notifySubscriptionStatus function available at client side. Rest of the push notificaitons will be notified over notifySystemBrowserChanges.

Resource Path	/sr/SystemBrowserSubscriptions /channelize/:requestId/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub connection Required
Response	HTTP 200 OK		
Examples	Request: POST api/sr/SystemBrowserSubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/448b3117-44ef-4d8f-baa2-		

	<pre> 7274aa3050af Post Data: ["System1.ApplicationView","System1.ManagementView"] Response: HTTP/1.1 200 OK notifySubscriptionStatus : { "Designations": ["System1.ApplicationView", "System1.ManagementView"], "Key": 2, "ErrorCode": 0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "_links": [{ "Rel": "unsubscribe", "Href": "api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904?subscriptionKey=%5b2%5d", "IsTemplated": false }] } </pre>
--	--

7.6.6.1 Subscription Status Notification

After subscribing for system browser, the subscription status will be sent only once.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. systemBrowserHub : legacy		
Function	notifySubscriptionStatus(status)		
	Property	Type	Description
	Status	Subscription Response	System Browser subscription object [→ 213]
Example	<pre> { "Designations": ["System1.ApplicationView", "System1.ManagementView"], "Key": 2, "ErrorCode": 0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor": "notifySystemBrowserChanges", "_links": [{ "Rel": "unsubscribe", "Href": "api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904?subscriptionKey=%5b2%5d", "IsTemplated": false }] } </pre>		

7.6.7 Deleting a Subscription for Changes in the System Browser views

7.6.7.1 Passing subscription keys in query parameter

Resource Path	/sr/systemBrowsersubscriptions/:connectionId		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	connectionId	String	Identifier of subscription Required
	subscriptionKey	Array of integer	Querystring parameter, which holds the JSON array of subscription keys
Response	UnsubscriptionStatus See [→ 72] Unsubscribe object which contains Key and ErrorCode based on whether the request is successful or not.		
Examples	Request: DELETE /api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904?subscriptionKey=%5b2%5d Response: <pre>[{ "ErrorCode": 0, "Key": 2 }]</pre>		

7.6.7.2 Passing subscription keys in request body

Resource Path	/sr/systemBrowsersubscriptions/withbody/:connectionId		
Category	Public		
HTTP Method	DELETE		
Parameters			
Body	Array of Subscription keys		
Query	Property	Type	Description
	connectionId	String	Identifier of subscription Required
Response	UnsubscriptionStatus See [→ 72] Unsubscribe object which contains Key and ErrorCode based on whether the request is successful or not.		
Examples	Request: DELETE /api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904 Body Data: <pre>["2"]</pre> Response: <pre>[</pre>		

	<pre>{ "ErrorCode": 0, "Key": 2 }</pre>
--	---

7.6.8 Update CNS Node Description

Resource Path	/src/systembrowser/{designation}		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	designation	string	Full view name; Designation id of node in systembrowser tree (Recommendation: Double UriEncode parameter prior to making it to part of URI.) Required
	cnsinformation	System- BrowserEditRep- resentation	Descriptor text for node Object
Response	HTTP 200 OK		
Examples	Request: api/systembrowser/System1.ApplicationView%3AApplicationView.Documents? api_key=defaultadmin%3Aa' Response: 200		

7.7 Value Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/values	Public	x	
/valuessubscriptions	Public	x	x

7.7.1 Retrieving a Value of an Object or Property

If no property is specified but only an object, the value of the default property is returned.

Resource Path	/values/:objectOrPropertyId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectOrPropertyId	string	Address of object or property Required
	readMaxAge	int	Optional query parameter with default value as null; When the value is 'older' than this number of milliseconds, it will first be re-read at the driver (0 implies always re-read). A negative value

			disables driver re-reading and always returns the value from process image.
Response	HTTP 200 OK An array of ValueDetails objects [→62]		
Examples	Request 1: Object Id +Property GET /api/values/GmsDevice_2_1504_4194306.Present_Value Response: <pre>[{ "DataType": "BasicFloat", "Value": { "Value": "35", "Quality": "9439544818971182337", "QualityGood": true, "Timestamp": "2017-12-20T08:56:06.904Z" }, "OriginalObjectOrPropertyId": "GmsDevice_2_1504_4194306.Present_Value", "ObjectId": "System1:GmsDevice_2_1504_4194306", "PropertyName": "Present_Value", "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:online.._value", "ErrorCode": 0, "IsArray": false }]</pre> Request 2: Designation + Property GET /api/values/System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value Response: <pre>[{ "DataType": "BasicFloat", "Value": { "Value": "35", "Quality": "9439544818971182337", "QualityGood": true, "Timestamp": "2017-12-20T08:58:05.037Z" }, "OriginalObjectOrPropertyId": "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value", "ObjectId": "System1:GmsDevice_2_1504_4194306", "PropertyName": "Present_Value", "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:online.._value", "ErrorCode": 0, "IsArray": false }]</pre> Request 3: Object Id +Property GET /api/values/GmsDevice_2_1504_4194306.Present_Value?readMaxAge=-1 Response: <pre>[{</pre>		

	<pre> "DataType": "BasicFloat", "Value": { "Value": "35", "Quality": "9439544818971182337", "QualityGood": true, "Timestamp": "2017-12-20T08:56:06.904Z" }, "OriginalObjectOrPropertyId": "GmsDevice_2_1504_4194306.Present_Value", "ObjectId": "System1:GmsDevice_2_1504_4194306", "PropertyName": "Present_Value", "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online.._value", "ErrorCode": 0, "IsArray": false }]</pre>
--	--

7.7.2 Retrieving Values for a List of Objects or Properties

Note : List of ObjectOrPropertyId should be of same SystemId or else it will throw Exception.

“Could not retrieve Values” (Distributed System Not Supported.)

Resource Path	/values		
Category	Public		
HTTP Method	POST		
Parameters			
Body	Array of ObjectOrPropertyId		
Query	readMaxAge	int	Optional parameter: default value as null; When the value is 'older' than this number of milliseconds, it will first be re-read at the driver (0 implies always re-read). A negative value disables driver re-reading and always returns the value from process image.
Response	HTTP 200 OK An array of ValueDetails objects [→62]		
Examples	Request: POST /api/values Post Data: <pre>["GmsDevice_2_1504_4194306.Present_Value", "System1:GmsDevice_2_1504_4194306.Present_Value", "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value" //Designation+Property]</pre> Response:		

```
[
  {
    "DataType": "BasicFloat",
    "Value": {
      "Value": "35",
      "Quality": "9439544818971181313",
      "QualityGood": true,
      "Timestamp": "2017-12-20T08:31:40.132Z"
    },
    "OriginalObjectOrPropertyId": "GmsDevice_2_1504_4194306.Present_Value",
    "ObjectId": "System1:GmsDevice_2_1504_4194306",
    "PropertyName": "Present_Value",
    "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online.._value",
    "ErrorCode": 0,
    "IsArray": false
  },
  {
    "DataType": "BasicFloat",
    "Value": {
      "Value": "35",
      "Quality": "9439544818971181313",
      "QualityGood": true,
      "Timestamp": "2017-12-20T08:31:40.132Z"
    },
    "OriginalObjectOrPropertyId": "System1:GmsDevice_2_1504_4194306.Present_Value",
    "ObjectId": "System1:GmsDevice_2_1504_4194306",
    "PropertyName": "Present_Value",
    "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online.._value",
    "ErrorCode": 0,
    "IsArray": false
  },
  {
    "DataType": "BasicFloat",
    "Value": {
      "Value": "35",
      "Quality": "9439544818971181313",
      "QualityGood": true,
      "Timestamp": "2017-12-20T08:31:40.132Z"
    },
    "OriginalObjectOrPropertyId": "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value",
    "ObjectId": "System1:GmsDevice_2_1504_4194306",
    "PropertyName": "Present_Value",
    "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online.._value",
    "ErrorCode": 0,
    "IsArray": false
  }
]
```

Request 2:

POST /api/values?readMaxAge=-1

Post Data:

[

	<pre>"GmsDevice_2_1504_4194306.Present_Value"] Response: [{ "DataType": "BasicFloat", "Value": { "Value": "35", "Quality": "9439544818971182337", "QualityGood": true, "Timestamp": "2017-12-20T08:56:06.904Z" }, "OriginalObjectOrPropertyId": "GmsDevice_2_1504_4194306.Present_Value", "ObjectId": "System1:GmsDevice_2_1504_4194306", "PropertyName": "Present_Value", "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online.._value", "ErrorCode": 0, "IsArray": false }]</pre>
--	---

7.7.3 Creating a Subscription for a Change of a Value (Obsolete)

Obsolete: Use channelized subscription instead.

Resource Path	/sr/valuessubscriptions/connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	connectionId	string	Identifier of subscription Required
	detailsRequired	bool	
Body	Array of ObjectOrPropertyId		
Response	HTTP 200 OK An array of Subscription objects [→56]		
Examples	Request: <pre>POST /api/sr/valuessubscriptions/977a6389-980c-41ca-84ab-2518fd0ca41a</pre> Post Data: <pre>["GmsDevice_2_1504_4194306.Present_Value", "System1:GmsDevice_2_1504_4194306.Present_Value", "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value"//Designation+Property]</pre> Response: <pre>[{ "Key": 1,</pre>		

	<pre> "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Val ue", "ErrorCode":0, "_links":[{ "Rel":"unsubscribe", "Href":"api/sr/valuessubscriptions/4999fe9e- 05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b1%5d", "IsTemplated":false }] }, { "Key":2, "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Val ue", "ErrorCode":0, "_links":[{ "Rel":"unsubscribe", "Href":"api/sr/valuessubscriptions/4999fe9e- 05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b2%5d", "IsTemplated":false }] }, { "Key":3, "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Val ue", "ErrorCode":0, "_links":[{ "Rel":"unsubscribe", "Href":"api/sr/valuessubscriptions/4999fe9e- 05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b3%5d", "IsTemplated":false }] } }] </pre>
--	--

7.7.3.1 Push Notification

After subscribing for changes of a value, the current value(s) will be sent in an initial notification. Afterwards, any change will be notified.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. valuesHub : legacy		
Function	notifyValues(data)		
	Property	Type	Description
	data	Array of ValueDetails	See ValueDetails [→62]

7.7.4 Creating Channelized Value Subscription

If channelized subscription is done ,subscription status will be pushed through notifySubscriptionStatus function available at client side. Value changes will be notified over notifyValues[→78].

Resource Path	/sr/valuessubscriptions/channelize/:requestId/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub connection Required
Response	HTTP 200 OK		
Examples	Request: <pre>POST /api/sr/valuessubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/977a6389-980c-41ca-84ab-2518fd0ca41a</pre> Post Data: <pre>["GmsDevice_2_1504_4194306.Present_Value", "System1:GmsDevice_2_1504_4194306.Present_Value", "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value" //Designation+Property]</pre> Response: <pre>HTTP/1.1 200 OK</pre> notifySubscriptionStatus : Notification 1: <pre>{ "Key":1, "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Value", "ErrorCode":0, objectOrPropertyId: "GmsDevice_2_1504_4194306.Present_Value", "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "_links":[{ "Rel":"unsubscribe", "Href":"api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b1%5d", "IsTemplated":false }] }</pre>		

	Notification 2: <pre>{ "Key":2, "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Value", "ErrorCode":0, objectOrPropertyId: "System1:GmsDevice_2_1504_4194306.Present_Value", "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "_links":[{ "Rel":"unsubscribe", "Href":"api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b2%5d", "IsTemplated":false }] }</pre> Notification 3: <pre>{ "Key":3, "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Value", objectOrPropertyId: "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2:.Present_Value", "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5" "ErrorCode":0, "_links":[{ "Rel":"unsubscribe", "Href":"api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b3%5d", "IsTemplated":false }] }</pre> notifyValues: Notification 1: <pre>{{ "DataType":"BasicFloat", "Value": {"Value":"35", "Quality":"9439544818971181313", "QualityGood":true, "Timestamp":"2019-04-10T10:21:02.527Z"}, "ErrorCode":0, "IsArray":false, "SubscriptionKey":1 }}</pre> Notification 2: <pre>{{ "DataType":"BasicFloat", "Value": {"Value":"40", "Quality":"9439544818971181313",</pre>
--	--

	<pre> "QualityGood":true, "Timestamp":"2019-04-10T10:21:02.527Z"}, "ErrorCode":0, "IsArray":false, "SubscriptionKey":2 }} Notification 3: [[{"DataType":"BasicFloat", "Value": {"Value":"90", "Quality":"9439544818971181313", "QualityGood":true, "Timestamp":"2019-04-10T10:21:02.527Z"}, "ErrorCode":0, "IsArray":false, "SubscriptionKey":3 }} </pre>
--	---

7.7.4.1 Subscription Status Notification

After subscribing for Values, for each key subscription status will be sent through notifySubscriptionStatus.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. valuesHub : legacy		
Function	notifySubscriptionStatus(status)		
	Property	Type	Description
	Status	Subscription Response	Subscription objects [→201]
Example	<pre> { "Key":1, "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Value", "ErrorCode":0, objectOrPropertyId: "GmsDevice_2_1504_4194306.Present_Value", "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifyValues", "_links":[{ "Rel":"unsubscribe", "Href":"api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b1%5d", "IsTemplated":false }] } </pre>		

7.7.1 Deleting a Subscription for a Change of a Value

7.7.1.1 Passing subscription keys in query parameters

Resource Path	/sr/valuessubscriptions/:connectionId		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	connectionId	String	Identifier of subscription Required
	subscriptionkey	Array of int	List of subscription-keys
Response	HTTP 200 OK An array of unsubscription objects. See Link [→57]		
Examples	Request: DELETE /api/sr/valuessubscriptions/9d4211db-a740-44c3-93b8-015df1a529ca?subscriptionKey=[2,3] Response: <pre>[{ "ErrorCode": 0, "Key": 2 }, { "ErrorCode": 0, "Key": 3 }]</pre>		

7.7.1.2 Passing subscription keys in request body

Resource Path	/sr/valuessubscriptions/withbody/:connectionId		
Category	Public		
HTTP Method	DELETE		
Parameters			
Body	Array of Subscription keys		
Query	Property	Type	Description
	connectionId	String	Identifier of subscription Required
Response	HTTP 200 OK An array of unsubscription objects. See Link [→57]		
Examples	Request: DELETE /api/sr/valuessubscriptions/9d4211db-a740-44c3-93b8-015df1a529ca Body Data: ["2","3"] Response: <pre>[{ "ErrorCode": 0, "Key": 2 }, { </pre>		

	<pre>"ErrorCode": 0, "Key": 3 }]</pre>
--	---

7.8 Property Value Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/propertyvalues	Public	x	

7.8.1 Retrieving Detailed Values for Object or Property Id

Retrieves properties with runtime values for an Object OR Property Id. This gives properties and attributes associated with the requested Object OR Property Id. If input is actually an Object Id then default property for the object id along with the attributes is returned in the response. If input is a property id then that specific property and attributes of that object id will be returned in the response. One optional Boolean type query string parameter readAllProperties is available in both the cases. If readAllProperties is set to True then all the properties along with the attributes are returned in the response for the requested Object OR Property Id

Resource Path	/propertyvalues/:objectOrPropertyId		
Category	Public		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectOrPropertyId	string	ObjectOrPropertyId Required
	readAllProperties	bool	If True then all the properties for the requested Object OR Property Id will be returned in the response. Default is False.
Response	HTTP 200 OK The Object<Type> see [→ 62]		
Examples	Request 1: GET api/propertyvalues/System1:HDB_ArchiveGroup_1 Response: { "ErrorCode": 0, "ObjectId": "System1:HDB_ArchiveGroup_1", "Attributes": { "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus", "ObjectId": "HDB_ArchiveGroup_1", "DisciplineDescriptor": "Management System", "DisciplineId": 0, "SubDisciplineDescriptor": "Unassigned", "SubDisciplineId": 0, "TypeDescriptor": "Server Software", "TypeId": 6900, "SubTypeDescriptor": "History Database", "SubTypeId": 6903, } }		

	<pre> "ManagedType": 14, "ManagedTypeName": "Station" }, "Properties": [{ "Order": 0, "PropertyName": "StatusPropagation.AggregatedSummaryStatus", "Descriptor": "Summary Status", "Type": "ExtendedEnum", "Usage": 7, "Value": { "Value": "0", "DisplayValue": "Normal", "Quality": "9439544818969084161", "QualityGood": true, "Timestamp": "2015-05-21T05:08:56.772Z" }, "Resolution": 0, "PropertyAbsent": false, "IsArray": false, "TextTable": TxG_PropagationSummaryStatus }] } </pre> Request 2: with scaled information GET api/propertyvalues/System2:GmsDevice_1_20811_1.Present_Value? Response : <pre> { "ErrorCode": 0, "ObjectId": "System2:GmsDevice_1_20811_1", "Attributes": { "DefaultProperty": "Present_Value", "ObjectId": "System2:GmsDevice_1_20811_1", "DisciplineDescriptor": "Building Automation", "DisciplineId": 50, "SubDisciplineDescriptor": "Unassigned", "SubDisciplineId": 0, "TypeDescriptor": "Sensor", "TypeId": 6700, "SubTypeDescriptor": "Unassigned", "SubTypeId": 0, "ManagedType": 80, "ManagedTypeName": "BACnetPoint", "ObjectModelName": "GMS_BACNET_EO_BA_AI_1", "ValidationRules": { "CommentRule": "Mandatory", "ReAuthentication": "NoNeed", "Configuration": 1, "IsFourEyesEnabled": false, "_links": [] } }, "Properties": [{ "Order": 2, </pre>
--	---

	<pre> "PropertyName": "Present_Value", "Descriptor": "Present Value", "Type": "ExtendedReal", "Usage": 7, "Value": { "Value": "25.55555555555556", "DisplayValue": "25.55555555555556", "Quality": "9439544818975376641", "QualityGood": true, "Timestamp": "2019-06-11T12:41:00.067Z" }, "Min": -572.7777777777778, "Max": 537.2222222222222, "UnitDescriptor": "°C", "UnitId": 62, "Resolution": 2, "PropertyAbsent": false, "IsArray": false, "ScaledInformation": { "Factor": 0.5555555555555556, "Index": 62, "Offset": -17.77777777777778 } }] } </pre> Request 3: using Designation GET /api/propertyvalues/System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;?readAllProperties=false Response: <pre> { "ErrorCode":0, "ObjectId":"System1:GmsDevice_2_1504_4194306", "Attributes":{ "DefaultProperty":"Present_Value", "ObjectId":"System1:GmsDevice_2_1504_4194306", "DisciplineDescriptor":"Building Automation", "DisciplineId":50, "SubDisciplineDescriptor":"Unassigned", "SubDisciplineId":0, "TypeDescriptor":"Command", "TypeId":1100, "SubTypeDescriptor":"Analog", "SubTypeId":1101, "ManagedType":80, "ManagedTypeName":"BACnetPoint" }, "Properties":[{ "Order":0, "PropertyName":"Present_Value", "Descriptor":"Present Value", "Type":"ExtendedReal", "Usage":7, "Value":{ </pre>
--	---

	<pre> "Value": "35", "DisplayValue": "35", "Quality": "9439544818971182337", "QualityGood": true, "Timestamp": "2017-12-20T09:15:11.96Z" }, "UnitDescriptor": "°F", "UnitId": 64, "Resolution": 0, "PropertyAbsent": false, "IsArray": false }] } </pre>
--	---

7.9 Properties Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/properties	Public	x	

7.9.1 Retrieving Detailed Values for Object or Property Id

Retrieves properties without any runtime values for an Object OR Property Id. If input is actually an Object Id then default property for the object id will be returned in the response. If input is a property id then that specific property will be returned in the response. Two optional query string parameters are available in this interface. First is requestType of type int, which specifies whether the request is to read Property Names, Properties, Attributes, Properties and Attributes. Second parameter is readAllProperties of type bool, if this is True then all the properties for the requested Object OR Property Id will be returned in the response.

Resource Path	/properties/:objectOrPropertyId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectOrPropertyId	string	ObjectOrPropertyId Required
	requestType	Int	Specifies the request type. See [→ 44]. Default is 2.
	readAllProperties	bool	If True then all the properties for the requested Object OR Property Id will be returned in the response. Default is False.
Response	HTTP 200 OK The Object<Type> see [→ 62]		
	requestType	Return Type	Description
	0	See [→ 67]	Property collections contain list of all property-names
	1	Object<null>	Property collections empty; object contains

			attributes
	2 (default)	Object<PropertyDetails>	Property collections populated; no attribute field
	3	Object<PropertyDetails>	Property collections populated; object contains attributes
Examples	Request1: Get Property Names GET api/properties/System1:HDB_ArchiveGroup_1?requestType=0 OR GET /api/properties/System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;?requestType=0 //using Designation Response1: <pre>{ "ObjectId": "System1:HDB_ArchiveGroup_1", "Properties": ["Export.Activated", "Max_Retention.Activated", "Max_Retention.Delay_Span", "Max_Retention.Delay_Unit", "Max_Retention.FirstDate", "Max_Retention.Invocation_Period", "Max_Retention.Invocation_Unit", "Max_Retention.Max_Count", "Min_Retention.Delay_Span", "Min_Retention.Delay_Unit", "Min_Retention.Min_Count", "StatusPropagation.AggregatedSummaryStatus"], "FunctionProperties": [] }</pre> Request2: Get Properties and Attributes GET api/properties/System1:HDB_ArchiveGroup_1?requestType=3 OR GET /api/properties/System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;?requestType=3 //using Designation Response2: <pre>[{ "ErrorCode": 0, "ObjectId": "System1:HDB_ArchiveGroup_1", "Attributes": { "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus", "ObjectId": "HDB_ArchiveGroup_1", "DisciplineDescriptor": "Management System", "DisciplineId": 0, "SubDisciplineDescriptor": "Unassigned", "SubDisciplineId": 0, "TypeDescriptor": "Server Software", "TypeId": 6900, "SubTypeDescriptor": "History Database", "SubTypeId": 6903, "ManagedType": 14,</pre>		

```

    "ManagedTypeName" : "Station"
  },
  "Properties": [
    {
      "Order": 0,
      "PropertyName":
"StatusPropagation.AggregatedSummaryStatus",
      "Descriptor": "Summary Status",
      "Type": "ExtendedEnum",
      "Usage": 7,
      "Resolution": 0,
      "PropertyAbsent": false,
      "IsArray": false
    }
  ]
}
]

```

Request 3: with scaled information

GET

api/properties/System2:GmsDevice_1_20811_1.Present_Value?requestType=3

Response 3 :

```

{
  "ErrorCode": 0,
  "ObjectId": "System2:GmsDevice_1_20811_1",
  "Attributes": {
    "DefaultProperty": "Present_Value",
    "ObjectId": "System2:GmsDevice_1_20811_1",
    "DisciplineDescriptor": "Building Automation",
    "DisciplineId": 50,
    "SubDisciplineDescriptor": "Unassigned",
    "SubDisciplineId": 0,
    "TypeDescriptor": "Sensor",
    "TypeId": 6700,
    "SubTypeDescriptor": "Unassigned",
    "SubTypeId": 0,
    "ManagedType": 80,
    "ManagedTypeName": "BACnetPoint",
    "ObjectModelName": "GMS_BACNET_EO_BA_AI_1",
    "ValidationRules": {
      "CommentRule": "Mandatory",
      "ReAuthentication": "NoNeed",
      "Configuration": 1,
      "IsFourEyesEnabled": false,
      "_links": []
    }
  },
  "Properties": [
    {
      "Order": 2,
      "PropertyName": "Present_Value",
      "Descriptor": "Present Value",
      "Type": "ExtendedReal",
      "Usage": 7,
      "Value": {
        "Value": "25.55555555555556",

```


	<pre> "DisplayValue": "25.55555555555556", "Quality": "9439544818975376641", "QualityGood": true, "Timestamp": "2019-06-11T12:41:00.067Z" }, "Min": -572.7777777777778, "Max": 537.2222222222222, "UnitDescriptor": "°C", "UnitId": 62, "Resolution": 2, "PropertyAbsent": false, "IsArray": false, "ScaledInformation": { "Factor": 0.5555555555555556, "Index": 62, "Offset": -17.77777777777778 } }] } </pre>
--	---

7.9.1.1 Detail Request Type

requestType must have one of the following values.

requestType	Description
0	Return property names only.
1	Return attributes of object only.
2	Return properties with detailed information; default.
3	Return properties with detailed information and attributes of object.

7.9.2 Retrieving Detailed Values for multiple Object or Property Ids bulk interface

This interface is same as that of 7.9.1 only thing is this provides bulk interface to read properties and attributes of multiple Object OR Property Ids in the same request using HTTP POST. Array of Object OR Property Ids is passed in the POST data. Rest all the configuration is same as that of 7.9.1.

Note : List of objectOrPropertyIds can be same or different SystemID.(**Distributed System is supported.**)

Resource Path	/properties		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestType	Int	Specifies the request type. See [→ 44]. Default is 2.
	readAllProperties	bool	If True then all the properties for the requested Object OR Property Id will be returned in the response.

			Default is False.
Body	Array of Object OR Property Ids		
Response	HTTP 200 OK The Object<Type> see [→ 62]		
	requestType	Return Type	Description
	1	Object<null>	Property collections empty; object contains attributes
	2	Object<PropertyDetails>	Property collections populated; no attribute field
	3	Object<PropertyDetails>	Property collections populated; object contains attributes
Examples	Request: Post api/properties?requestType=3 Request Body: ["System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Unit"] OR //Using Designation + Property ["System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span", "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Unit"] Response: [{ "ErrorCode": 0, "ObjectId": "System1:HDB_ArchiveGroup_1", "Attributes": { "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus", "ObjectId": "HDB_ArchiveGroup_1", "DisciplineDescriptor": "Management System", "DisciplineId": 0, "SubDisciplineDescriptor": "Unassigned", "SubDisciplineId": 0, "TypeDescriptor": "Server Software", "TypeId": 6900, "SubTypeDescriptor": "History Database", "SubTypeId": 6903, "ManagedType": 14, "ManagedTypeName": "Station" }, "Properties": [{ "Order": 3, "PropertyName": "Min_Retention.Delay_Span", "Descriptor": "Min Retention Span", "Type": "ExtendedInt", "Usage": 11,		

	<pre> "Min": 1, "Resolution": 0, "PropertyAbsent": false, "IsArray": false }], { "ErrorCode": 0, "ObjectId": "System1:HDB_ArchiveGroup_1", "Attributes": { "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus", "ObjectId": "HDB_ArchiveGroup_1", "DisciplineDescriptor": "Management System", "DisciplineId": 0, "SubDisciplineDescriptor": "Unassigned", "SubDisciplineId": 0, "TypeDescriptor": "Server Software", "TypeId": 6900, "SubTypeDescriptor": "History Database", "SubTypeId": 6903, "ManagedType": 14, "ManagedTypeName": "Station" }, "Properties": [{ "Order": 4, "PropertyName": "Min_Retention.Delay_Unit", "Descriptor": "Min Retention Unit", "Type": "ExtendedEnum", "Usage": 11, "Max": 7, "Resolution": 0, "PropertyAbsent": false, "IsArray": false }] }] </pre>
--	---

7.9.1 Retrieving an Image for a Property

Retrieve an icon for a property. A couple of important notes:

- 1) If the property has not been assigned an image the return will still be OK (200) but the return will be an empty string.
- 2) The only supported format for the image is Base64-encoded, PNG with a size of 32 pixels wide by 32 pixels high. This may change in the future.

Resource Path	/properties/:propertyId/icon		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	propertyId	string	The fully-qualified property name, which includes the system information. For example:

			System1:GmsDevice_1_8166_1.Present_Value Required
Response	HTTP 200 OK The icon as a Base64 encoded PNG of size 32x32		
Examples	Request: GET api/properties/System1:GmsDevice_1_8166_1.Present_Value/icon Response: iVBORw0KGgoAAAANSUhEUgAAAAEAAAABCAyAAAFFcSJAAAAAXNSR0IArs4c6 QAAAAARnQU1BAACxjwv8YQUAAAAJcEhZcwAADsIAAA7CARUoSoAAAAANSURBVB hXY/j//78GAakAyb6sRXrAAAAE1FTkSuQmCC		

7.10 Command Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/commands	Public	x	
/commandssubscriptions	Public	x	x

7.10.1 Retrieving a List of Commands for a Provided Property

Some commands might be non executable at this moment.

Resource Path	/commands/:propertyId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	propertyId	string	PropertyId Required
	commandId	string	Optional filter for a specific commandId
	enabledCommandsOnly	bool	If set to true only currently enabled commands will be returned (default false).
	clientType	string	If set commands can be filtered for specific clients (All, Headless, Headful)
Response	HTTP 200 OK An array of PropertyCommand objects See [→ 52]		
Examples	Request 1: GET api/commands/System1%3AHDB_ArchiveGroup_1.Min_Retention.Delay_Span Response: [{ "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "ErrorCode": 0, "Commands": [{		

	<pre> "Configuration":2 "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "Descriptor": "Set", "Parameters": [{ "Name": "Value", "DataType": "ExtendedInt", "DefaultValue": "0", "Order": 0, "Descriptor": "", "Min": 1, "Application": 0, "EnumerationTexts": [] }], "Id": "Write", "IsDefault": false, "_links": [{ "Rel": "self", "Href": "api/commands/System1%3aHDB_ArchiveGroup_1.Min_Retention.D elay_Span/Write", "IsTemplated": false }] }] }]] </pre> Request 2: using Designation + Property GET /api/commands/System1.ManagementView:ManagementView.Manage mentSystem.Servers.Server.HistoryDatabase.ArchiveSchedules .ActivityLog;.Min_Retention.Delay_Span Response: <pre> [{ "PropertyId":"System1.ManagementView:ManagementView.Manage mentSystem.Servers.Server.HistoryDatabase.ArchiveSchedules .ActivityLog;.Min_Retention.Delay_Span", "ErrorCode":0, "Commands":[{ "PropertyId":"System1.ManagementView:ManagementView.Manage mentSystem.Servers.Server.HistoryDatabase.ArchiveSchedules .ActivityLog;.Min_Retention.Delay_Span", "Descriptor":"Set", "Parameters":[{ "Name":"Value", "DataType":"ExtendedInt", "DefaultValue":"4", "Order":0, "Descriptor":"", "Min":1, "Application": 0, "EnumerationTexts":[] }] }] }] </pre>
--	---

	<pre>], "Id": "Write", "IsDefault": false, "Configuration": 0, "_links": [{ "Rel": "self", "Href": "api/commands/System1.ManagementView%3aManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog%3b.Min_Retention.Delay_Span/Write", "IsTemplated": false }] } }] </pre>
--	---

7.10.2 Retrieving Lists of Commands for a List of Properties

Resource Path	/commands		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	propertyId	string	PropertyId Required
	commandId	string	Optional filter for a specific commandId
	enabledCommandsOnly	bool	If set to true only currently enabled commands will be returned (default false).
	clientType	string	If set commands can be filtered for specific clients (All, Headless, Headful)
Body	Array of PropertyId		
Response	HTTP 200 OK An aArray of PropertyCommand objects See [→ 52]		
Examples	Request 1: Post : /api/commands/ Request Body: <pre> ["System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Unit"] </pre> Response: List of Commands fetched for two different PropertyId: <pre> [{ "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "ErrorCode": 0, "Commands": [</pre>		

	<pre> { "Configuration":2 "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "Descriptor": "Set", "Parameters": [{ "Name": "Value", "DataType": "ExtendedInt", "DefaultValue": "0", "Order": 0, "Descriptor": "", "Min": 1, "Application": 0, "EnumerationTexts": [] }], "Id": "Write", "IsDefault": false, "_links": [{ "Rel": "self", "Href": "api/commands/System1%3aHDB_ArchiveGroup_1.Min_Retention.D elay_Span/Write", "IsTemplated": false }] }], }, { "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Unit", "ErrorCode": 0, "Commands": [{ "Configuration":1 "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Unit", "Descriptor": "Set", "Parameters": [{ "Name": "Value", "DataType": "ExtendedEnum", "DefaultValue": "0", "Order": 0, "Descriptor": "", "Max": 7, "Application": 0, "EnumerationTexts": [{ "Descriptor": "seconds", "Value": 0 }, { "Descriptor": "minutes", "Value": 1 }] }] }] }] </pre>
--	---

```

    }
  ]
}
],
"Id": "Write",
"IsDefault": false,
"_links": [
  {
    "Rel": "self",
    "Href":
"api/commands/System1%3aHDB_ArchiveGroup_1.Min_Retention.D
elay_Unit/Write",
    "IsTemplated": false
  }
]
}
]
}
]

```

Request 2: Using Designation + Property

Post: /api/commands/

Request Body:

```

[
  "System1.ManagementView:ManagementView.ManagementSystem.Ser
vers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;
.Min_Retention.Delay_Span",
  "System1.ManagementView:ManagementView.ManagementSystem.Ser
vers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;
.Min_Retention.Delay_Unit"
]

```

Response:

```

[
  { "PropertyId": "System1.ManagementView:ManagementView.Manag
ementSystem.Servers.Server.HistoryDatabase.ArchiveSchedule
s.ActivityLog;.Min_Retention.Delay_Span",
    "ErrorCode": 0,
    "Commands": [
      {
        "PropertyId": "System1.ManagementView:ManagementView.Manag
ementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules
.ActivityLog;.Min_Retention.Delay_Span",
        "Descriptor": "Set",
        "Parameters": [
          {
            "Name": "Value",
            "DataType": "ExtendedInt",
            "DefaultValue": "4",
            "Order": 0,
            "Descriptor": "",
            "Min": 1,
            "Application": 0,
            "EnumerationTexts": []
          }
        ],
        "Id": "Write",
        "IsDefault": false,
        "Configuration": 0,
        "_links": [
          {

```


	<pre> "Rel": "self", "Href": "api/commands/System1.ManagementView%3aManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog%3b.Min_Retention.Delay_Span/Write", "IsTemplated": false }] }] }, { "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Unit", "ErrorCode": 0, "Commands": [{ "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Unit", "Descriptor": "Set", "Parameters": [{ "Name": "Value", "DataType": "ExtendedEnum", "DefaultValue": "4", "Order": 0, "Descriptor": "", "Max": 7, "Application": 0, "EnumerationTexts": [{ "Descriptor": "seconds", "Value": 0 }, { "Descriptor": "minutes", "Value": 1 }, { "Descriptor": "hours", "Value": 2 }, { "Descriptor": "days", "Value": 3 }, { "Descriptor": "months", "Value": 4 }, { "Descriptor": "years", "Value": 5 }, { "Descriptor": "week", "Value": 6 }, { "Descriptor": "quarter", </pre>
--	--

	<pre> "Value":7 }] }], "Id":"Write", "IsDefault": false, "Configuration":0, "_links":[{"Rel":"self", "Href":"api/commands/System1.ManagementView%3aManagementVi ew.ManagementSystem.Servers.Server.HistoryDatabase.Archive Schedules.ActivityLog%3b.Min_Retention.Delay_Unit/Write", "IsTemplated":false }] }] }] </pre>
--	--

7.10.3 Executing a Command

Resource Path	/commands/:propertyId/:commandId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	propertyId	string	PropertyId Required
	commandId	string	Command identifier Required
	commandInput	CommandInputForExecutionValidateParam See [→ Error! Bookmark not defined.]	command parameter details (Name, Value pair) sent from client if required.
Body	An array of NameValue objects of Type(CommandInputForExecutionValidateParam)		
Response	HTTP 200 OK		
Examples	Request: POST /api/commands/System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span/Write OR //Using Designation + Property POST /api/commands/System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span/Write Request Body: <pre> [{ "Name": "Value", "DataType": "ExtendedInt", "Value": "3", </pre>		

	<pre> "MultiLangComments": ["comment1", "comment2", "comment3"], >Password": "password", SuperName": "supervisorusername", SuperPassword": "supervisorpassword" }] </pre> Response: HTTP 200 OK
--	---

7.10.4 Bulk Commanding

Resource Path	/commands/:commandId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	commandId	string	Command identifier Required
Body	bulkCommandInput	BatchCommand Execution See [→ 218]	command parameter details (Name, Value pair) sent from client if required along with array of property ids of same type.
User Validation	User Validation is at parameter level.		
Response	HTTP 200 OK An array of BatchExecution objects See Link [→ 218]		
Examples	Request: POST : /api/commands/Write Request Body: <pre> { "CommandInputForExecution": [{ "Name": "Value", "DataType": "ExtendedInt", "Value": "3", "MultiLangComments": ["comment1", "comment2", "comment3"], "Password": "password1", "SuperName": "supervisorusername", "SuperPassword": "supervisorpassword" }], "PropertyIds": ["System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span", "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog.Min_Retention.Delay_Span", //Designation + Property "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.EventLog; .Mi </pre>		

	<pre> n_Retention.Delay_Span" //Designation + Property] } </pre> Response: <pre> HTTP 200 OK [{ "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "ErrorCode": 0 }, { "PropertyId": "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span", "ErrorCode": 0 }, { "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;Min_Retention.Delay_Span", "ErrorCode": 0 }, { "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.EventLog;Min_Retention.Delay_Span", "ErrorCode": 0 }] </pre>
--	--

7.10.5 Bulk Commanding

Resource Path	/validateproperty/commands/:commandId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	commandId	string	Command identifier Required
Body	bulkCommandInput	BatchCommand ExecutionValidateProperty See [→ 218]	command parameter details (Name, Value pair) sent from client if required along with array of property ids of same type.
User Validation	User Validation is at Property level.		
Response	HTTP 200 OK An array of BatchCommandExecutionValidateProperty objects See Link [→ 218]		
Examples	Request: <pre>POST : api/commands/validateproperty/write</pre> Request Body: <pre> { "CommandInputForExecution": [{ "Name": "Value", "DataType": "ExtendedInt", </pre>		

	<pre> "Value": "3" }], "PropertyIds": ["System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span", "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span", //Designation + Property "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.EventLog;.Min_Retention.Delay_Span" //Designation + Property], "Comments": { "MultiLangText": ["comment1", "comment2", "comment3"] }, "Password": "password1", "SuperName": "supervisorusername", "SuperPassword": "supervisorpassword" } } Response: HTTP 200 OK [{ "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "ErrorCode": 0 }, { "PropertyId": "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span", "ErrorCode": 0 }, { "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span", "ErrorCode": 0 }, { "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.EventLog;.Min_Retention.Delay_Span", "ErrorCode": 0 }] </pre>
--	--

7.10.6 Bulk Commanding with different pair of property values and command id

Resource Path	/commands/propertycommand
---------------	---------------------------

Category	Public		
HTTP Method	POST		
Body	An array of propertyCommandInput	PropertyCommandInput Execution See [→219]	Array of property command object. Each property command contains commandid ,propertyid and optional command parameter details
User Validation	User Validation is at parameter level.		
Response	HTTP 200 OK An array of BatchExecution objects See Link [→ 218]		
Examples	Request: POST : /api/commands/propertycommand Request Body: <pre>[{ "CommandInputForExecution": [{ "Name": "Value", "DataType": "ExtendedUint", "Value": "1500" }], "PropertyId": "LogViewer.ViewSize", "CommandId": "Write" }, { "PropertyId": "LogViewer.SuppressAlarms", "CommandId": "Enable" }]</pre> Response: HTTP 200 OK <pre>[{ "PropertyId": "LogViewer.ViewSize", "ErrorCode": 0 }, { "PropertyId": "LogViewer.SuppressAlarms", "ErrorCode": 0 }]</pre>		

7.10.7 Bulk Commanding with different pair of property values and command id

Resource Path	/validateproperty/commands/propertycommand		
Category	Public		
HTTP Method	POST		
Body	An array of propertyCommandInput	PropertyCommandInput Execution See [→219]	Array of property command object. Each property command contains commandid ,propertyid and optional command parameter details
User Validation	User Validation is at Property level.		

Response	HTTP 200 OK An array of BatchExecution objects See Link [→ 218]
Examples	Request: POST : /api/commands/validateproperty/propertycommand Request Body: <pre>[{ "CommandInputForExecution": [{ "Name": "Value", "DataType": "ExtendedUInt", "Value": "1500" }], "PropertyId": "LogViewer.ViewSize", "CommandId": "Write" "Comments": { "CommonText": "Common comment for all the languages" }, "Password": "loggedinuserpassword", "SuperName": "SuperUserName", "SuperPassword": "SuperUserPassword" }, { "CommandInputForExecution": [{ "Name": "Value", "DataType": "ExtendedUInt", "Value": "1500" }], "PropertyId": "LogViewer.SuppressAlarms", "CommandId": "Enable" "Comments": { "CommonText": "Common comment for all the languages" }, "Password": "loggedinuserpassword", "SuperName": "SuperUserName", "SuperPassword": "SuperUserPassword" }]</pre> Response: HTTP 200 OK <pre>[{ "PropertyId": "LogViewer.ViewSize", "ErrorCode": 0 }, { "PropertyId": "LogViewer.SuppressAlarms", "ErrorCode": 1, "ErrorMessage": "command execution failed!" }]</pre>

7.10.8 Creating a Subscription for a Change of a Command (Obsolete)

Obsolete: Use channelized subscription instead.

Resource Path	/sr/commandssubscriptions/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	connectionId	string	Identifier of subscription Required
	propertyIdList	String[]	Array of Properties from POST data, which will be subscribed for Command Change notification
Body	Array of PropertyId		
Response	HTTP 200 OK An array of Subscription objects See Link [→ 46]		
Examples	Request: Post :api/sr/commandssubscriptions/87a73827-f25b-41e8-9225-2f08316ea0fb Request Body: <pre>["System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span", "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span" //Designation + Property]</pre> Response: <pre>[{ "Key":5, "PropertyId":"System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span", "ErrorCode":0, "_links":[{ "Rel":"unsubscribe", "Href":"api/sr/commandssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b5%5d", "IsTemplated":false }] }, { "Key":6, "PropertyId":"System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span", "ErrorCode":0, "_links":[{ "Rel":"unsubscribe", "Href":"api/sr/commandssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b6%5d", "IsTemplated":false }] }]</pre>		

	}
	]

7.10.9 Push Notification

After subscribing for changes of a command(s) **all** command(s) will be sent in an initial notification. Afterwards any change will be notified.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. commandsHub : legacy		
Function	notifyCommands(data)		
	Property	Type	Description
	data	Array of PropertyCommand	See PropertyCommand [→ 216]

7.10.9.1 Creating Channelized Command Subscription

If subscription status will be pushed through notifySubscriptionStatus function available at client side. Command changes will be notified over notifyCommands.

Resource Path	/sr/commandssubscriptions/channelize/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	connectionId	string	Identifier of subscription Required
	propertyIdList	String[]	Array of Properties from POST data, which will be subscribed for Command Change notification
Body	Array of PropertyId		
Response	HTTP 200 OK		
Examples	Request: Post :api/sr/commandssubscriptions/ channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/87a73827-f25b-41e8-9225-2f08316ea0fb Request Body: <pre>["System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span", "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span" //Designation + Property]</pre> HTTP/1.1 200 OK notifySubscriptionStatus: Notification 1: <pre>{ "Key":5, "PropertyId":"System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span", "SubscriptionId":"System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span", "ErrorCode":0,</pre>		

	<pre> "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor": "notifyCommands", "_links": [{ "Rel": "unsubscribe", "Href": "api/sr/commandssubscriptions/aa82215b- b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b5%5d", "IsTemplated": false }] } Notification 2:{ "Key": 6, "PropertyId": "System1.ManagementView:ManagementView.Manage mentSystem.Servers.Server.HistoryDatabase.ArchiveSchedules .ActivityLog;.Min_Retention.Delay_Span", "SubscriptionId": "System1.ManagementView:ManagementView.Ma nagementSystem.Servers.Server.HistoryDatabase.ArchiveSched ules.ActivityLog;.Min_Retention.Delay_Span", "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor": "notifyCommands", "ErrorCode": 0, "_links": [{ "Rel": "unsubscribe", "Href": "api/sr/commandssubscriptions/aa82215b- b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b6%5d", "IsTemplated": false }] }] </pre>
--	---

7.10.9.2 Subscription Status Notification

After subscribing for Commands, for each key subscription status will be sent through notifySubscriptionStatus.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. commandsHub : legacy		
Function	notifySubscriptionStatus(status)		
	Property	Type	Description
	Status	Subscription Response	Subscription objects See Link [→ 201]
Example	<pre> { "Key": 6, "PropertyId": "System1.ManagementView:ManagementView.Management System.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Mi n_Retention.Delay_Span", "SubscriptionId": "System1.ManagementView:ManagementView.Managem entSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog; .Min_Retention.Delay_Span", "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor": "notifyCommands", "ErrorCode": 0, "_links": [{ </pre>		

	<pre>"Rel": "unsubscribe", "Href": "api/sr/commandssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b6%5d", "IsTemplated": false }] }</pre>
--	--

7.10.9.3 Deleting a Subscription for a Change of a Command.

7.10.9.3.1 Passing subscription keys in query parameter

Resource Path	/sr/commandssubscriptions/:connectionId		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	connectionId	String	Identifier of subscription Required
	subscriptionkey	Array of int	List of subscription keys
Response	HTTP 200 OK An array of unsubscription objects. See Link [→56]		
Examples	Request: Delete api/sr/commandssubscriptions/a43e4363-296e-405d-977b-36dec9ddbafc?subscriptionKey=[1,2] Response: <pre>{ "ErrorCode": 0, "Key": 1 }, { "ErrorCode": 0, "Key": 2 }</pre>		

7.10.9.3.2 Passing subscription keys in request body

Resource Path	/sr/commandssubscriptions/withbody/:connectionId		
Category	Public		
HTTP Method	DELETE		
Parameters			
Body	Array of Subscription keys		
Query	Property	Type	Description
	connectionId	String	Identifier of subscription Required
Response	HTTP 200 OK An array of unsubscription objects. See Link [→56]		
Examples	Request: api/sr/commandssubscriptions/a43e4363-296e-405d-977b-36dec9ddbafc Body Data: ["1", "2"]		

	Response: <pre>{ "ErrorCode": 0, "Key": 1 }, { "ErrorCode": 0, "Key": 2 }</pre>
--	--

7.11 Trend Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/trendseriesinfo	Public	x	
/trendseries	Public	x	

7.11.1 Retrieving a List of All Trend Collector Objects

Retrieve a list of all Trend Collector objects of a given system. It includes all Online and Offline objects as following

1. Trendlog
2. Trendlog Multiple
3. Trend Log Definitions
4. Trended property with VL flag set to TRUE

Resource Path	/trendseriesinfo/:systemId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
Response	HTTP 200 OK An array of TrendCollector objects [→64]		
Examples	Request: GET /api/trendseriesinfo/1 Response: <pre>[{ "ObjectId": "System1:GmsDevice_1_29129_4194305", "PropertyName": "Present_Value", "CollectorObjectOrPropertyId": "System1:GmsDevice_1_29129_113246209", "TrendseriesId": "System1:GmsDevice_1_29129_113246209.general.Data:_origina l.2._value" }, { "ObjectId": "System1:GmsDevice_1_29129_4194305", "PropertyName": "Present_Value", "CollectorObjectOrPropertyId": "System1:GmsDevice_1_29129_113246210", "TrendseriesId": "System1:GmsDevice_1_29129_113246210.general.Data:_origina l.2._value" }]</pre>		

7.11.2 Retrieving a List of Trend Collector Objects

Retrieve a list of Trend Collector objects. An object ID of a trend object (e.g. TrendLog/TrendLogMultiple) or a property ID of a trended object is to be provided.

Resource Path	/trendseriesinfo/objectOrPropertyId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectOrPropertyId	string	ObjectOrPropertyId Required
Response	HTTP 200 OK An array of TrendCollector objects [→64]		
Examples	Request: GET /api/trendseriesinfo/System1:GmsDevice_2_1500_83886081 OR //Using Designation GET /api/trendseriesinfo/System1.ApplicationView:ApplicationView.Trends.OfflineTrends.TestNet.Hardware.Devices_1500.Locall_IO.TL_1; Response: <pre>[{ "ObjectId": "System1:GmsDevice_2_1500_1", "PropertyIndex": 1, "PropertyName": "Present_Value", "CollectorObjectOrPropertyId": "System1:GmsDevice_2_1500_83886081", "TrendseriesId": "System1:GmsDevice_2_1500_83886081.general.Data:_offline.._value" }]</pre>		

7.11.3 Retrieving TrendViewDefinition object details

Retrieve a TrendViewDefition object details. An object ID of a trend view definition.

Resource Path	/trendseriesinfo/tvd/:objectId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectId	string	ObjectId of Trend View Definition object Required
Response	HTTP 200 OK A TrendViewDefinitionRepresentation object [→221]		
Examples	Request: GET api/trendseriesinfo/tvd/ApplicationView_Trends_TrendViewDefinitions_sampletvd OR GET api/trendseriesinfo/tvd/System1.ApplicationViewApplicationView.Trends.TrendViewDefinitions.sampletvd; Response:		

	<pre> { "AxisX": { "AutoScale": true, "AxisType": 0, "ScaleMax": 100, "ScaleMin": 0, "ShowTitle": false, "ShowZeroLine": false, "Title": "" }, "AxisYLeft": { "AutoScale": true, "AxisType": 1, "ScaleMax": 100, "ScaleMin": 0, "ShowTitle": false, "ShowZeroLine": false, "Title": "" }, "AxisYRight": { "AutoScale": true, "AxisType": 2, "ScaleMax": 100, "ScaleMin": 0, "ShowTitle": false, "ShowZeroLine": false, "Title": "" }, "DescriptionTvd": "", "ObjectIdOfTvd": "System1:ApplicationView_Trends_TrendViewDefinitions_sampl etvd", "IdTvd": "a6393de5-454d-4f41-8e13-043c2f01dfdb", "NameTvd": "sampletvd", "NumberDisplayedSamplesPerTrendSerie": 100, "RemoveOnlineTrendLogOfDeletedTrendSerie": false, "TimeRange": { "ValidTimeRange": 1, "AbsoluteTimeRange": { "From": "2019-06-24T10:07:33.075Z", "To": "2019-06-24T10:12:33.075Z" }, "RelativeTimeRange": { "AnchorMode": 1, "AnchorTime": "2019-06-24T10:12:30.986Z", "NumberOfTimeUnits": 1, "TimeUnit": 3 } }, "TitleLeft": "", "TitleRight": "", "TitleTop": "", "TsdCollectionInfo": { "TrendSeriesDefinitions": [{ "ReservedDateTime": "1777-07-07T07:07:07Z", "AccessDenied": false, </pre>
--	---

	<pre> "ArchiveOn": true, "AxisAttachment": "ToLeft", "ChartLineStyle": "Full", "ChartLineType": "Line", "Color": "ffff0000", "DescriptionTlField": "", "DescriptionTlGms": "Log Viewer-ViewSize", "DescriptionTo": "", "DescriptionToProperty": "View Size", "ObjectIdOfTrendLog": "", "ObjectIdOfTrendLogOnline": "System1:TLO_LogViewer_ViewSize", "ObjectIdOfTrendedObject": "System1:LogViewer.ViewSize:_original.._value", "Index": -1, "IsLineVisible": true, "LineWidth": 1, "MaxTimeStamp": "0001-01-01T00:00:00Z", "MinTimeStamp": "9999-12-31T18:29:59.999Z", "ShowDataLabels": false, "ShowMarkers": false, "ShowQualityIcons": false, "Type": "Online", "UnitText": "", "ValueType": "Float" }] }, "TvObjectType": 0, "TvCovType": "Auto", "_links": [] } </pre>
--	---

7.11.4 Create/Update TrendViewDefinition

Api to Modify Existing Tvd or Create new tvd. If Existing tvd is to be modified, then the TrendViewDefinitionUpdateRepresentation TvdoId must be set. If new Tvd is to be created, then TvdoId must be empty and CNSDescription and Location parameters must be set.

Resource Path	/trendseriesinfo/tvd		
Category	Public		
HTTP Method	PUT		
Parameters	Property	Type	Description
	tvdUpdate	TrendViewDefinitionUpdateRepresentation	Trend View Definition Update object Required
Response	HTTP 200 OK A TrendViewDefinition object [221]		
Examples	Request: PUT api/trendseriesinfo/tvd Request Body: {		

	<pre> "AxisX": { "AutoScale": true, "ScaleMax": 100, "ScaleMin": 0, "ShowTitle": false, "ShowZeroLine": false, "Title": "" }, "AxisYLeft": { "AutoScale": true, "ScaleMax": 500, "ScaleMin": 0, "ShowTitle": false, "ShowZeroLine": false, "Title": "" }, "AxisYRight": { "AutoScale": true, "ScaleMax": 100, "ScaleMin": 0, "ShowTitle": false, "ShowZeroLine": false, "Title": "" }, "ObjectIdOfTvd": "", "CNSDescription": "tvd4", "Location": "System1.ApplicationView:ApplicationView.Trends.TrendViewDefinitions", "NumberDisplayedSamplesPerTrendSerie": 200, "RemoveOnlineTrendLogOfDeletedTrendSerie": false, "TimeRange": { "ValidTimeRange": 2, "AbsoluteTimeRange": { "From": "2020-01-23T09:23:49.969Z", "To": "2020-02-07T10:01:48.329Z" }, "RelativeTimeRange": { "AnchorMode": 1, "AnchorTime": "2020-01-27T09:23:13Z", "NumberOfTimeUnits": 1, "TimeUnit": 3 } }, "TitleLeft": "Title for Left", "TitleRight": "Title for Right", "TitleTop": "Top name", "TsdCollectionInfo": { "TrendSeriesDefinitions": [{ "ArchiveOn": false, "AxisAttachment": "ToLeft", "ChartLineStyle": "Full", "ChartLineType": "Line", "Color": "ffff00f0", "ObjectIdOfTrendLog": "System1.GmsDevice_1_90_83886081", </pre>
--	--

	<pre> "ObjectIdOfTrendLogOnline": "", "ObjectIdOfTrendedObject": [System1:GmsDevice_1_90_1", "IsLineVisible": true, "LineWidth": 1, "ShowDataLabels": false, "ShowMarkers": true, "ShowQualityIcons": false, "MarkerType": "Circle", "Type": "Offline" }] }, "TvCovType": "Auto" } </pre> Response: A TrendViewDefinition Object[↗221]
--	---

7.11.5 Delete TrendViewDefinition

Resource Path	/trendseriesinfo/tvd/:objectId		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	objectId	string	ObjectId of Trend View Definition object Required
Response	HTTP 200 OK A TrendViewDefinition object [↗221]		
Examples	Request: <pre>DELETE api/trendseriesinfo/tvd/ System1:ApplicationView_Trends_TrendViewDefinitions_tvd42</pre> Response: HTTP 200 OK No Content		

7.11.6 Delete TrendLogOnline

Resource Path	/trendseriesinfo/tsdOnline/:objectId		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	objectId	string	ObjectId of Trend Log Online Required
Response	HTTP 200 OK		
Examples	Request: <pre>DELETE api/trendseriesinfo/tvd/ System1:TLO_bac1_Hardware_proj1device_90_Local_IO_AI_1_Present_Value</pre>		

	Response : HTTP 200 OK No Content
--	---

7.11.7 Retrieving Borders of a Trend Series

Resource Path	/trendseries/:trendseriesId/borders		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	trendseriesId	string	Trend series identifier Required
Response	HTTP 200 OK A TrendBorder object [→64]		
Examples	Request: GET api/trendseries/System1%3AGmsDevice_1_1546_113246209.general.Data%3A_original.1._value/borders Response: { "From": "2015-05-13T12:00:34.37Z", "To": "2015-05-13T12:06:57.9Z" }		

7.11.8 Retrieving a List of Trends

Retrieves historical time-series, consisting of values, quality attributes and time-stamps.

Resource Path	/trendseries/:trendseriesId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	trendseriesId	string	Trend series identifier Required
	from	DateTime	Start time of trend series Required
	to	DateTime	End time of trend series Required
	intervals	ushort	If provided, the result contains the min. and max. value per interval.
	addDescriptor	bool	If true, adds localized value (if applicable) to result. Default: false
Response	HTTP 200 OK A TrendSeries object [→64]		
Examples	Request: GET api/trendseries/System1:GmsDevice_1_29129_83886081.general.Data:_offline._value?from=2015-05-13T12:05:53.000Z&to=2015-05-13T12:05:54.000Z&intervals=5&description=true Response: { "Id":		

	<pre> "System1:GmsDevice_1_29129_83886081.general.Data:_offline. ._value", "SeriesPropertyId": "System1:GmsDevice_1_29129_16777217.Present_Value", "Series": [{ "Value": "1", "DisplayValue": "ACTIVE", "Quality": "8589934592", "QualityGood": true, "Timestamp": "2015-05-13T12:05:52.59Z" }, { "Value": "1", "DisplayValue": "INACTIVE", "Quality": "8589934592", "QualityGood": true, "Timestamp": "2015-05-13T12:05:55.59Z" }] } </pre>
--	--

7.11.9 Retrieving a List of Trend Aggregates

For good samples of history data retrieves mathematical aggregations for Standard Deviation and Least Square Regression.

Trend aggregation is not Distribution enabled. It only serves for local system only.

Resource Path	/trendseries/:trendseriesId/:aggregateUnit		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	trendseriesId	string	Trend series identifier Required
	from	DateTime	Start time of trend series Required
	to	DateTime	End time of trend series Required
	aggregateUnit	ushort	Unit on which aggregation will take place. See [117]. Default is 7
	length	int	Default Value = 1000
	qualityMask	long	Mask tells you which bits you have to take into account
	qualityValue	long	Value tells you what value the bits in consideration need to have
	correctedValues	bool	Flag that signals use of corrected values in Trends (Default Value=True)
	addDescriptor	bool	If true, adds localized value (if applicable) to result. Default: false
Response	HTTP 200 OK A TrendAggregateSeries object [→220]		

Examples	Request: <pre>GET api/trendseries/System1%3AGmsDevice_1_29128_83886081.general.Data%3A_offline.._value/7?from=2015-10-27T06:01:04.510Z&to=2015-11-27T11:44:41.660Z&length=1000&qualityMask=0&qualityValue=0&addDescriptor=false</pre> Response: <pre>{ "Id": "System1:GmsDevice_1_29128_83886081.general.Data:_offline._value", "SeriesPropertyId": "System1:GmsDevice_1_29128_1.Present_Value", "Series": [{ "AvgX": 1445945963123.153, "AvgY": 65.28828828828829, "Count": 333, "FromValueDescriptor": "65", "FromValue": 65, "FromTime": "2015-10-27T06:01:05.12Z", "Min": 45, "Max": 80, "StDevX": 287308.05418783723, "StDevY": 8.742276612722005, "SumX": 481500005720010, "SumXX": 6.962229895146511e+26, "SumXY": 481549949961870, "SumY": 83, "SumYY": 1444883, "ToTime": "2015-10-27T06:17:41.13Z", "ToValueDescriptor": "61", "ToValue": 61, "VarX": 82545918001.2012, "VarY": 76.42740037334613 }, ] }</pre>
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7.11.9.1 Aggregate Unit

Aggregate Unit must have one of the following values.

Id	Description
0	Year
1	Quarter
2	Month
3	Week
4	Day
5	Hour
6	Minute
7	Second
8	Millisecond

7.12 History Logs Service

This service provides a way to retrieve data for available log tables.

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/historylogs	Public	x	

7.12.1 Retrieving Log Tables

Provides a list of all available log tables with other details which can be used in subsequent requests for data retrieval and filtering. This method returns tables for a specified system.

Resource Path	/historylogs/:systemId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
Response	HTTP 200 OK List of LogTable objects. See [→224]		
Examples	Request: GET api/historylogs/1 Response: <pre>[{ "Name": "ActivityLogTable", "Descriptor": "Activities", "Description": "Logs Activity Data", "HasChildColumns": false, "HasConditionFilter": true, "HasNameFilter": true, "HasTimeFilter": true, "HasTimeFilterDefinedBySource": false, "HasAndOperatorOnly": true, "HasMaxAge": false, "HasSingleNameFilter": true, "SupportedTypes": ["*"] }, { "Name": "AlarmLogTable", "Descriptor": "Events", "Description": "Logs Alarm Data", "HasChildColumns": false, "HasConditionFilter": true, "HasNameFilter": true, "HasTimeFilter": true, "HasTimeFilterDefinedBySource": false, "HasAndOperatorOnly": true, "HasMaxAge": false, "HasSingleNameFilter": true, "SupportedTypes": [</pre>		

	<pre> " * "] }] </pre>
--	--

7.12.2 Retrieving Columns of a Log Table

Provides a list columns for a requested log table from the specified system. Each column is associated with other details which can be used in subsequent requests for displaying and filtering data.

Resource Path	/historylogs/systemId/tableName		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	tableName	string	Name of table for which columns need to be retrieved Required
	isParentColumns	bool	If TRUE then parent table columns will be retrieved for the requested table. Otherwise child table columns will be retrieved. (Default is TRUE)
Response	HTTP 200 OK List of LogColumn objects. See [→225]		
Examples	Request: GET api/historylogs/1/ActivityLogTable Response: <pre> [{ "Name": "UserName", "Descriptor": "User Name", "DataType": "String", "ErrorSupport": false, "IsArray": false, "IsDefault": true, "IsHidden": false, "IsSortable": true, "IsEnum": false, "IsFilterable": true }, { "Name": "Attachment", "Descriptor": "Attachment", "DataType": "String", "ErrorSupport": false, "IsArray": false, "IsDefault": true, "IsHidden": false, "IsSortable": true, "IsEnum": false, "IsFilterable": false }] </pre>		

7.12.3 Retrieving Condition Filter Operators for a Column

Provides a list of Condition Filter operators supported by the requested column. This operators then can be used to compose a Condition Filter.

Resource Path	/historylogs/:systemId/:tableName/operators/:columnName		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	tableName	string	Name of the Table Required
	columnName	string	Name of column for which Condition Filter operators will be retrieved Required
Response	HTTP 200 OK LogConditionFilter object. See [→225]		
Examples	Request: GET api/historylogs/1/ActivityLogTable/operators/LogType Response: <pre>{ "Operators": ["=", "<>"], "ErrorInfo": ["This is Error1", "This is Error2"] }</pre>		

7.12.4 Retrieving Enum Values for a Column

Provides a list of enum values which are associated with the requested column if column is of type enum. These enum values then can be used to compose a Condition Filter. This list will be a global list from all systems i.e. union of all the values from all systems.

Resource Path	/historylogs/:systemId/:tableName/enumvalues/:columnName		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	tableName	string	Name of the table Required
	columnName	string	Name of column for which enum values will be retrieved Required
Response	HTTP 200 OK LogEnumValues object. See [→225]		

Examples	Request: GET api/historylogs/1/ActivityLogTable/enumvalues/LogType Response: <pre>{ "EnumValues": ["Unknown", "Reno", "Trend"], "ErrorInfo": ["This is Error1", "This is Error2"] }</pre>
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7.12.5 Retrieving Data for Table

Provides page wise data for the requested table, based on specified filters and sorting criteria. User also needs to send a list of interested columns. If no columns are provided data for all columns will be returned.

A call without a snapshotId creates a snapshot with all filters and sorting parameters applied. Call with a snapshotId returns data of this snapshot without applying new filters and sorting criteria.

Resource Path	/historylogs/:systemId/:tableName		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	tableName	string	Name of table Required
	snapshotId	string	User defined Snapshot id. Provided Snapshot Id can be used to cancel report processing. If SnapshotId is provided by client then same Id will be received in snapshotId as response else client will receive system generated snapshotId
	sortColumnData	string	This is a JSON serialized string of Array of LogSortInfo objects. For LogSortInfo see [→225] This array conveys 3 different information for sorting. 1. ColumnName on which sorting needs to be performed 2. LogSortType either Ascending or Descending See [→225] 3. Order in which elements are present in the array will be considered as sort priority. i.e. first element has the highest priority
	nameFilter	string	This is JSON serialized string of a string array. Each element in the

			array is a Name Filter which will be used to filter data. See [→123]
	conditionFilter	string	This is Condition Filter string which will be used to filter data. See [→124]
	fromDate	DateTime	From date time for the Time Filter Note: fromDate and toDate both should be present to Time filter the data
	toDate	DateTime	To date time for the Time Filter Note: fromDate and toDate both should be present to Time filter the data
	isDefinedBySourceTimeFilter	bool	Flag which denotes whether Time Filter defined by source should be applied If TRUE then Time Filter received in the request will be ignored and instead Time Filter defined by source will be used to filter data (Default is FALSE)
	clientTimeSpan	string	Client's timezone UTC offset converted to timespan. Used to localize datetime data as per client's time zone.
	page	int	Current page number (Default is 1)
	size	int	Number of records per page
	isChildDataRequired	bool	Flag which denotes that user is also interested in child data, if supported by the requested table (Default is FALSE)
Body	columnList	ColumnList	JSON serialized object of type LogColumnList. See [→226]
Response	HTTP 200 OK LogTabData object. See [→226]		
Examples	Request: Post : api/historylogs/1/ActivityLogTable?size=2&isChildDataRequired=true Request Body: <pre>{ "Parent": ["UserName", "Workstation", "DpeName1", "Action", "MessageText"], "Child": ["ParentId", "RDbId", "RowNum", "DpeName", "Id", "LogType"] }</pre>		

	Response: <pre>{ "TableName": "ActivityLogTable", "Size": 2, "Page": 1, "Result": [{ "UserName": "DefaultAdmin", "Workstation": "_#WEBCLIENT#_", "DpeName1": "System1:Gms_Security_Users", "Action": "Login" }, { "UserName": "root", "Workstation": "AAEINPU742011D", "DpeName1": "System1:Gms_Security_Users", "Action": "Automatic Logoff" }], "SnapshotId": "ceffdb59-6b5d-4b2a-a3e5-e127867ed299:fe671773-1d76-4f13-8630-a338263565ca", "Total": 2919, "_links": [{ "Rel": "discardsnapshot", "Href": "api/historylogs/1/ActivityLogTable/ceffdb59-6b5d-4b2a-a3e5-e127867ed299%3afe671773-1d76-4f13-8630-a338263565ca", "IsTemplated": false }, { "Rel": "page", "Href": "api/historylogs/1/ActivityLogTable?snapShotId=ceffdb59-6b5d-4b2a-a3e5-e127867ed299%3afe671773-1d76-4f13-8630-a338263565ca&page=1&size=2", "IsTemplated": false }, { "Rel": "next", "Href": "api/historylogs/1/ActivityLogTable?snapShotId=ceffdb59-6b5d-4b2a-a3e5-e127867ed299%3afe671773-1d76-4f13-8630-a338263565ca&page=2&size=2", "IsTemplated": false }], "ErrorInfo": ["This is error 1", "This is error 2"] }</pre>
--	---

7.12.5.1 Name Filter

- Name filter allows to filter data on the basis of Designation of the object
- If no Name filter is specified, data will be retrieved for all objects of current system.

- While specifying Name filter, following wild cards are supported: * and ?
- To retrieve child nodes of an object, Name filter string should end with .* (<dot><star>)

7.12.5.2 Condition Filter

- Condition filter can be provided on multiple columns
- Multiple conditions on same or different columns can be ANDed
- If Condition filter is available on invalid column, data is zero records
- Condition filter specification formats for different types of columns:

1. String column

```
<'Column Descriptor'>[space]<Operator>[space]<"Value">
e.g. 'Location' = "\"text\""
```

Column Descriptor is in single quotes and Value is in double quotes. If an input string contains double quotes (""), it needs to be escaped with backslash(\).

2. Integer/float/double column

```
<'Column Descriptor'>[space]<Operator>[space]<Value>
```

Column Descriptor is in single quotes and Value is NOT in double quotes.

3. Date Time column

```
<'Column Descriptor'>[space]<Operator>[space]<Value>
```

Column Descriptor is in single quotes and Value is NOT in double quotes. Here date time value format should be based on client/logged in User's culture.

4. Enum (Test Group) column (type of String)

```
<'Column Descriptor'>[space]<Operator>[space]<"Value">
```

Column Descriptor is in single quotes and Value is in double quotes.

5. GmsDuration column

```
<'Column Descriptor'>[space]<Operator>[space]<"Value">
```

Column Descriptor is in single quotes and Value is in double quotes. Gms duration is special kind of case for which value should be specified in specific format. Its fixed format is DD:HH:MM:SS.fff

- Specifying multiple values for a column:

```
<'Column Descriptor'>[space]<Operator>[space]{<Value 1>;[<value 2>;[value N]]}
e.g., Discipline = {Fire ; HVAC}
```

This is nothing but OR operator supported between conditions for a single column. This format is supported only for equal-to(=) operator.

- Condition filter specification with Array values:
Some columns support display of array data. To specify condition filter on these columns, <-(IN) operator shall be used.

7.12.6 Discarding a Snapshot of a Log Table

Discards a snapshot of log table from memory. The data of this snapshot cannot be accessed afterwards.

Resource Path	/historylogs/:systemId/:tableName/:snapshotId		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	tableName	string	Name of table Required
	snapshotId	string	Id of snapshot which is to be discarded Required
Response	HTTP 200 OK LogDiscardedSnapshot object. See [→226]		
Examples	Request: DELETE api/historylogs/1/ActivityLogTable/ceffdb59-6b5d-4b2a-a3e5-e127867ed299%3afe671773-1d76-4f13-8630-a338263565ca Response: { "TableName": "ActivityLogTable", "SnapshotId": "c4005c4b-2d00-4a9c-b22a-750d55fba609:591ce21f-11a2-42e6-b875-21ec59abea70", "IsSnapshotDiscarded": true }		

7.12.6.1 Discard Snapshot strategy used by the WSI manager

Every Snapshot can consists of millions of records from the log table, it's a very heavy object to maintain. Maintaining multiple such snapshots for every session can intern increase the load of WSI in terms of Memory and Performance. So WSI have to have its own strategy to discard any snapshots which are not in use.

For this following configurable parameters will be used:

Property	Value	Description
MaxSnapshotsForSingleSession	5	Maximum number of Snapshots that a single session can have at any time
MaxSnapshotInactive Time	10 minutes	Maximum time in Minutes for which snapshot will be maintained by WSI if it is not used. After 10 minutes of unused time Snapshot will be discarded
MaxVirtualMemorySizeForProcess	bytes	Default Value: 4294967296 bytes (4 GB) Memory band for WSI manager. If WSI is running out of this memory band then before creating a new snapshot, an existing Snapshot from the system will be discarded first.

For discarding Snapshots following strategy will be used:

Greedy or fair share:

1. First check if current session is having any snapshots associated with it. If YES then the Snapshot with the oldest LastUsedTime will be identified to discard.
2. If current session is not having any Snapshot associated with it. Then the session for which maximum number of Snapshots are present will be identified first. And from that session Snapshot with the oldest LastUsedTime will be identified to discard.

Below are the 4 different cases in which force discard of snapshot will be called:

1. Request to create a new Snapshot and WSI manager is running out of decided Memory band
 - In this case till the time sufficient memory is not available and WSI runs within the decided memory band one by one Snapshots will be discarded by applying above mentioned strategy.
2. Request to create a new Snapshot and limit for maximum number of Snapshots is reached for that session
 - In this case applying above strategy one snapshot will be discarded first and after that request to create a new Snapshot will be processed
3. Snapshot is idle (not used) for 10 minutes
 - Any Snapshot for any session will be discarded if it is not in use for 10 minutes
4. User logs out/ Session timeout
 - If user logs out from WSI or if session timeout is called then all the snapshots for that session if any will be discarded.

7.12.7 Validating Condition Filter

Validates the Condition Filter which will be used to filter the data. If filter is not valid then a to-do message is provided.

Resource Path	/historylogs/:systemId/:tableName/validateconditionfilter/:conditionFilter		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	tableName	string	Name of the Table Required
	conditionFilter	string	Condition filter string to validate Required
Response	HTTP 200 OK LogConditionFilterValidation object. See [→226]		
Examples	Request: GET api/historylogs/1/EventLogTable/validateconditionfilter/'Action'%20=%20%22Add%20Camera Response: <pre>{ "ErrorPosition": 11, "IsValidConditionFilter": false,</pre>		

	<pre> "ErrorMessageDescriptor": "The symbol/term could not be recognized: \"Add Camera\", "ToDoMessageDescriptor": "Correct the condition filter expression as per rules:\r\nCondition filter name: 'HdbId', 'HRdbId', 'Event ID', 'Event Category', 'Event Time', ..., \r\nCondition filter operator: =, <, <=, >, >=, <=, \r\nCondition filter data: textual data (e.g. '\"Any text\"'), numeric data (e.g. '42'), date time data (e.g. '07/05/1976 14:07:00'), #COM, N/A, #ERR, null, \r\nBoolean operator: AND, OR, NOT, ;, \r\nSymbol: (,), {, }, " } </pre>
--	---

7.12.8 Retrieving Child Data for a Parent Table

Retrieves child data for a parent table. To retrieve child data a call 'Retrieving Data for Table' with isChildDataRequired set to true is required first. User also needs to send a list of interested columns for child data. If no columns are provided data for all columns will be returned. See [→**Error! Bookmark not defined.**]

Resource Path	/historylogs/:systemId/:tableName/:snapshotId/:parentId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	tableName	string	Name of table Required
	snapshotId	string	Snapshot Id for which child data will be retrieved Required
	parentId	string	Parent Id for which child data needs to be retrieved Required
Body			
Response	HTTP 200 OK LogChildData object. See [→226]		
Examples	Request: POST api/historylogs/1/EventLogTable/78b71424-bcce-44e1-9d7e-ffdfc11a0179:f6139cbd-5854-467f-be0e-227cab6b88fa/235 Request Body: Response: <pre> { "Result": [{ "RowNum": "2", "DpeName": "System1:ManagementView_ManagementSystem_Servers_Server.So ftwareRunning", "Id": "235" }, { "RowNum": "1", "DpeName": "System1:ManagementView_ManagementSystem Servers Server.So </pre>		

	<pre> ftwareRunning", "Id": "354", "LogType": "Property Changes" }], "Total": 2, "Page": 1, "Size": 100000, "_links": [{ "Rel": "discardsnapshot", "Href": "api/historylogs/1/EventLogTable/78b71424- bcce-44e1-9d7e-ffdfc11a0179%3af6139cbd-5854-467f-be0e- 227cab6b88fa", "IsTemplated": false }, { "Rel": "page", "Href": "api/historylogs/1/EventLogTable/78b71424- bcce-44e1-9d7e-ffdfc11a0179%3af6139cbd-5854-467f-be0e- 227cab6b88fa/235?page=1&size=100000", "IsTemplated": false }], "ErrorInfo":["This is error 1", "This is error 2"] } </pre>
--	--

7.12.9 Cancel Report Processing

Cancels in-process report generation for the SnapshotId that was provided in Retrive Data Pagewise Request [121]. If cancellation is successful then Retrive Data Pagewise Request will return Http status Bad Request(400) with below error:

Id":2402098

Error:"htlog_cancelled"

Details:"Request preliminary canceled by the user."}

Resource Path	/historylogs/systemId:snapshotId		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	systemId	uint	ID of a system Required
	snapshotId	string	Id of snapshot which is to be Cancel Required
Response	HTTP 200 OK LogDiscardedSnapshot object. See [→226]		
Examples	Request: DELETE https://aaeinpun001661.ad001.siemens.net:444/WSIApp/api/historylogs/1/sdsfsdfsdfffff3333 Response: HTTP 200 OK		

7.13 Diagnostics Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/diagnostics	Public		

7.13.1 Test Reachability of the Web Service Interface

A test whether or not the Web Service Interface replies to requests. This service also returns Server's UTC datetime as json response.

Resource Path	/diagnostics		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	clientDatetime	string	Client's UTC datetime. Optional
Response	HTTP 200 OK [→237]		
Examples	Request: GET /NBWSITest/NorthboundWebApiClient/api/diagnostics/ Response: HTTP 200 OK		

7.14 Language Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/languages	Public	x	

7.14.1 Retrieving the Language of the Logged in User

Resource Path	/languages		
Category	Public		
HTTP Method	GET		
Response	HTTP 200 OK A Language object [→64]		
Examples	Request: GET /api/Languages?api_key=username:password Response: { "Descriptor": "English (United States)", "Code": "en-US" }		

7.15 Image Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/images	Internal	x	

7.15.1 Retrieving an Image


NOTE:

In order to successfully process this request, the image needs to be present in a feasible format and size in the system.

Resource Path	/images:/imageId		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	imageId	string	Image name (file name without extension). Note that while XAML files can contain multiple images, only one image per file is supported. Required
	path	string	Path within a library. As of this writing, the path must not include the file extension: only XAML files are supported. Required
	format	string	Image format (png or svg) Required
	width	int	Image width in pixels
	height	int	Image height in pixels
	encodeAsBase64	bool	If true, the image is returned as a base64-encoded string.
Response	HTTP 200 OK The returned data depends on encodeAsBase64.		
	encodeAsBase64	Return Type	Description
	false (default)	image (image media type; see [10] [→ 19])	Image in the requested format
	true	string	base64-encoded string
Examples	Request: /api/images/Evt_Discp_Infrastr_None_001?format=png&width=1&height=1&path=System1:libraries%5CGlobal_Base_HQ_1%5Cicons&encodeAsBase64=true Response: iVBORw0KGgoAAAANSUheUgAAAAEAAAABCAyAAAAfFcSJAAAAAXNSR0IArs4c6QAAARnQU1BAACxjwv8YQUAAAJcEhZcwAADSIAAA7CARUoSoAAAAANSURBVbhXY/j//78GAakAyb6sRXrAAAAE1FTkSuQmCC		

7.16 Tables Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/tables	Partially Internal	x	

7.16.1 Retrieving a Table or an Entry of a Table

Resource Path	/tables/:tableName/:tableEntryId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Table identifier (table name) [131] Required
	id	string	Table entry identifier (key of row of table)
Response	HTTP 200 OK The returned data depends whether or not id is provided.		
	id	Return Type	Description
	Not provided	Array of KeyValue<keyname,string>	List of all key-value pairs in this table
	Provided	string	Localized value of this column
Examples	Request1: id Not provided GET api/tables/disciplines Response1: { "0": "Management System", "20": "Building Infrastructure", "50": "Building Automation", "100": "Fire", "150": "Security", "200": "Energy Management", "250": "Notification" } Request2: id provided GET api/tables/disciplines/100 Response2: "Fire"		

7.16.1.1 Table Names

TableId must have one of the following Name:

Name	Description
Categories	Categories defined for the System
Disciplines	Disciplines supported by Manage System
ObjectTypes	Object types used in Subsystems
Timefilter	Timefilter supported in Management System
Units	Supported units in Management System

7.16.2 Retrieving a Subtable of a Table

Resource Path	/tables/:tableName/subtables/:subtableName		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Table identifier (table name) Required
	subtableName	string	Subtable identifier (limited to "icons" or "colors") Required
	systemId	Int	System Identifier (query parameter) Optional
Response	HTTP 200 OK The returned data depends on the subtable id. Note that the example below is for the "icons" subtable id.		
	Subtable Id	Return Type	Description
	icons	Key-value pairs (int, string) containing the index and relative path to the icon	Icons are only available from the Disciplines table
	colors	Key-value pairs (int, Dictionary<int, Color>) indicating the valid colors	Colors are only available from the Categories table
Examples	Request: GET api/tables/disciplines/subtables/icons Response: <pre>{ "0": "System1:libraries\\Global_Base_HQ_1\\icons\\Evt_Discp_System_None_001.xml Evt_Discp_System_None_001", "20": "System1:libraries\\Global_Base_HQ_1\\icons\\Evt_Discp_Infras tr_None_001.xml Evt_Discp_Infrastr_None_001", "50": "System1:libraries\\Global_Base_HQ_1\\icons\\Evt_Discp_BA_Non e_001.xml Evt_Discp_BA_None_001" }</pre>		

7.16.3 Retrieving a Subtable of a Table for given row of table

Resource Path	/tables/:tableName/:id/subtables/:subtableName		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Table identifier (table name) Required
	id	int	Table entry identifier (key of row of table) Required
	subtableName	string	Subtable Identifier Required

Response	HTTP 200 OK		
	The returned data depends on the subtable id. Note that the example below is for the "colors" subtable id.		
	Subtable Id	Return Type	Description
	icons	Key-value pairs (int, string) containing the index and relative path to the icon	Icons are only available from the Disciplines table
Examples	colors	Key-value pairs (int, Dictionary<int, Color>) indicating the valid colors	Colors are only available from the Categories table See Object [→ 210]
	Request: GET api/tables/categories/1/subtables/colors Response: <pre>{ "1": "255, 255, 0", "2": "255, 255, 0", "3": "255, 255, 0", "4": "0, 0, 0", "5": "255, 255, 0", "6": "239, 105, 38", "7": "201, 43, 39", "8": "167, 73, 27", "9": "147, 40, 16", "10": "152, 150, 154", "11": "90, 93, 96", "12": "230, 148, 113", "13": "250, 189, 155" }</pre>		

7.16.4 Retrieving Subgroups with filter

This method help retrieves subgroups of the Disciplines OR ObjectTypes. Filter is also available for retrieving specific Disciplines, SubDisciplines OR ObjectTypes, SubObjectTypes. This filter parameter is query string parameter. If no SubDisciplines/SubObjectTypes are specified while requesting Disciplines/ObjectTypes respectively then all the SubDisciplines/SubObjectTypes which comes under the requested Discipline/ObjectType will be returned. Otherwise only the requested SubDisciplines/SubObjectTypes will be returned.

Resource Path	/tables/:tableName/subgroups		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Table identifier (table name) Required
	filter	string	This is query string parameter. Filter for Disciplines, SubDisciplines OR ObjectTypes, SubObjectTypes Required
Response	HTTP 200 OK Based on the request is for Disciplines OR ObjectTypes response will be returned. - Disciplines : See [→ 63] - ObjectTypes : See [→ 63]		

Examples	Request1: Disciplines and SubDisciplines request <pre>GET api/tables/disciplines/subgroups?filter={"0":[0,1],"100":[101,102]}</pre> Response1: <pre>[{ "DisciplineId": 0, "DisciplineDescriptor": "Management System", "SubDisciplines": [{ "Id": 0, "Descriptor": "Unassigned" }, { "Id": 1, "Descriptor": "Applications" }] }, { "DisciplineId": 100, "DisciplineDescriptor": "Fire", "SubDisciplines": [{ "Id": 101, "Descriptor": "Evacuation" }, { "Id": 102, "Descriptor": "Extinguishing" }] }]</pre> Request2: ObjectTypes and SubObjectTypes request <pre>GET api/tables/objecttypes/subgroups?filter={"1000":[1001,1002],"1100":[1101,1102]}</pre> Response2 : <pre>[{ "ObjectTypeId": 1000, "ObjectTypeDescriptor": "Client Software", "SubObjectTypes": [{ "Id": 1001, "Descriptor": "Control Room" }, { "Id": 1002, "Descriptor": "Mobile" }] }, { </pre>
----------	---

	<pre>"ObjectId": 1100, "ObjectTypeDescriptor": "Command", "SubObjectTypes": [{ "Id": 1101, "Descriptor": "Analog" }, { "Id": 1102, "Descriptor": "Binary" }] }</pre>
--	--

7.16.5 Retrieving Image from Global Text Table

**NOTE:**

If the request is for a PNG image, the image file must be in a format that is either PNG or can be converted to PNG. If the request is for an SVG image, no conversion can be made, so the image file must be SVG.

Resource Path	/tables/global/:tableName/icons/:entryId		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Well-known global text table name [→ 238] Required
	entryId	int	Entry within the table; only values greater than or equal to zero are valid. Required
	format	string	Image format(s) in the order in which to attempt to load them – separated by a comma (currently only “png” and “svg” are supported). The default is “svg,png” but if only one format is acceptable, the string should be “png” or “svg”. Optional
	width	int	Image width in pixels. The default is 32. Optional
	height	int	Image height in pixels. The default is 32. Optional
Response	HTTP 200 OK.		
Examples	Request: /api/tables/global/Disciplines/icons/0 Response: One of the following: 1) Base64 encoded PNG: with the specified dimensions. 2) SVG: the entire contents of the SVG file 3) Blank: there is no icon assigned to the entry, it could not be loaded, or the specified icon was not in one of the supported formats.		

7.16.5.1 Well-Known Global Text Table Names

Global text tables are tables that are found on all online systems in a distributed environment. Within DCC, there is a limited number of tables that are considered "well-known global text tables:" they are specified below.

Name	Description
Disciplines	Disciplines supported by the system.
Subdisciplines	Subdisciplines supported by the system; a subdiscipline will normally refer to a discipline.
ObjectTypes	Object types used in the system.
ObjectSubtypes	Object subtypes used in the system; a subtype will normally refer to an object type.
Categories	Categories defined for the system.

7.16.6 Retrieving Text From Global Text Table

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. The sample response below includes the color property on some (but not all) of the entries. See [→ 228] for further discussion of the color property.

Resource Path	/tables/global/:tableName/text		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Well-known global text table name [→ 238] Required
Response	HTTP 200 OK A list of objects containing value and text properties [→ 227]		
Examples	Request: /api/tables/global/Disciplines/text Response: <pre>[{ "Value": 0, "Color": 4290764270, "Text": "Management System" }, { "Value": 20, "Text": "Building Infrastructure" }, { "Value": 50, "Color": 4290764270, "Text": "Building Automation" }, { "Value": 100, "Color": 4290764270, "Text": "Fire" }, { "Value": 150, "Text": "Security" }, { "Value": 200,</pre>		

	<pre>"Text": "Energy Management" }, { "Value": 250, "Text": "Notification" }]</pre>
--	--

7.16.7 Retrieving all Language Text from Global Text Table

The returned table will contain text for every language that is installed; in the example below there are three languages (en-US, de-DE, fr-CA).

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. The sample response below includes the color property on some (but not all) of the entries. See [→ 228] for further discussion of the color property.

Resource Path	/tables/global/:tableName/langText		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Well-known global text table name [→ 238] Required
Response	HTTP 200 OK An array of objects containing a value, text for each language, and optionally a color [→ 227]		
Examples	Request: /api/tables/global/Categories/langText Response: [{ "Value": 1, "Color": 4290764270, "LangText": ["Status", "Zustand", "État", "Status"] }, { "Value": 2, "Color": 4292203223, "LangText": ["Fault", "Störung", "Défaut", "Fault"] }, { "Value": 3, "Color": 4294501476, "LangText": ["Trouble", "Problem", "Problème", "Trouble"] },]],		

7.16.8 Retrieving Nested Text From Global Text Table

The returned table will contain entries from the specified text table and any subtext. The example below is of the Disciplines table: the Subdisciplines table considers the Disciplines table as a 'parent' and an entry in the Subdisciplines table normally refers to an entry in the Disciplines table.

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. The sample response below includes the color property on some (but not all) of the entries. See [→ 228] for further discussion of the color property.

Resource Path	/tables/global/: tableName /nestedText		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Well-known global text table name [→ 238] Required
Response	HTTP 200 OK The return is an array of TextTableEntry objects [→ 227]		
Examples	Request: /api/tables/global/Disciplines/nestedText Response: <pre>[{ "Value": 0, "Text": "Management System", "Color": 4290764270, "SubText": [{ "Value": 0, "Color": 4290764270, "Text": "Unassigned" }, { "Value": 1, "Text": "Applications" }, { "Value": 2, "Color": 4290764270, "Text": "Communication" }, { "Value": 3, "Text": "Supervision" }, { "Value": 4, "Color": 4290764270, "Text": "System Settings" }] }, { "Value": 20, "Text": "Building Infrastructure", "SubText": [</pre>		

	<pre>{ "Value": 0, "Text": "Unassigned" }, { "Value": 21, "Color": 4290764270, "Text": "Elevators" }, { "Value": 22, "Color": 4290764270, "Text": "Escalators" }, { "Value": 23, "Text": "Data Network" }, { "Value": 24, "Text": "I/O Monitoring" }], { ... more entries ... }]</pre>
--	--

7.16.9 Retrieving Text From a Local Text Table

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. The sample response below includes the color property on some (but not all) of the entries. See [→ 228] for further discussion of the color property.

Resource Path	/tables/local/:systemId:/tableName/text		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	int	The id of the system from which the text table is to be retrieved; zero is not a legal id but can be used to indicate the "local" system. Required
	tableName	string	Local text table name Required
Response	HTTP 200 OK A list of objects containing value, text, and optionally color properties. [→ 227]		
Examples	Request: /api/tables/local/0/TxG_Discipline/text Response: [{ "Value": 0, "Color": 4290764270, "Text": "Management System" }, { "Value": 20,		

	<pre> "Text": "Building Infrastructure" }, { "Value": 50, "Color": 4290764270, "Text": "Building Automation" }, { "Value": 100, "Color": 4290764270, "Text": "Fire" }, { "Value": 150, "Text": "Security" }, { "Value": 200, "Text": "Energy Management" }, { "Value": 250, "Text": "Notification" }] </pre>
--	---

7.16.10 Retrieving All Language Text From a Local Text Table

The returned table will contain text for every language that is installed; in the example below there are three languages (en-US, de-DE, fr-CA).

Note that if a color has been assigned to the text table entry it will be included in the returned JSON string. The sample response below includes color in some of the entries to illustrate this fact. See [→ 228] for further discussion of the color property.

Resource Path	/tables/local/:systemId/: tableName /langText		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	int	The id of the system from which the text table is to be retrieved; zero is not a legal id but can be used to indicate the "local" system. Required
	tableName	string	Local text table name Required
Response	HTTP 200 OK A list of objects that includes value, text for every installed language, and optionally a color (if one has been assigned to the entry [→ 227])		
Examples	Request: /api/tables/local/0/TxG_Categories/langText Response: <pre> [{ "Value": 1, "Color": 4290764270, "LangText": [</pre>		

	<pre>"Status", "Zustand", "État", >Status"] }, { "Value": 2, "Color": 4292203223, "LangText": ["Fault", "Störung", "Défaut", "Fault"] }, { "Value": 3, "Color": 4294501476, "LangText": ["Trouble", "Problem", "Problème", "Trouble"] }, }, }</pre>
--	---

7.16.11 Retrieving an Image for an Entry of a Table

Resource Path	/tables/local/:systemId/:tableName/icons/:tableEntryId		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Identifies a text table (such as "TxG_ValidationProfile") Required
	tableEntryId	uint	Table entry identifier (key of row of table)
Response	HTTP 200 OK The returned data will be an empty string if no image has been assigned to the entry.		
Examples	Request: GET api/tables/local/0/TxG_ObjectType/icons/100 Response: iVBORw0KGgoAAAANSUheUgAAACAAAAAgCAYAAABzenr0AAAAAXNSR0IArs4c6QAAARnQUlBAACxjwv8YQUAAAJcEhZcwAADsMAAA7Dacdvq		

7.16.12 Retrieving an Entry of a Table

Note that if a color has been assigned to the text table entry it will be included in the returned JSON string. The sample response below includes color to illustrate this fact, but color is not guaranteed to appear in all cases. See [→ 228] for further discussion of the color property.

Resource Path	/tables/local/:systemId/:tableName/text/:tableEntryId		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	tableName	string	Identifies a text table (such as "TxG_ValidationProfile") Required
	tableEntryId	uint	Table entry identifier (key of row of table)

Response	HTTP 200 OK An object containing the value, text, and optionally a color. [→ 227]
Examples	Request: GET api/tables/local/0/TxG_ValidationProfile/text/0 Response: <pre>{ "Value": 0, "Text": "Disabled", "Color": 4278190080 }</pre>

7.17 Systems Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/systems	Public	x	
/systemsSubscriptions	Public	x	x

7.17.1 Retrieving list of installed languages (same order) as installed in project

Resource Path	/systems/languages
Category	Public
HTTP Method	GET
Response	HTTP 200 OK A SystemsLanguages object [→ 228]
Examples	Request: GET /api/systems/languages Response: <pre>{ "ArrayIndex": 0, "Descriptor": "English (United States)", "Code": "en-US" }, { "ArrayIndex": 1, "Descriptor": "French (France)", "Code": "fr-FR" }, { "ArrayIndex": 2, "Descriptor": "German (Germany)", "Code": "de-DE" }</pre>

7.17.2 Retrieving list of all available systems with list of installed languages.

Resource Path	/systems
Category	Public
HTTP Method	GET

Response	HTTP 200 OK A SystemsRepresentation object [→ 228]
Examples	Request: GET / api/systems Response: <pre>{ "Systems": [{ "Name": "System14", "Id": 14, "IsOnline": true }, { "Name": "System15", "Id": 15, "IsOnline": true }], "Languages": [{ "ArrayIndex": 0, "Descriptor": "English (United States)", "Code": "en-US" }, { "ArrayIndex": 1, "Descriptor": "German (Germany)", "Code": "de-DE" }], "IdLocal": 14, "IsDistributed": true }</pre>

7.17.3 Retrieving information of a specific system

Resource Path	/systems/{systemid}
Category	Public
HTTP Method	GET
Response	HTTP 200 OK A SystemsRepresentation object [→ 228]
Examples	Request: GET / api/systems/1 Response: <pre>{ "Systems": [{ "Name": "System14", "Id": 14, "IsOnline": true }], "Languages": [{ "ArrayIndex": 0,</pre>

	<pre> "Descriptor": "English (United States)", "Code": "en-US" }, { "ArrayIndex": 1, "Descriptor": "German (Germany)", "Code": "de-DE" }] } </pre>
--	---

7.17.4 Retrieving information of a local system

Resource Path	/systems/local
Category	Public
HTTP Method	GET
Response	HTTP 200 OK A SystemsRepresentation object [→ 228]
Examples	Request: GET /api/systems/local Response: <pre> { "Systems": [{ "Name": "System2", "Id": 2, "IsOnline": true }], "Languages": [{ "ArrayIndex": 0, "Descriptor": "English (United States)", "Code": "en-US" }, { "ArrayIndex": 1, "Descriptor": "German (Germany)", "Code": "de-DE" }, { "ArrayIndex": 2, "Descriptor": "French (France)", "Code": "fr-FR" }] } </pre>

7.17.5 Get service status

This will return the current status ('isConnected') of all services. If user has scope of multiple system, it will return the service statuses of multiple systems.

Note: Currently, only the status for Report Manager (id 58) is returned.

Resource Path	/systems/services
Category	Public

HTTP Method	GET
Response	HTTP 200 OK A List< ServicesNotificationRepresentation > object
Examples	Request: GET / api/systems/services Response: [{ "SystemId":4, "ServiceId":58, "IsConnected":false }]

7.17.6 Create channelize Systems subscription

If subscription response and notifications are needed on signalr channel.

Subscription response will be pushed through notifySubscriptionStatus function available at client side. Sound changes will be notified via notifySounds.

Resource Path	/api/sr/systemssubscriptions/channelize/:requestId:/connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub connection Required
Response	HTTP 200 OK		
Examples	Request: POST api/sr/systemssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b notifySubscriptionStatus: { "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifySystems" }		

7.17.7 Delete Subscription for a Systems

Resource Path	/api/sr/ systemssubscriptions /{connectionId}		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	connectionId	String	Identifier of subscription Required
Response	HTTP 200 OK.		

Examples	Request: DELETE /api/sr/systemssubscriptions/e3d90a35-1f7a-4ec2-8b3b-2fba71b76bc7 Response: HTTP/1.1 200 OK
----------	--

7.17.8 Push Notification

After subscribing for event sounds, the current sound file with a link to download it will be sent in an initial notification. Afterwards, any change will be notified.

Type	Name (case sensitive)		
Hub	norisHub		
Function	notifySystems (data)		
	Property	Type	Description
	data	SystemsRepresentation	System details [→ 228]



NOTE:

If in JSON response the value of “FileName” and “_links” are empty then no sound has to be played.

7.17.8.1 Subscription Status Notification

After subscribing for event sounds, the subscription status will be sent only once.

Type	Name (case sensitive)		
Hub	1. norisHub 2. eventSoundsHub (lagacy)		
Function	notifySubscriptionStatus(status)		
	Property	Type	Description
	Status	Subscription Response	See Subscription Status [210]
Example	<pre>{ "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifySystems" }</pre>		

7.17.9 Create channelize Services subscription

This allows subscription of services according to system. If the system id is within the scope of the logged in user and if the service id is valid, then user will be subscribed to that service id for that particular system and will get notifications when the state of the service changes.

Currently, the only supported Service ID is 58 (for Report Manager). Other services will not be subscribe.

Resource Path	/api/sr/systemssubscriptions/channelize/:requestId/:connectionId/services/
---------------	--

Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub connection Required
Body	Services	ServicesRegistrationRepresentation[]	Contains a list of the systems and their corresponding services which the client wants to subscribe to. Required
Response	HTTP 200 OK as List<ServicesSubscriptionsRepresentation>		
Examples	Request: POST api/sr/systemssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b/services Body: serviceIds [{"SystemId":1,"Ids":[58,60,78]},{"SystemId":4,"Ids":[58,60]},{"SystemId":5,"Ids":[58,61,62]}] Response: [{"SystemId":1,"ServiceId":58,"ErrorCode":1},{"SystemId":1,"ServiceId":60,"ErrorCode":1},{"SystemId":1,"ServiceId":78,"ErrorCode":1},{"SystemId":4,"ServiceId":60,"ErrorCode":1},{"SystemId":5,"ServiceId":61,"ErrorCode":1},{"SystemId":5,"ServiceId":62,"ErrorCode":1},{"SystemId":4,"ServiceId":58,"ErrorCode":0},{"SystemId":5,"ServiceId":58,"ErrorCode":0}] notifySubscriptionStatus: ErrorCode: 0, RequestFor: "notifyServices",RequestId: "9565ca41-8556-4dbb-94cc-1b89451a5db5"		

Note: For a particular connectionId, subscription request can be sent only once. If more systems or services are to be included in the subscription, then unsubscribe and subscribe again.
If no valid combination of systemid and service id is passed in the request, it will return an error with 400 code.

7.17.9.1 Services Notification

After successful subscription, an **initial notification** will be generated with the current statuses of the subscribed services. After that, a notification will be generated whenever the state of the subscribed service id(s) **changes**.

Type	Name (case sensitive)
Hub	1. norisHub

Function	notifyServices (status)		
	Property	Type	Description
	Status	List<ServicesNotificationRepresentation> in case of initial notification. ServicesNotificationRepresentation in case of change in service status(for subsequent notifications)	See ServicesNotificationRepresentation
Example	<pre> ▼ {C: "d-165E5373-B,0 C,7 D,1", M: [{H: "NorisHub", M: "notifyServices", A: [...]}]} C: "d-165E5373-B,0 C,7 D,1" ▼ M: [{H: "NorisHub", M: "notifyServices", A: [...]}] ▼ 0: {H: "NorisHub", M: "notifyServices", A: [...]} ▼ A: [...]</pre>		

7.17.10 Unsubscribe Services

This will unsubscribe all services which have been subscribed for a particular connection id.

Resource Path	/api/sr/ systemssubscriptions /services/{connectionId}		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	connectionId	String	Identifier of subscription Required
Response	HTTP 200 OK.		
Examples	Request: DELETE /api/sr/systemssubscriptions/services/e3d90a35-1f7a-4ec2-8b3b-2fba71b76bc7 Response: HTTP/1.1 200 OK		

7.18 Products Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/Products	Public	x	

7.18.1 Retrieving Product Information

Retrieve a list of all Trend Collector objects of a given system. It includes all Online

Resource Path	/Products
Category	Public
HTTP Method	GET
Response	HTTP 200 OK ProductInfo object [→ 229Error! Bookmark not defined.]
Examples	Request: GET /api/Products Response: { "ProductNameShort": "Desigo CC", "ProductName": "Desigo™ CC", "VendorName": "Siemens Switzerland Ltd.", "ProductCopyRights": "© Siemens Switzerland Ltd, 2016", "ProductVersion": "V3.0", "ServerVersion": "1.0.0.0", "SupportLink": "www.siemens.com/SBT", "links": [{ "Rel": "self", "Href": "api/products", "IsTemplated": false }, { "Rel": "downloadfile", "Href": "api/repository/ProductPictures/files?names=entrypage_image.JPG", "IsTemplated": false }, { "Rel": "downloadfile", "Href": "api/repository/ProductPictures/files?names=SplashScreenImage.jpg", "IsTemplated": false }] }

7.19 Files Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/files/{fileType}	Partially Internal		
/files/{systemId}/{relativeFilePath}		X	

7.19.1 Retrieving Files of specific type

Resource Path	/files/{fileType}		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	fileType	BasicFiles See [→ 229]	Supported file types Required
	names	String[]	This is query string parameter. Contains only the name of the files. or array of file names.
Response	HTTP 200 OK		
Examples	Request:		

	GET /api/files/0?names=entrypage_image.JPG Response: Returns stream content down to the client, chunk by chunk
--	---

7.19.2 Retrieve Files from system

Currently access is provided for "libraries" folder path

Resource Path	/files/{systemId}/{relativeFilePath}		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	systemId	uint	System identifier Required
	relativeFilePath	String	Relative file path.e.g. see below Required
Response	HTTP 200 OK		
Examples	Request: GET / api/files/1/libraries\BA_Cooling_HQ_1\Symbols\DYN_2D_Chiller_None_Analog_C entral_001.ccs Response: Returns byte array content.		

7.19.3 Retrieve Documents from system

Documents can be retrieved by passing designation of Systembrowser tree nodes

Resource Path	/files/documents/{designation}		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	designation	string	Designation id of node in systembrowser tree (Recommendation: Double UrlEncode parameter prior to making it to part of URI.) Required
Response	HTTP 200 OK Content-Type header set to particular filetype		
Examples	Request: GET / api/files/documents/System1.ApplicationView%253AApplicationView.Documents.p dfSample Response: Returns byte array content.		

7.20 Graphics Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/graphics/{objectId}	Internal	x	
/graphics/itemIds/{objectDesignation}	Internal	x	
/graphics/items/{objectId}	Internal	x	
/children/itemIds/{objectDesignation}	Internal	x	

This service provides the following functionality:

- To determine whether an object is relevant to a client that processes graphics. By relevant we mean: (a) the object itself is one of the known graphic types (such as a graphic or a graphic viewport), (b) the object is related to a graphic type (such as an object that is found in a graphic), or (c) one of the object's ancestors is either a graphic type or is related to a graphic type. This functionality is generally referred to as the "preselect" operation.
- To retrieve the list of graphic objects related to this object. If the object is one of the graphic types it will be returned; if it is not one of the graphic types, the list will contain all graphic types related to the object and all graphic types related to all ancestors of the object. This functionality is generally referred to as the "select" operation.
- To retrieve a specific graphic object, such as a graphic or graphic viewport. This functionality is generally referred to as the "load" operation.
- To retrieve the first child (of a node in a view) that is either graphical or has a related graphics.

Note that the term *ancestor* in this context means objects higher in a hierarchy, up to and including the root object. During a preselect or a select operation all hierarchies on the system where the object is defined are checked.

7.20.1 Check if object is graphical

Determine if an object, an object it references, or one of its ancestors is graphical in nature. For example, an object will be considered "graphical in nature" if it is a DCC graphic, template, or viewport (within a graphic). The object itself may not be graphical but can refer to another object that is graphical (such as a non-graphical object that appears on a graphic).

Resource Path	/graphics/{objectId}		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectId	String	Object identifier, including system information Required
Response	HTTP 200 Ok HTTP 204 No Content		
Examples	Request: GET /api/graphics2/System1:ObjectName Response: Returns status code 200 if the object (or something it references) or one of its ancestors is graphical in nature; otherwise code 204 is returned.		

7.20.2 Retrieving graphic item ids

Retrieve all graphic items ids related to an object or its ancestors: this may include the object itself. Note a couple of things about the graphic items:

- 1) Graphic items that reference a graphical template will include the designation of the 'owning' object (see 'Context' in example below).
- 2) There will be no duplicate references to a graphic.
- 3) There will be no duplicate references to a template for the same 'owning' object. For example: if an object is determined to be an ancestor in two or more situations, only one of the situations will be reflected in the results.

Resource Path	/graphics/itemIds/{objectDesignation}
Category	Internal
HTTP Method	GET

Parameters	Property	Type	Description
	objectDesignation	String	The “fully assembled” name of an object within a hierarchy. This name is also commonly referred to as the “designation.” The name specifies the system, hierarchy, and position of the object within the hierarchy. Required
Response	HTTP 200 OK (one or more GraphicalItems [→ 230] will be returned) HTTP 204 No Content (there are no graphic items associated with the object)		
Examples	Request: This example is a loop object found within a BACnet device (8166) on a BACnet network (“bacnetNetwork”). GET /api/graphics/itemIds/System1:ManagementView:ManagementView.FieldNetworks.bacnetNetwork.Hardware.8166.Local_IO.LP_1 Response: Returns a list of related graphic items, possibly including the object itself (if it is graphical in nature). <pre>[{ "Context": "System1:ManagementView:ManagementView.FieldNetworks.bacnetNetwork.Hardware.8166.Local_IO.LP_1", "ObjectId": "System1:GraphicsTemplate_TEM_Object_Loop_None_None_001", "ManagedType": "Graphic Template", "DisplayName": "TEM_Object_Loop_None_None_001" }, { "Context": "System1:ManagementView:ManagementView.FieldNetworks.bacnetNetwork.Hardware.8166.Local_IO.LP_1", "ObjectId": "System1:Graphics_AnotherGraphic", "ManagedType": "Graphic", "DisplayName": "SomeGraphic" }]</pre>		

7.20.3 Retrieving a graphical item

Retrieve the content for a graphical item. The content will be in the form of XML containing SVG with two sections: <graphic> and <libraries>. The <graphic> section will include the definition of the graphical item, while the <libraries> section will contain any symbols referenced by the item – note that a symbol may refer to other symbols, and these will also be in this section.

Resource Path	/graphics/items/{objectId}		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectId	String	Object identifier; this must include the system information. For example: “System1:MyGraphic” Required
Response	HTTP 200 OK HTTP 204 No Content		
Examples	Request: GET /api/graphics/items/System1:MyGraphicalObject Response: <pre><?xml version="1.0" encoding="utf-8"?> <!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 1.1//EN" ""> <graphicItems version="0.9"> <graphic> <svg> ... graphical item is defined here ... </svg> </graphic> </graphicItems></pre>		

	<pre><libraries> <svg> ... symbol definition ... </svg> <svg> ... another symbol definition ... </svg> </libraries> </graphicItems></pre>
--	---

7.20.4 Retrieving a child item

Retrieve the first “interesting” child of a node, where “interesting” means:

- 1) The child is a graphical object itself
- 2) The child has a related template
- 3) The child has a related graphical object (such as being in a graphic)

Note that the method requires a fully qualified view node name and that the children of that node are searched for one that is interesting.

Resource Path	/graphics/children/itemIds/{objectDesignation}		
Category	Internal		
HTTP Method	GET		
Parameters	objectDesignation	String	Object designation (a fully qualified node name). For example: "System.View:View.Level" Required
	generations	UInt32	The number of generations to search (downward). The range is 1 to 20; if the value is not provided a default of 1 is used. Optional
	searchOptions	UInt64	Search options as a bitmask. Currently only one option (value 0x1) has been defined: to return a single object (the alternative is to return all interesting children). As of right now this parameter is ignored and only the first interesting child is returned. Optional
	types	UInt32	The types of objects to consider 'graphical.' As of right now this parameter is ignored; the types to be considered 'graphical' for this method are custom graphics and templates. Optional
Response	HTTP 200 OK (a browser object will be returned) HTTP 204 No Content (no interesting child was found)		
Examples	Request: GET /api/graphics/children/itemIds/System.View:View.Level Response: { "HasChild": false, "ViewId": 11, "ViewType": 4, "Attributes": { "DefaultProperty": "Present_Value", "ObjectId": "System1:GmsDevice_1_8166_1", "DisciplineDescriptor": "Building Automation", "DisciplineId": 50, "SubDisciplineDescriptor": "Unassigned", "SubDisciplineId": 0, "TypeDescriptor": "Sensor", "TypeId": 6700, "SubTypeDescriptor": "Unassigned", "SubTypeId": 0, "ManagedType": 80, "ManagedTypeName": "BACnetPoint", "ObjectModelName": "GMS_BACNET_EO_BA_AI_1", } }		

```

"ValidationRules": {
  "CommentRule": "Optional",
  "ReAuthentication": "NoNeed",
  "Configuration": 0,
  "IsFourEyesEnabled": false,
  "_links": []
},
},
"Location": "System.View:View.Level.Analog Input 1",
"SystemId": 1,
"Name": "AI_1",
"Descriptor": "Analog Input 1",
"Designation": "System.View:View.Level.AI_1",
"ObjectId": "System1:GmsDevice_1_8166_1",
"_links": [
  {
    "Rel": "properties",
    "Href": "api/properties/System1%253AGmsDevice_1_8166_1?requestType=0",
    "IsTemplated": false
  }
]
}
}

```

7.21 EventCategorySound Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/api/eventcategorysounds	Public	x	
api/sr/eventcategorysoundssubscriptions	Public	x	x

7.21.1 Get current event sound



NOTE:

If in JSON response the value of "FileName" and "_links" are empty then no sound has to be played.

Resource Path	api/eventcategorysounds		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	disableCategories	query:(string)	Filter to disable categories. List of disable category. i.e. category Ids for which sound information is not required Optional
Response	HTTP 200 OK File details object See [→230]		
Examples	Request: GET / api/eventcategorysounds?disableCategories=[1,2] Response: <pre> { "FileName": "0dfe9579-2c20-4cad-9a07-7bc14f4164a5.wav", "_links": [{ "Rel": "DownloadSoundFile", "Href": </pre>		

	<pre>"api/files/1/Libraries%5cGlobal_Events_HQ_1%5cMediaLibrary%5c0dfe9579-2c20-4cad-9a07-7bc14f4164a5.wav", "IsTemplated": false }] }</pre>
--	---

7.21.2 Subscribe for event sound

Resource Path	/api/sr/eventcategorysoundssubscriptions/{connectionId}		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	connectionId	string	Identifier of subscription Required
Response	HTTP 200 OK		
Examples	Request: POST api/sr/eventcategorysoundssubscriptions/c6c95ece-133b-480b-af1f-067a173e0f5b Response: HTTP/1.1 200 OK		

7.21.3 Delete Subscription for a event sounds changes.

Resource Path	/api/sr/eventcategorysoundssubscriptions/{connectionId}		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	connectionId	String	Identifier of subscription Required
Response	HTTP 200 OK.		
Examples	Request: DELETE /api/sr/eventcategorysoundssubscriptions/e3d90a35-1f7a-4ec2-8b3b-2fba71b76bc7 Response: HTTP/1.1 200 OK		

7.21.3.1 Push Notification

After subscribing for event sounds, the current sound file with a link to download it will be sent in an initial notification. Afterwards, any change will be notified.

Type	Name (case sensitive)		
Hub	1. norisHub : recommended 2. eventSoundsHub : lagacy		
Function	notifySounds (data)		
	Property	Type	Description
	data	FileRepresentation	File details . See [→230]

**NOTE:**

If in JSON response the value of "FileName" and "_links" are empty then no sound has to be played.

7.21.4 Create channelize event sound subscription

If subscription response and notifications are needed on signalr channel.

Subscription response will be pushed through notifySubscriptionStatus function available at client side. Sound changes will be notified via notifySounds.

Resource Path	/api/sr/eventcategorysoundssubscriptions/channelize/:requestId/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub connection Required
Response	HTTP 200 OK		
Examples	Request: POST api/sr/eventcategorysoundssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b notifySubscriptionStatus: <pre>{ "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":" notifySounds" }</pre>		

7.21.4.1 Subscription Status Notification

After subscribing for event sounds, the subscription status will be sent only once.

Type	Name (case sensitive)		
Hub	2. norisHub 3. eventSoundsHub (lagacy)		
Function	notifySubscriptionStatus(status)		
	Property	Type	Description
	Status	Subscription Response	See Subscription Status [210]
Example	<pre>{ "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifySounds" }</pre>		

7.22 Settings Service

Settings are stored on a “per user” basis.

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/settings/{settingId}	Internal	x	

7.22.1 Read setting

Read a setting; the setting value is returned ad the body of the response.

Resource Path	/settings/:settingId		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	settingId	String	Identifies the setting to be retrieved. Required
Response	HTTP 200 OK HTTP 204 No Content		
Examples	Request: GET /api/settings/MySettingId Response: Returns status code 200 if the setting exists; otherwise code 204 is returned.		

7.22.2 Write setting

Write a setting; the setting id is required as part of the URL while the content must be provided as part of the body of the request (a null or empty string is not valid and will result in a Bad Request response code). Note that a setting does not have to exist prior to writing it.

Resource Path	/settings/:settingId		
Category	Internal		
HTTP Method	PUT		
Content Type	application/x-www-form-urlencoded		
Parameters	Property	Type	Description
	settingId	String	Identifies the setting to be written. Required
Body	The content of the setting to be stored for the current user.		
Response	HTTP 200 OK HTTP 204 No Content HTTP 400 Bad Request		
Examples	Request: PUT /api/settings/MySettingId Post Data: A string Response: Returns status code 200		

7.22.3 Delete setting

Delete a setting for the current user; note that deleting a non-existent setting returns success (200).

Resource Path	/settings/:settingId		
Category	Internal		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	settingId	String	Identifies the setting to be deleted. Required
Response	HTTP 200 OK		
Examples	Request: DELETE /api/settings/MySettingId Response: Returns status code 200		

7.23 Users Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/users/password	Internal		

7.23.1 Change Password

Change password by providing user details.



NOTE:

If user has a valid token and password is changed, token remains valid.

If the new password does not follow the password policy (configured in SMC), then password will not be allowed to be changed and HTTP 400 error will be returned.

Resource Path	api/users/password		
Category	Internal		
HTTP Method	PUT		
Body	Property	Type	Description
	userName	string	Username Required
	currentPassword	string	Current password Required
	newPassword	string	New password Required
Response	HTTP 200 OK HTTP 400 Bad Request		
Examples	Request: PUT /api/users/password Body: (new password follows policy) <pre>{ "userName": "GmsUser", "currentPassword": "XXXX", "newPassword": "ZZZZ" }</pre> Response: HTTP 200 OK Body: (new password does not follow policy)		

	<pre>{ "userName": "GmsUser", "currentPassword": "aaaa", "newPassword": "bbbb" }</pre> Response: HTTP 400 Bad Request <pre>{ Details: "Password does not meet policy requirements." Error: "password_not_compatible" Id: 2400004 }</pre>
--	--

7.23.2 Get Login Information

Get login information for the requested user. This is open API, i.e. it does not require any authentication. Provided a user, it will return account type of that user as integer. Based on this account type client of this API (especially Flex Client) can take decision on how to accept the password by showing correct UI to the end user. For any user which is not present in the DesigoCC, AccountType will be returned as DesigoCC (AccountType = 0).

If the user is of type OpenID (AccountType = 3) then the additional query parameter redirectUri needs to be sent in the request along with the client's login page uri as its value. If this redirectUri value is null or is not in a valid Uri format, then AccountType will be returned as DesigoCC (AccountType = 0).

Following are possible account type values:

0 = DesigoCC Account

1 = Windows Account

2 = Software Account (Not used for login from UI as of now)

3 = OpenID Account

Incase if the account type is OpenID (AccountType = 3), and the provided redirectUri is in valid Uri format then additional value of OpenID Login Uri will be sent in the same response as shown in the below example. In all the other cases this will be null.

When the OpenID configuration is missing in the system, or provided redirect Uri is not in valid format then fallback will happen for account type from OpenID to DesigoCC even though the user is correctly mapped to OpenID account in the system. Because if the necessary configuration is missing then we cannot prepare LoginUri which is mandatory for the OpenID authentication.

Resource Path	api/users/logininfo/{username}		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	username	string	Username to login with. (Recommendation: Double UriEncode parameter prior to making it to part of URI.) Required
	redirectUri	string	Redirect Uri post login. Optional
Response	HTTP 200 OK HTTP 400 Bad Request HTTP 503 Service Unavailable		
Example1	AccountType of the user is DesigoCC/Windows Request: GET /api/users/test Response: HTTP 200 OK <pre>{ "UserName": "test", "AccountType": 0 }</pre>		
Example2	AccountType of the user is OpenID, but the OpenID configuration is missing in DesigoCC system. Request: GET /api/users/test.test%40test.com?redirectUri=https%3A%2F%2Fdf1i3r5c%3A444%2FFlexApp_5_1%2Floginpage Response: HTTP 200 OK <pre>{ "UserName": "test.test@test.com", "AccountType": 0 }</pre>		
Example3	AccountType of the user is OpenID, OpenID configuration is present in the DesigoCC system. Request: GET /api/users/test.test%40test.com?redirectUri=https%3A%2F%2Fdf1i3r5c%3A444%2FFlexApp_5_1%2Floginpage Response: HTTP 200 OK <pre>{ "UserName": "test.test@test.com", "AccountType": 3, "OpenIdLoginUri": "https://MyAuth0Domain.com/authorize?client_id=R4Q34ZgNZPl11jFslXqNJOFMw3k0kF3V&response_type=code&scope=openid profile email&login_hint= test.test@test.com&state= test.test@test.com _39a36aaf-b07a-4892-aa5f-04b5115ed13b&redirect_uri=https%3a%2f%2fdf1i3r5c%3a444%2fFlexApp_5_1%2floginpage&connection=main-tenant-oidc" }</pre>		

In case of users of type OpenID (AccountType = 3), the client will be performing login with the help of the OpenIdLoginUri received as shown in above response. When this login Uri is visited, the client will be redirected to ID server for authentication. And after the authentication is done, it will be redirected to the login page of the client(redirect Uri sent in above request).

The Logout Uri will be sent to client as a part of Login response. After performing the logout, client will be redirected to the redirect Uri submitted in above request.

7.23.3 Get Public Key

Gets public key in PEM format and a session key.

SessionKey: Session key is uniquely generated key, which is associated with and identifies corresponding public-private key pair.

PublicKey: Public key is RSA asymmetric algorithm key, which to be used for data encryption purposes.

Resource Path	api/users/getpublickey
Category	Internal
HTTP Method	GET
Response	HTTP 200 OK HTTP 400 Bad Request
Examples	Request: GET /api/users/getpublickey Response: {"SessionKey": "2023-02-27 05:10:36Z_3d7e2ff8-f731-47a3-8065-d5909ef4e228", "PublicKey": "-----BEGIN PUBLIC KEY----- \nMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEApz1NEQC3U FI0vZb2JT6f\nSk4cGwlhFnPrKM8x8FSVQRFChdgxRtBKSjphb+G6J +n6FSXoNK7/LbNGPK5dTJC\nQKDmI7H56P60tdWdjatxdAH4Ti2DEin 2fFXE/hVXplFaA+sI33rERILhBGEXPITL\naoGwQ+YciRYUNwkzcPm FIQrOxADXH6xrShusoIh4gFCN17ZBBQexx1vhLoj/zVC\nTpyjRZpm/ wZO2ywp15FeYLCM1ZTq2UiUKdlp4m922Qi2sbFV8Dv+ld/oV2ST2b26 \nHyMwpMhH0f9Bi5RgcjuYCI3LMsw9kkO+Z6sE98qwJOY+mnAo2yN+a OBAKDyo8ztE\nFQIDAQAB\n -----END PUBLIC KEY-----"} HTTP 200 OK

For encryption of data following steps are to be followed:
1.

Import the RSA key with hash as SHA-256.

<https://developer.mozilla.org/en-US/docs/Web/API/SubtleCrypto/importKey>

2. Encrypt the data using the public key imported.

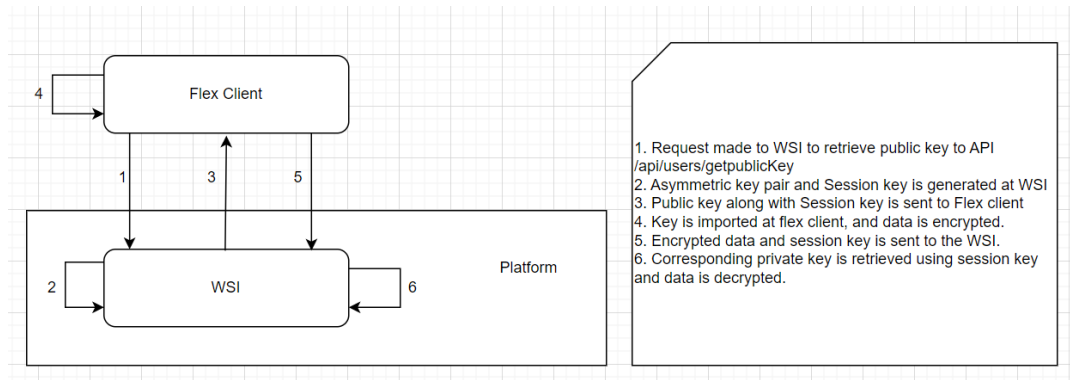
<https://developer.mozilla.org/en-US/docs/Web/API/SubtleCrypto/encrypt>

3. Convert the resulting byte array into string.

4. Encode the resulting string in base64 format.

<https://developer.mozilla.org/en-US/docs/Web/API/btoa>

Steps are depicted in the image below:



Send encrypted string (in base 64 format) and session key to the server for decryption.

Session key and public key gets expired in 5 minutes, so whole operation of encryption and decryption needs to be completed in 5 minutes, if key is expired you need to again make the request to get the public key and perform subsequent operations.

Also, this public key is one time use key, so you can perform encryption decryption using single key only once, if you need to perform these operations again you need to raise request for the new public key.

7.24 Scheduler Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/scheduler	Internal	x	

7.24.1 Gets BACnet schedule object based on the schedule object id

Resource Path	/scheduler/schedule/:objectId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectId	string	Required
Response	HTTP 200 OK Object of WeeklySchedule [->231]		
Examples	Request: GET /api/scheduler/schedule/System1%253AGmsDevice_1_1800_71303169 Response : <pre>{ "DeviceId": "System1:GmsDevice_1_1800_33556232", "Name": "System1:GmsDevice_1_1800_71303169", "Setup": { "DefaultValue": "1.1", "StartDate": { "Date": "0001-01-01T00:00:00Z" }, "EndDate": { "Date": "0001-01-01T00:00:00Z" } } }</pre>		

	<pre> }, "Priority": 16, "DataType": "BACnetDataTypeReal", "ScheduleMin": -3.402823e+38, "ScheduleMax": 3.402823e+38, "TextGroupId": "TxG_EngineeringUnits", "Unit": 64 },], "Outputs": ["Sys- tem1:GmsDevice_1_1800_8388609.Present_Value"], "ScheduleType": "BACnet", "Schedules": [{ "Day": "monday", "Entries": [{ "Time": "07:00:00", "Value": 18.5 }, { "Time": "21:00:00", "Value": 15.5 }] }],], "Exceptions": [{ "Priority": 6, "Mode": 3, "CalendarException": { "CalendarObjectId": "Sys- tem1:GmsDevice_1_1800_25165825", "CalendarEntries": [{ "StartDate": { "Date": "0001-01-01T00:00:00Z", "Days": [6], "Months": [1], "DaysInMonth": [1], "Years": [2005] }, "Mode": 0 }, { "StartDate": { "Date": "0001-01-01T00:00:00Z", "Days": [4], "Months": [6], "DaysInMonth": [27], "Years": [2019] } }] } }] </pre>
--	---

	<pre>] }, "Mode": 0 }], "CalendarName": "" }, "ExceptionTimes": [{ "Time": "00:00:00", "Value": 13.78 }, { "Time": "08:32:32.3200000", "Value": 66.6 }, { "Time": "10:56:56.5600000", "Value": 13.78 }, { "Time": "12:11:11.1100000", "Value": 66.6 }, { "Time": "23:59:59.5900000", "Value": 13.78 }] }] } </pre>
--	---

7.24.2 Save BACnet schedule values

Resource Path	/scheduler/schedule/save		
Category	Public		
HTTP Method	POST		
	Property	Type	Description
Body	schedule	WeeklySchedule [->231]	
Response	HTTP 200 OK true or false as body		
Examples	Request: POST /api/scheduler/schedule/save <pre> { "DeviceId": "System1:GmsDevice_1_1800_33556232", "Name": "System1:GmsDevice_1_1800_71303169", "Setup": { "DefaultValue": "1.1", "StartDate": { "Date": "0001-01-01T00:00:00Z" }, "EndDate": { "Date": "0001-01-01T00:00:00Z" }, "Priority": 16, "DataType": "BACnetDataTypeReal", "ScheduleMin": -3.402823e+38, </pre>		

	<pre> "ScheduleMax": 3.402823e+38, "TextGroupId": "TxG_EngineeringUnits", "Unit": 64 }, "Outputs": ["Sys- tem1:GmsDevice_1_1800_8388609.Present_Value"], "ScheduleType": "BACnet", "Schedules": [{ "Day": "monday", "Entries": [{ "Time": "09:54:00", "Value": 18.5 }, { "Time": "21:00:00", "Value": 15.5 }], "OutputValuesActive": [], "OutputValuesInactive": [] }], "Exceptions": [{ "Priority": 6, "Mode": 3, "CalendarException": { "CalendarObjectId": "Sys- tem1:GmsDevice_1_1800_25165825", "CalendarEntries": [{ "StartDate": { "Date": "0001-01-01T00:00:00Z", "Days": [6], "Months": [1], "DaysInMonth": [1], "Years": [2005] }, "Mode": 0 }, { "StartDate": { "Date": "0001-01-01T00:00:00Z", "Days": [4], "Months": [6], "DaysInMonth": [27], "Years": [2019] } }] } }], </pre>
--	---

	<pre> "Mode": 0 },], "CalendarName": "" }, "ExceptionTimes": [{ "Time": "00:00:00", "Value": 13.78 }, { "Time": "08:32:32.3200000", "Value": 66.6 }, { "Time": "10:56:56.5600000", "Value": 13.78 }, { "Time": "12:11:11.1100000", "Value": 66.6 }, { "Time": "23:59:59.5900000", "Value": 13.78 }] }] } } </pre> Response: 200 OK true
--	---

7.24.3 Save BACnet schedule exceptions

Resource Path	/scheduler/exception/save		
Category	Public		
HTTP Method	POST		
Body	Property	Type	Description
	schedule	WeeklySchedule [->231]	
Response	HTTP 200 OK		
Examples	Request:		

7.24.4 Save BACnet default value for schedule

Resource Path	/scheduler/schedule/saveDefaultValue		
Category	Public		
HTTP Method	GET		
Query	Property	Type	Description
	defaultValue	string	Required
	objectId	string	
Response	HTTP 200 OK		

Examples	Request:
----------	-----------------

7.24.5 Retrieve BACnet calendar list for schedule object id

Resource Path	/scheduler/calendarList/objectId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectId	string	Required
Response	HTTP 200 OK		
Examples	Request:		

7.24.6 Retrieve calendar object based on calendar object Id

Resource Path	/scheduler/calendar/objectId		
Category	Public		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectId	string	Required
Response	HTTP 200 OK Object of Calendar [->230]		
Examples	Request: GET /api/scheduler/calendar/System1%253AGmsDevice_1_1800_25165825 Response: 200 OK { "ScheduleCalendarType": 0, "CalendarEntries": [{ "StartDate": { "Date": "0001-01-01T00:00:00Z", "Months": [1], "DaysInMonth": [1], "Years": [2005] }, "Mode": 0 }], "AppliedSchedules": ["System1:GmsDevice_1_1800_71303169"], "ContainerObjectId": "System1:GmsDevice_1_1800_33556232", "Device": 1800 }		

	}
--	---

7.24.7 Save BACnet calendar

Resource Path	/scheduler/calendar/save		
Category	Public		
HTTP Method	POST		
Body	Property	Type	Description
	calendar	Calendar [->230]	
Response	HTTP 200 OK true or false in body		
Examples	Request: POST /api/scheduler/calendar/save Body: { "ScheduleCalendarType": 0, "CalendarEntries": [{ "Mode": 0, "StartDate": { "Years": [2005], "Months": [1], "DaysInMonth": [1] } }, { "Mode": 0, "StartDate": { "Years": [2019], "Months": [6], "DaysInMonth": [27] } }], "AppliedSchedules": ["System1:GmsDevice_1_1800_71303169"], "ContainerObjectId": "System1:GmsDevice_1_1800_33556232", "Device": 1800, "ObjectId": "System1:GmsDevice_1_1800_25165825" }		

	Response: 200 OK true
--	-----------------------------

7.25 Objects Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/objects	Internal	x	

7.25.1 Get available Objects types

Resource Path	/objects/:designation		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	designation	string	Full view name; Designation id of node in systembrowser tree (Recommendation: Double UrlEncode parameter prior to making it to part of URI.) Required
	includeChildInfo	bool	If true, response will include child object type information Default: false
Response	HTTP 200 OK An array of ChildObjectType objects See [→ 232]		
Examples	Request: api/Objects/System1.ApplicationView%3AApplicationView.Trends.TrendViewDefinitions Response: { "Designation": "System1.ApplicationView:ApplicationView.Trends.TrendViewDefinitions", "Description": "Trends Folder", "ObjectType": "_GmsTrendsFolder", "IsGenericDeleteAllowed": false, "ChildObjects": [{ "IsGenericCreateAllowed": false, "MaxChild": 4294967295, "ExistingChildCount": 0, "Description": "Trend View Definition", "ObjectType": "GMS_TVD", "IsGenericDeleteAllowed": true }, { "IsGenericCreateAllowed": true, "MaxChild": 4294967295, "ExistingChildCount": 0, "Description": "Trends Folder", "ObjectType": "_GmsTrendsFolder", }] }		

	<pre> "IsGenericDeleteAllowed": false }] } </pre>
--	--

7.25.2 Create Objects

Resource Path	/objects		
Category	Internal		
HTTP Method	POST		
Body	Property	Type	Description
	childObject	ObjectRepresentation. See [→ 232]	Child node details which user needs to create.
Response	HTTP 200 OK BrowserNode objects See [→ 232]		
Examples	Request: api/Objects Body Parameter: <pre> { "Designation": "System1.ApplicationView:ApplicationView.Trends.TrendViewDefinitions", "NameChildNode": "TestFolderTVD1", "ObjectModelName": "_GmsTrendsFolder", "Descriptor": { "CommonText": "TestFolderTVDNode", "MultiLangText": ["TestFolderTVDNodeLang1", "TestFolderTVDNodeLang1"] } } </pre> Response: <pre> { "HasChild": false, "SystemId": 1, "ViewId": 10, "ViewType": 1, "Name": "TestFolderTVD1", "Descriptor": "TestFolderTVDNode", "Designation": "System1.ApplicationView:ApplicationView.Trends.TrendViewDefinitions.TestFolderTVD1", "ObjectId": "System1:TestFolderTVD1", "Attributes": { "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus", "ObjectId": "System1:TestFolderTVD1", "DisciplineDescriptor": "Management System", "DisciplineId": 0, "SubDisciplineDescriptor": "Unassigned", "SubDisciplineId": 0, "TypeDescriptor": "View Element", "TypeId": 8000, "SubTypeDescriptor": "User Defined Folder", "SubTypeId": 8016, "ManagedType": 59, "ManagedTypeName": "TrendsFolder", "ObjectModelName": "_GmsTrendsFolder", "ValidationRules": { </pre>		

	<pre>"CommentRule": "Optional", "ReAuthentication": "NoNeed", "Configuration": 0, "IsFourEyesEnabled": false, "_links": [] } }, "Location": "System1.Application View:Applications.Trends.Trend View Definitions.string", "_links": [{ "Rel": "properties", "Href": "api/properties/System1%253ATestFolderTVD1?requestType=0", "IsTemplated": false }] }</pre>
--	--

7.25.3 Delete Objects

Resource Path	/objects/:designation		
Category	Internal		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	designation	string	
Response			
Examples	Request: DELETE api/Objects/System1.ApplicationView%3AApplicationVi ew.Trends.TrendViewDefinitions.TestFolderTVD1 Response: Returns status code 200		

7.25.4 Get Information text

Resource Path	/objects/servicetext/:objectId		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	objectId	string	Object id Recommendation: Double UrlEncode parameter prior to making it to part of URI
Response	HTTP 200 OK Array of ObjectInfoRepresentation[→233]		
Examples	Request: GET api/Objects/servicetext/System1:ManagementView_Mana gementSystem_Servers_Server Response: Returns status code 200 [{ "ObjectId": "System1:ManagementView_ManagementSystem_Servers_Server",		

	<pre> "ServiceText": { "InformationText": "", "Memo": "" }, { "Error": { "Id": 2402709, "Error": "obj_invalid_object", "Details": "Failed to resolve passed object id to a valid object" }, "ObjectId": "InvalidObjectId" }] </pre>
--	--

7.25.5 Put Memo text

Resource Path	/objects/servicetext/:objectId/memo		
Category	Internal		
HTTP Method	PUT		
Parameters	Property	Type	Description
	objectId	string	Object id Recommendation: Double UrlEncode parameter prior to making it to part of URI
Body	memodetail	ServiceTextMemoRepresentation [→233]	Memo details
Response	HTTP 200 OK Array of ObjectInfoRepresentation[→233]		
Examples	Request: PUT api/Objects/servicetext/System1:ManagementView_ManagementSystem_Servers_Server/memo BODY <pre> { "Memo": "Sample Memo" } </pre> Response: Returns status code 200		

7.26 Related Items Service

The purpose of the related items API is to access items that are associated to another object. For example, an object in a graphic is said to be `related to` that graphic. In most situations the relationship between that object and the graphic is discovered by requesting the graphic – but it is often useful to know where that object is used (such as a graphic or re report) or if there is something external (such as a file or a website) that has a connection to the object.

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/relateditems	Public		

7.26.1 Read Related Items

Return related items for a list of objects. Please keep in mind that:

- The system information for each object is optional and if not provided, the WSI will assume that the object is on the "local" system. If the object does not exist in that system the list of related items for that object will be empty.
- The list of objects must all be from the same system. If they are not, the entire call will fail with an HTTP error code return of 400. **(Distributed System Not Supported will throw error response "Object ids are from different systems")**
- A related item is just another object in the system and can appear in multiple locations within a hierarchy or within multiple hierarchies. Each related item will contain a BrowserNode for each location, where the BrowserNode has the object and view information.
- A small feature implemented to address duplicate Report creation in flex Response Data of objectId filtered based on sourceType Enum Value.

Unknown = 0,
System = 1,
User = 2,
Dynamic = 3

Scenarios

-> If source type not selected it will display all data.

-> If source type selected and has data it is displaying data of particular source-Type only.

-> If source type selected and don't have data it shows related item list empty.

Resource Path	api/relateditems
Category	Internal
HTTP Method	POST
Parameters	None
Body	Array of Object Ids
Response	HTTP 400 Bad Request HTTP 200 OK The top level object contains an array of results, one element for each incoming object id. Each element holds the object id, an error code, and an array of RelatedItem objects for that object id. For the full description of a RelatedItem go here .
Example	Request: POST /api/relateditems Body: <pre>["SystemName:Object1", "SystemName:Object2"]</pre> Response: HTTP 200 OK <pre>{ RelatedResults: [{ ObjectId: "SystemName:Object1", ErrorCode: 0, RelatedItems: [<related item 1 for Object1>] }] }</pre>

	<pre> "ItemDescriptor": "", "Reference": "", "Mode": 1, "SourceType": 1, "Nodes": [{ <BrowserNode: see [→ 55]> }, { <BrowserNode: see [→ 55]> }] }, <related item 2 for Object1> { "ItemDescriptor": "", "Reference": "", "Mode": 2, "SourceType": 1, "Nodes": [{ <BrowserNode: see [→ 55]> }] }, <... more related items for Object1>] }, { ObjectId: "SystemName:Object2", ErrorCode: 0, RelatedItems: [<related item 1 for Object2> { "ItemDescriptor": "", "Reference": "", "Mode": 1, "SourceType": 1, "Nodes": [{ <BrowserNode: see [→ 55]> }, { <BrowserNode: see [→ 55]> }] }, <related item 2 for Object2> { "ItemDescriptor": "", "Reference": "", "Mode": 2, "SourceType": 1, "Nodes": [{ <BrowserNode: see [→ 55]> }] }, <... more related items for Object2>] }] } </pre>
--	--

7.27 Access rights Service

Application rights assigned to user group can be retrieved by this service.

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/accessrights	Internal	x	x

7.27.1 Retrieve access rights

Resource Path	/accessrights
Category	Internal
HTTP Method	GET
Response	HTTP 200 OK Array of AccessRightResponseRepresentation object [→Error! Bookmark not defined.]
Examples	Request: GET /api/accessrights Response: Returns status code 200 with applications and their respective access. <pre> { "ApplicationRights": [{ "Name": "BACnet Configuration", "Id": 6, "Operations": [{ "Name": "Show", "Id": 192 }, { "Name": "Configure", "Id": 194 }] }] } </pre>

7.27.2 Retrieve user roles

Resource Path	/accessrights/roles
Category	Internal
HTTP Method	GET
Response	HTTP 200 OK Array of UserRoleRepresentation object
Examples	Request: GET /api/accessrights/roles Response: Returns status code 200 with applications and their respective access. <pre> { "RolesMapping": [{ "RoleId": 101, "RoleState": false, "RoleName": "DefaultAdmins", "RoleDescription": "Default Administrator Group" }, { "RoleId": 100, </pre>


```
        "RoleState": false,  
        "RoleName": "DefaultAdmins1",  
        "RoleDescription": "Default Administrator Group"  
    }  
]  
}
```

7.27.3 Push Notification

Subscription for any changes in application rights configuration for logged in user will be notified.

Type	Name (case sensitive)		
Hub	norisHub		
Function	notifyAccessRights (data)		
	Property	Type	Description
	data	AccessRightsRepresentation	See AccessRightsRepresentation [→ 236]

7.27.3.1 Create channelize Access rights subscription

Subscription response will be pushed through notifySubscriptionStatus function available at client side. Application rights changes will be notified via notifyAccessRights.

Resource Path	/api/sr/accessrightssubscriptions/channelize/:requestId /:connectionId?includedetails=:includedetails		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub connection Required
	includedetails	bool	Default set to false, This flag will tell server to send all application rights as soon as something changes on server.Setting this flag to false, will result in empty object in push notification stating something is changed. Optional [Setting it to true could be resource heavy if changes are done too frequently]
Response	HTTP 200 OK		
Examples	Request: POST /api/sr/accessrightssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b?includedetails=false notifySubscriptionStatus: { "ErrorCode":0, 		

	<pre>"RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor": "notifyAccessRights" }</pre>
--	--

7.27.3.2 Create User Roles Subscription

Resource Path	/api/sr/accessrightssubscriptions/channelize/:requestid/:connectionId?includedetails=includedetails/userroles		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub connection Required
	includedetails	bool	Default set to false, Setting this flag to false, will result in empty object in push notification stating something is changed. [Currently setting it to true does not change the response. This might be implemented in the future.]
Response	HTTP 200 OK		
Examples	Request: POST /api/sr/accessrightssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b?includedetails=false/userroles notifySubscriptionStatus: <pre>{ "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"userRoles" }</pre>		

7.27.3.3 Delete Subscription for a Access Rights

Resource Path	/api/sr/ accessrightssubscriptions /{connectionId}		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	connectionId	String	Identifier of subscription Required
Response	Object of UnsubscriptionStatus See Link [→237]		
Examples	Request: DELETE api/sr/ accessrightssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1 Response: HTTP/1.1 200 OK		

7.27.3.4 Delete Subscription for User Roles

Resource Path	/api/sr/ accessrightssubscriptions /{connectionId}/userroles		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	connectionId	String	Identifier of subscription Required
Response	HTTP 200 OK		
Examples	Request: DELETE api/sr/ accessrightssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1/userroles Response: HTTP 200 OK		

7.28 Operating Procedures Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/operatingprocedures	Internal	x	x

7.28.1 Retrieve Operating procedure

Resource Path	/operatingprocedures
Category	Public
HTTP Method	GET
Response	HTTP 200 OK Object of Operating procedure [→235]
Examples	Request: GET /api/operatingprocedures? id=1.832765.2.23.2.131675.637082217374440000.0 Response: Returns status code 200. <pre>{ "AlertCount": 1, "AlertSource": "System1", "AlertTime": "2019-11-07T11:03:04.325Z", "Id": "1.832765.2.23.2.131675.637082217374440000.0", "InTreatmentBy": "#DefaultAdmin", "IsClosed": true, "ResetSteps": 0, "Sequential": true, "Steps": [{ "HasConfirmedExecution": true, "IsCompleted": true, "Operator": "Admin", "RuntimeStatus": "Runtime text", "Status": "Failed", "StepId": "System1:GraphicsStep", "StepName": "GraphicsStep" }], "TemplateName": "GraphicsOperation" }</pre>

7.28.2 Update Operating procedure step

Resource Path	/operatingprocedures/{id}/step		
Category	Internal		
HTTP Method	PUT		
Parameters	Property	Type	Description
	id	string	Required Operating Procedure id
Body	operatingStep	OperatingProcedureStep	Required Object of Operating Procedure Step All properties are not updatable, please refer Operating procedure step [→236]

Response	HTTP 200 OK Object of Operating procedure step [→236]
Examples	Request: PUT api/operatingprocedures/1.832765.2.23.2.131675.6370 82217374440000.0/step BODY: <pre>{ "HasConfirmedExecution": true, "IsCompleted": true, "Operator": "Admin", "RuntimeStatus": "Runtime text", "Status": "Failed", "StepId": "System1:GraphicsStep" }</pre> Response: Returns status code 200 <pre>{ "HasConfirmedExecution": true, "IsCompleted": true, "Operator": "Admin", "RuntimeStatus": "Runtime text", "Status": "Failed", "StepId": "System1:GraphicsStep", "StepName": "GraphicsStep" }</pre>

7.28.3 Push Notification

After subscribing for Operating Procedures initial notification will be sent.
Afterwards any change will be notified.

Type	Name (case sensitive)		
Hub	norisHub		
Function	notifyOperatingProcedures (data)		
	Property	Type	Description
	data	OperatingProcedureSubscriptionRepresentation	See OperatingProcedureSubscriptionRepresentation [→ 236]

7.28.3.1 Create channelize Operating procedures subscription

Subscription response will be pushed through notifySubscriptionStatus function available at client side. Operating procedure changes will be notified via notifyOperatingProcedures.

Resource Path	/api/sr/operatingproceduressubscriptions/channelize/:requestId/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required

	connectionId	String	Identifier of subscription hub connection Required
	id	string	Operating procedure id Optional
Response	HTTP 200 OK		
Examples	Request: POST api/sr/operatingproceduresubscriptions /channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b notifySubscriptionStatus: <pre>{ "Id": "c6c95ece-133b-480b-af1f-067a173e0f5b", "Key": "1a", "ErrorCode": 0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor": "notifyOperatingProcedures" }</pre>		

7.28.3.2 Subscription Status Notification

After subscribing for OperatingProcedure, for each key subscription status will be sent through notifySubscriptionStatus.

Type	Name (case sensitive)		
Hub	norisHub		
Function	notifySubscriptionStatus(status)		
	Property	Type	Description
	Status	OperatingProcedureSubscriptionKey Response	Subscription objects See Link [→237]
Example	<pre>{ "Id": "c6c95ece-133b-480b-af1f-067a173e0f5b", "Key": "1a", "ErrorCode": 0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor": "notifyOperatingProcedures", "_links": [{ "Rel": "unsubscribe", "Href": "api/sr/operatingproceduresubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=1a", "IsTemplated": false }] }</pre>		

7.28.3.3 Delete Subscription for a Operating procedure

Resource Path	/api/sr/operatingproceduresubscriptions/{connectionId}		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description

	connectionId	String	Identifier of subscription Required
	subscriptionKey	String	Query Parameter with subscription key Required
Response	Object of UnsubscriptionStatus See Link [→237]		
Examples	Request: DELETE api/sr/operatingproceduresubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=1a Response: HTTP/1.1 200 OK <pre>{ "Key": "1a", "ErrorCode": 0 }</pre>		

7.29 Licenses Service

Settings are stored on a “per user” basis.

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/licenses	Public	x	

7.29.1 Read License Mode

Read current Licences Mode; the License mode information is returned in the body of the response.

Resource Path	/licenses
Category	Public
HTTP Method	GET
Response	Object of License Mode See Link [→237]
Examples	Request: GET /api/licenses Response: Returns status code 200. <pre>{ "LicenseModeName": "Siemens Developer License", "LicenseModeValue": 4, "ExpirationTime": 0 }</pre>

7.29.2 Read License Options

Read current Licences Mode; the License mode information is returned in the body of the response.

Resource Path	/licenses/ <u>licenseoptions</u>
Category	Public
HTTP Method	GET
Response	Object of License Option See Link [→8.24]

Examples	Request: GET /api/licenses/licenseoptions Response: Returns status code 200. <pre>[{ "Id": "sbt_gms_intOnly_allfeature", "Required": 0, "Remaining": 4, "Available": 1, "Description": "Unlimited use for Development Pur- poses" }, { "Id": "sbt_gms_ext_sm", "Required": 0, "Remaining": -1, "Available": -1, "Description": "System Manager Application" }]</pre>
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7.30 Multimonitor Service

Multimonitor Configuration assigned to user group can be retrieved by this service.

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/multimonitors	Internal	x	

7.30.1 Retrieve Multi Monitor Configuration

Description	- Returns the default multi-monitor configuration for the specified client. - The client is identified by client certificate or optionally by the StationIdentifier parameter		
Resource Path	/multimonitors		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	stationIdentifier	String	- Identifier for the particular station preferably hostname of any particular machine - or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier. Optional
Response	Object of Multimonitor Service		

Examples	Request Url
	/api/multimonitors/user?stationIdentifier=stat1&api_key=defaultadmin%3AGaurav%4026
	Response Body
	<pre>{ "Configuration": "conf1", "OverrideSettings": true, "HostName": "host19" }</pre>

7.30.2 Retrieve MultiMonitor Configuration For User

Description	• Returns the multi-monitor configuration for the specified client and logged in user. • The client is identified by client certificate or optionally by the clientid parameter		
Resource Path	/multimonitors		
Category	Internal		
HTTP Method	GET		
Parameters	Property	Type	Description
	stationIdentifier	String	• Identifier for the particular station preferably hostname of any particular machine • or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier. • Optional
Response	Object of Multimonitor Service		
Examples	Request Url		
	/api/multimonitors/user?stationIdentifier=z&api_key=defaultadmin%3A9		
	Response Body <pre>{ "Configuration": "y", "OverrideSettings": false, "HostName": "x"} </pre>		

7.30.3 Add/Update Multi Monitor Configuration

Description	- Stores the default multi-monitor configuration for the specified client. - The client is identified by client certificate or optionally by the clientid parameter - Saving checks, if the application right 'Multi-Monitor – Configure' is set for the logged in user.		
Category	Internal		
HTTP Method	POST		
Parameters	Property	Type	Description
	Configuration	string	Json object for configuration of MultiMonitor Settings
	OverrideSettings	bool	Any user can overrule the default multi monitor configuration with its own setting. This setting is used by the client to enable/disable the respective UI controls which support the multi monitor configuration. I.e. if it is set to false, the user (except the user which has the application right 'Configure multi monitor') cannot: - Open an additional system manager. - Undock a dedicated event manager. - Close an additional system or event manger. Move windows.
	HostName	string	HostName of the client where multimonitor configuration being used.
	stationIdentifier	String	- Identifier for the particular station preferably hostname of any particular machine - or it can unique id. - If valid Client Certificate is present then it is given higher precedence than stationIdentifier. Optional
Response	Object of Multimonitor Service		
Examples	Request Url /api/multimonitors/user?stationIdentifier=stat1&api_key=defaultadmin%3AGurav%4026 Response Body <pre>{ "StationIdentifier": "stat1", "ErrorCode": 0 }</pre>		

7.30.4 Add/Update MultiMonitor Configuration For User

Description	- Stores the multi-monitor configuration for the specified client and logged in user. - The client is identified by client certificate or optionally by the clientid parameter - Saving checks if either one of the following options is met - No default multi-monitor configuration is available for the specified client. - A default multi-monitor configuration is available for the specified client and the flag 'OverrideSettings' is set. - Note, that the 'Multi-Monitor – Configure' application right is not applied/needed!		
Category	Internal		
HTTP Method	POST		
Parameters	Property	Type	Description
	Configuration	string	Json object for configuration of MultiMonitor Settings
	OverrideSettings	bool	Any user can overrule the default multi monitor configuration with its own setting. This setting is used by the client to enable/disable the respective UI controls which support the multi monitor configuration. I.e. if it is set to false, the user (except the user which has the application right 'Configure multi monitor') cannot : - Open an additional system manager. - Undock a dedicated event manager. - Close an additional system or event manger. Move windows.
	HostName	string	HostName of the client where multimonitor configuration being used.
stationIdentifier		string	- Identifier for the particular station preferably hostname of any particular machine - or it can unique id. - If valid Client Certificate is present then it is given higher precedence than stationIdentifier. Optional
Response	Object of Multimonitor Service		
Examples	Request Url /api/multimonitors/user?stationIdentifier=z&api_key=defaultadmin%3A9 Response Body <pre>{ "UserName": "DefaultAdmin", "StationIdentifier": "z", "ErrorCode": 0 }</pre>		

7.30.5 Delete MultiMonitor Configuration

Description	Delete the default multi-monitor configuration for the specified client.		
Resource Path	/multimonitors		
Category	Internal		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	stationIdentifier	String	- Identifier for the particular station preferably hostname of any particular machine - or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier. Optional
Response	Object of Multimonitor Service		
Examples	Request Url /api/multimonitors/user?stationIdentifier=stat1&api_key=defaultadmin%3AGaurav%4026 Response Body <pre>{ "StationIdentifier": "stat1", "ErrorCode": 0 }</pre>		

7.30.6 Delete MultiMonitorConfiguration For User

Description	Deletes the multi-monitor configuration for the specified client and logged in user.		
Resource Path	/api/sr/ accessrightssubscriptions /{connectionId}/userroles		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	stationIdentifier	String	- Identifier for the particular station preferably hostname of any particular machine - or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier. Optional
Response	Object of Multimonitor Service		
Examples	Request Url /api/multimonitors/user?stationIdentifier=a&api_key=defaultadmin%3A9 Response Body <pre>{ "UserName": "DefaultAdmin", "StationIdentifier": "a", "ErrorCode": 0 }</pre>		

7.31 Validation Service

Resource	Service Category	Requires Authentication	Requires SignalR (Push Notifications)
/validation	Public	x	

7.31.1 Command Validation Operation

This method checks the incoming properties (and the command group) for the most restrictive validation requirements.

The return will also include the contents of a text group if all objects have been configured for "pre-defined" comments

if not all are configured that way, then the text group is not provided.

Resource Path	/api/validation/getCommandValidationOperation		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	PropertyIds	String[]	List of Property Ids to validate for commanding. Required
	Group Number	Int	Group Number of Command to execute. Required
Response	Object of ValidateOpResponse See Link 8.1.4		
Examples	Request: POST /api/validation/getCommandValidationOperation Response: Returns status code 200 <pre>{ "IncrVersion": true, "Log": true, "PredefinedComments": [{ "Value": 0, "Text": "string", "Color": 0, "SubText": [{}] }], "CommentRule": "string", "ReAuthentication": "string", "IsFourEyesEnabled": true, "_links": [{ "Rel": "string", "Href": "string", "IsTemplated": true }] }</pre>		

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7.31.2 Edit Validation Operation

This method checks the incoming collection of objects (DplIdentifiers) for the most restrictive validation requirements.

The return will also include the contents of a text group if all objects have been configured for "pre-defined" comments

if not all are configured that way, then the text group is not provided.

Resource Path	/api/validation/getEditValidationOperation		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	objectIds	String[]	List of object Ids to validate to edit object. Required
Response	Object of ValidateOpResponse See link 8.1.4		
Examples	Request: POST /api/validation/getEditValidationOperation Response: Returns status code 200 <pre>{ "IncrVersion": true, "Log": true, "PredefinedComments": [{ "Value": 0, "Text": "string", "Color": 0, "SubText": [{}] }], "CommentRule": "string", "ReAuthentication": "string", "IsFourEyesEnabled": true, "_links": [{ "Rel": "string", "Href": "string", "IsTemplated": true }] }</pre>		

7.31.3 Validate Credential

Validates the User and Supervisor Credentials.

Resource Path	/api/validation/validateCredential
Category	Public
HTTP Method	POST

Parameters	Property	Type	Description
	objectIds	String[]	List of Object Ids to validate for provided credentials. Required
	CheckCredentials	int	0 validate both user and supervisor 1 validate only user 2 validate supervisor only. Required
	UserPassword	string	Password of logged in user.
	SupervisorName	string	Name of Supervisor.
	SupervisorPassword	string	Password of Supervisor
Response	HTTP 200		
Examples	Request: POST /api/validation/validateCredential Response: Returns status code 200		

7.32 Tags Service

7.32.1 Retrive all available tags for objects

Description	• Returns all available tags for objects. •		
Resource Path	/tags		
Category	Internal		
HTTP Method	POST		
Parameters	Property	Type	Description
	ObjectIds	String[]	List of ObjectIds
	Key	String[]	List of Keys
Response	Object of TagsResponseRepresentation		
Examples	Request URL		
	<code>/api/tags?api_key=defaultadmin%3Aa</code>		
	Request Body <pre>{ "ObjectIds": ["GmsDevice_1_8817_4194305"], "Keys": ["key01"] }</pre> Response Body <pre>{ "ObjectTagInfos": [{ "ErrorCode": 0, "ObjectId": "System1:GmsDevice_1_8817_4194305", "OriginalObjectId": "GmsDevice_1_8817_4194305", "KeyValueInfo": { "key01": "Team1" } }] }</pre>		

7.32.2 Write tags for an object

Description	• This API is useful for updating tags for an object • In the "tagsRequestRepresentation" parameter enter single key value for an object for which tags are to be added
Resource Path	/tags
Category	Internal

HTTP Method	PUT		
Parameters	Property ()	Type	Description
	tagsRequestRepresentation	TagsWriteRequestRepresentation	In the "tagsRequestRepresentation" parameter enter single key value for an object for which tags are to be added
Response	Object of TagsResponseRepresentation		
Examples	Request URL		
	<code>/api/tags?api_key=defaultadmin%3Aa</code>		
	Request Body		
	<pre>{ "WriteInfos": [{ "ObjectId": "System1:ApplicationView_Documents", "KeyValueInfo": {"key03": "desigocc"} }] }</pre>		
	Response Body		
	<pre>{ "ObjectTagInfos": [{ "ErrorCode": 0, "ObjectId": " System1:ApplicationView_Documents ", "OriginalObjectId": " System1:ApplicationView_Documents " }] }</pre>		

7.32.3 Traversal in System browser shall be in upward direction

Description	• Traversal in Systembrowser in upward direction to retrieve all tags with respect to provided information. • In the "traversalRequest" parameter enter the object id whose parent nodes have tags assigned to them, view id.	
Resource Path	/api/tags/traversal	
Category	Internal	
HTTP Method	POST	
Parameters	Property	Type
	traversalRequest	TagsTraversalRequestRepresentation
Response	Object of TagsTraversalResponseRepresentation	

Examples	Request URL
	/api/tags?api_key=defaultadmin%3Aa Request Body <pre>{ "TraversallInfos": [{ "ObjectId": "GmsDevice_1_8817_83886081", "ViewId": 11, "ObjectModelName": "GMS_Aggregator" }], "Keys": [] }</pre> Response Body <pre>{ "TagsTraversallInfos": [[{ "Node": { "SystemId": 1, "Name": "Building", "Descriptor": "Building", "Designation": "System1.UserView:UserView.Campus.Building", "ObjectId": "System1:GMS_Aggregator_UserView_UserView_Campus_Building", "_links": [] }, "ErrorCode": 0, "ObjectId": "System1:GMS_Aggregator_UserView_UserView_Campus_Building", "KeyValueInfo": { "keyinfoRootNode": "Val_RootNode", "keyinfoRootNode": "RootNodeValue" } }, { "Node": { "SystemId": 1, "Name": "UserView", "Descriptor": "UserView", "Designation": "System1.UserView:UserView", "ObjectId": "System1:GMS_Aggregator_UserView_UserView", "_links": [] }, "ErrorCode": 0, "ObjectId": "System1:GMS_Aggregator_UserView_UserView", "KeyValueInfo": { "key1231": "Pun1", "key1232": "Pun2", "key1233": "Pun3", "key1234": "Pun4", "key1235": "Pun5" } }]] }</pre>

7.32.4 Tags Query

Description	In the "tagQuery" parameter enter query for which all the information based on the entered query is to be retrieved		
Resource Path	/api/tags/query		
Category	Internal		
HTTP Method	POST		
Parameters	Property	Type	Description
	tagQuery	String	Query tag for which user wated to retrive all available tags
Response	Object of TagsResponseRepresentation		
Examples	Request URL		
	/api/tags/query?api_key=defaultadmin%3Aa		
	"key1" "key4 AND key5" >>>>> Return all DPs containing both the tags "key1 AND NOT key5" >>>>> Return all DPs containing the first tag but not the second tag "key1 OR key5" >>>>> Return all DPs containing either of the tag (or both)		
	Response Body All available tags should be retrieved for the object based on the query provided.		

7.33 User Sessions Service

7.33.1 Terminate Session

Description	To terminate session based on sessionId		
Resource Path	/api/sessions		
Category	Internal		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	systemId	integer	SystemId of system
	sessionId	string	sessionId of particular session which you want to terminate
Response	Boolean		
Examples	Request Url		
	/api/sessions/1/7ba1ccf37db248e78ddfcc1b32369b42?api_key=defaultadmin%3Aa		
	Response Body : True		
	Response code : 200		

7.34 User Sessions Subscriptions Service

7.34.1 Subscribe for User Sessions

Description	To get an updated list of active user sessions		
Resource Path	/api/sr/SessionsSubscriptions/channelize/{systemId}/{connectionId}/{requestId}		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	systemId	Integer	System Identifier
	connectionId	string	Identifier of subscription Required
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required
Response	HTTP 200 OK		
Examples	Request: POST api/sr/SessionsSubscriptions/channelize/1/78fe62d5-c523-4fff-a548-bebda08eaa87/a10470b4-8e07-9cec-89b0-4c70e282ad7a Response: HTTP/1.1 200 OK		

7.34.2 Subscription Status Notification

After subscribing for user sessions, the subscription status will be sent only once.

Type	Name (case sensitive)		
Hub	1. norisHub		
Function	notifySubscriptionStatus(status)		
	Property	Type	Description
	Status	Subscription Response	See Subscription Status SubscriptionStatusWithKey
Example	<pre>{ "ErrorCode":0, "Key": 6 "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifySessionChanges" }</pre>		

7.34.3 Session Change Notification

Type	Name (case sensitive)		
Hub	1. norisHub		
Function	notifySessionChanges (data)		
	Property	Type	Description
	SessionRepresentations	List<SessionRepresentation>	List of active sessions detail see definition
	SubscriptionKey	Integer	Subscription Key
Example	<pre>{ "ErrorCode":0, "Key": 6 "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifySessionChanges" }</pre>		

7.34.4 Delete Subscription for a User Sessions

Resource Path	/api/sr/SessionsSubscriptions/{systemId}/{connectionId}		
Category	Public		
HTTP Method	DELETE		
Parameters	Property	Type	Description
	systemId	integer	System Identifier
	connectionId	String	Identifier of subscription Required
	subscriptionKey	String	Query Parameter with subscription key Required
Response	HTTP 200 OK.		
Examples	Request: DELETE api/sr/SessionsSubscriptions/1/5fff25b8-36b0-4e30-893c-4e75c2cc8ae4?subscriptionKey=4&api_key=defaultadmin%3Aa Response: HTTP/1.1 200 OK		

8 Objects and Data Types



NOTE:

In case a property has an empty or null value it may be omitted.

8.1 Common

8.1.1 Link

Models a hyperlink to another resource.

Property	Type	Description
Href	string	Hyperlink to a resource (URI [RFC3986] or URI template [RFC6570])
IsTemplated	bool	If true, Href contains a placeholder in its link.
Rel	string	Link relation type (RFC5988)

8.1.2 Page<Name, Type>

Property	Type	Description
_links	Array of Link	See Link [→ 50]
Events	Array of <Type>	List of events See Link [→204]
Page	int	Number of current page (first page = 1)
Size	int	Number of items per page
Total	Int	Total number of items
CreationTimeFrom	DateTime	
CreationTimeTo	DateTime	

8.1.3 KeyValue<Name, Type>

Property	Type	Description
<Name>	<Type>	Name and type depend on current instance

8.1.4 NameValue

Property	Type	Description
DataType	ApiDataType	Data type of the property
Name	string	Name of the property
Value	string	Value of the property

8.1.5 Subscription

Property	Type	Description
_links	Array of Link	See Link [→ 199]
Key	int	Subscription key is specific for the requested ID
ErrorCode	int	If value is >0, an error occurred.
PropertyId	string	Address of a property

8.1.6 UnSubscription

Property	Type	Description
Key	int	Subscription key is specific for the requested ID
ErrorCode	int	If value is >0, an error occurred.

8.1.7 Value

Property	Type	Description
Value	string	Raw value
DisplayValue	string	Value for display purpose (textual representation of the raw value)
Timestamp	DateTime	Timestamp of this value
QualityGood	bool	
Quality	string	See Quality Flags [→ 200]
IsPropertyAbsent	bool	Driver reported that property is not present Note: Only set if true and not available for trends

8.1.7.1 Quality Flags

Quality is a string representation of a 64-bit bit-set. The string contains a ulong numeric value. Bits not listed are not in use.

Bit	Description	QualityGood	Effect on Value
0	variable active/inactive	TRUE	value ok
1	default value set explicitly	TRUE	value ok
2	default value set automatically	TRUE	value ok
3	WinCC OA value range violated	FALSE	value not ok
4	value of the variable "out of range"	FALSE	value not ok
5	explicit invalid	FALSE	value not ok
6	invalid set by an interface driver	FALSE	value not ok
8	default value invalid	TRUE	value ok
9	set during general query	TRUE	value ok
10	set during an individual query	TRUE	value ok
11	interface driver active	TRUE	value ok
12	value corrected	TRUE	value ok
13	value condensed or compressed, used (HDB & RAIMA) with archiving/compression.	TRUE	value ok
14	corrected value condensed	TRUE	value ok
15	additional correction value	TRUE	value ok
16	compressed value invalid	TRUE	value ok
17	source time invalid (corrected by the Event Manager)	TRUE	value ok
19	disable last value storage	TRUE	value ok
20	value changed	TRUE	value ok
21	value up (or set to same value)	TRUE	value ok
24	for internal use only	TRUE	value ok
32	driver out of service	TRUE	value ok
33	driver alarm	TRUE	value ok
34	driver fault	TRUE	unreliable
35	driver overridden	TRUE	value ok
36	driver subscribed	TRUE	value ok
37	driver property absent	FALSE	value not ok
40	trend: time shift	FALSE	no value, is event

41	trend: log enabled	FALSE	no value, is event
42	trend: error	FALSE	no value, is event
43	trend: purge	FALSE	no value, is event
44	trend: rollover	FALSE	no value, is event
45	trend: value is status	FALSE	value is status/error
46	trend: log interrupted	FALSE	no value, is event
48	trend: start logging	TRUE	value ok
49	trend: value reduced	TRUE	value ok
50 ¹	priority 1 (0 bit; 2 ⁰)	TRUE	value ok
51 ¹	priority 2 (1 bit; 2 ¹)	TRUE	value ok
52 ¹	priority 4 (2 bit; 2 ²)	TRUE	value ok

¹ Bit 50 / Bit 51 / Bit 52 are to be combined into a single value (priority 0..7)

8.1.8 ColorRepresentation

Property	Type	Description
A	byte	Color code
R	byte	Red color code
G	byte	Green color code
B	byte	Blue color code

8.1.1 CommandsSubscription

Property	Type	Description
_links	Array of Link	See Link [→ 199]
Key	int	Subscription key is specific for the requested ID
ErrorCode	int	If value is >0, an error occurred.
PropertyId	string	Address of a property
RequestId	string	Channelized subscription request identifier Note: Only available for channelize requests
RequestFor	string	Notification function name where client will receive further updates Note: Only available for channelize requests
SubscriptionId	string	Original requested string for subscription

8.1.2 ValuesSubscription

Property	Type	Description
_links	Array of Link	See Link [→ 199]
Key	int	Subscription key is specific for the requested ID

ErrorCode	int	If value is >0, an error occurred.
PropertyId	string	Address of a property
RequestId	string	Channelized subscription request identifier Note: Only available for channelize requests
RequestFor	string	Notification function name where client will receive further updates Note: Only available for channelize requests
OriginalObjectOrPropertyId	string	Original requested string for subscription
ObjectOrPropertyId	string	Original requested string for subscription
AttributeId	string	Object Name
ObjectId	string	Object Id only of requested ObjectOrPropertyId
PropertyName	string	PropertyName of requested ObjectOrPropertyId

8.1.3 Attributes

Property	Type	Description
DefaultProperty	string	Default property of object-model
DisciplineDescriptor	string	Localized name of discipline
DisciplineId	int	ID of discipline
FunctionDefaultProperty	string	Default property of assigned function
FunctionName	string	
ManagedType	uint	
ManagedTypeName	string	
ObjectModelName	string	
ObjectId	string	
SubDisciplineDescriptor	string	Localized name of subdiscipline
SubDisciplineId	int	ID of subdiscipline
SubTypeDescriptor	string	Localized name of subtype
SubTypeId	int	ID of subtype
TypeDescriptor	string	Localized name of type
TypeId	int	ID of type

8.1.4 ApiDataType

Name	Description	BaseType
None		
BasicChar		char
BasicUInt		uint
BasicInt		int
BasicFloat		double
BasicBool		boolean
BasicBit32		uint
BasicString		string
BasicTime		DateTime
BasicObjectOrPropertyId		string
BasicLangText		string
BasicBlob		string
ExtendedBool	Extends BasicBool with a language text for true/false.	
ExtendedInt	Extends BasicInt with Min, Max, Default, Unit.	

ExtendedUInt	Extends BasicUInt with Min, Max, Default, Unit.	
ExtendedReal	Extends BasicFloat with Min, Max, Default, Unit.	
ExtendedEnum	Extends BasicUInt with enumeration texts.	
ExtendedBitString	Extends BasicUInt with bit set texts.	
ExtendedDateTime	Extends BasicString with semantics for a DateTime string format.	
ExtendedApplSpecific	Extends BasicString with the knowledge that the string has a dedicated meaning for some application. Typically, the string will be XML-encoded.	
ExtendedAny	Extends BasicFloat with the permission to switch its type.	
ExtendedComplex	Extends BasicBlob.	
ExtendedDuration	Extends BasicUInt with Min, Max, Default and time duration specific configurations.	

8.1.5 Subgroups

Property	Type	Description
Id	UInt	Id of the SubGroup
Descriptor	string	Descriptor of the SubGroup

8.1.6 StatusNotification

Property	Type	Description
ErrorCode	int	Error code of operation: 0 – Success 1 - Error
RequestId	string	Unique Identifier passed by client in subscription request
RequestFor	string	Client Method where push notification will be served

8.2 Token Service

8.2.1 Login

Property	Type	Description
access_token	string	Token to be provided for requesting protected resources.
expires_in	int	Seconds until token expires at the latest NOTE: Token will expire sooner in case of inactivity; see Authentication [→ 28].
token_type	string	Only Bearer is supported at this time.
user_name	string	Logged in user name
user_descriptor	string	Localized user descriptor in user language
user_profile	string	name of currently logged in user's profile
flex_user_profile	string	name of currently logged in flex-user profile
user_inactivity_timeout	string	Inactivity timeout of user's group (uint) in minutes.
password_reminder_counter	string	Days remaining for password expiry
personally_used	string	"true" : machine is personally used "false" : machine is not personally used for provided client certificate

password_reminder_counter: this value gives counter i.e., number of days. Non-zero value of this counter gives number of days left before password expires. If the value is Zero then, the number of days left before password expires, are greater than the set number of days in "Reminder for password expiration" property of passwordpolicy of that System. The values of this counter in login response can be interpreted as shown in following table.

Response	Interpretation
password_reminder_counter = "0"	Password is active, no need to change it.
password_reminder_counter= "10"	Password will be expired in 10 days; it can be changed proactively before it expires.
{ "Id":2400003, "Error":"password_expired", "Details":"Logon rejected. Password of this account is expired and needs to get changed first." }	Password of the account is already expired and need to be changed.

8.3 Event Service

8.3.1 Event

Property	Type	Description
_links	Array of Link	See Link [→ 199]
CategoryDescriptor	string	Localized textual representation of CategoryId
CategoryId	int	Alarm category
Cause	string	Cause came with alarm (localized)
Commands	Array of EventComm and	List of available commands. See Event Command [→ 208]
CreationTime	DateTime	Time when alarm created.
Deleted	bool	Flag whether an event was deleted. This is for server sent notifications only.
Direction	EventDirection	See EventDirection [→ 206]
EventId	uint	Event counter (non-unique)
EventText	string	Service text attached to an event.
Id	string	Desigo CC alarm identifier → Invariant across sessions
InfoDescriptor	string	Additional information to an event
SrcDescriptor	string	Tree node descriptor OBSOLETE : use DescriptionList instead
SrcDesignation	string	Concatenated tree node names OBSOLETE : use DesignationList instead
SrcDisciplineDescriptor	string	Localized textual representation of DisciplineId
SrcDisciplineId	int	Alarm discipline
SrcLocation	string	Concatenated tree-node descriptors OBSOLETE :use DescriptionLocationsList instead
SrcName	string	Tree node name
SrcObservedPropertyId	string	Observed property which lets SrcPropertyId trigger an event
SrcPropertyId	string	Source of an event
SrcState	SourceState	See SourceState [→ 205]
SrcSystemId	uint	Unique system identifier

SrcViewDescriptor	string	View descriptor
SrcViewName	string	View name
State	EventState	See EventState [→ 205]
SuggestedAction	SuggestedAction	See SuggestedAction[→207]
NextCommand	NextCommand	See NextCommand [→207]
SourceDesignationList	Array of Event Designation List	List of node name. See EventDetailsList [→209]
DescriptionLocationsList	Array of EventsLocations	List of Concatenated tree-node descriptors See EventDetailsList [→209]
DesignationList	Array of EventsDesignation	List of Concatenated tree node names See EventDetailsList [→209]
DescriptionList	Array of EventsDescription	List of Tree node descriptor See EventDetailsList [→209]
MessageText	Array of string	Information Texts of events
SrcAlias	string	Event Source Alias
OperatingProcedureId	string	Operating procedure id associated with event
Sound	string	Path of sound file associated with event
AutomaticTreatmentData	AutomaticTreatmentData	See Automatic Treatment Data[→208]

8.3.1.1 SourceState

SrcState must have one of the following values.

SrcState	Description
Active	Object is active
Quiet	Object is quiet

8.3.1.2 EventState

State must have one of the following values.

State	Description
UnprocessedWithTimer	Unprocessed with Timer for user action
Unprocessed	Unprocessed
ReadyToBeResetWithTimer	Ready to be reset with timer for user action
ReadyToBeReset	Ready to be reset
ReadyToBeClosed	Ready to be closed
WaitingOPCompletion	Waiting for completion
Acked	Acknowledged
Closed	Closed

8.3.1.3 EventDirection

Direction must have one of the following values:

Direction	Description
In	IN alarm
Out	OUT alarm
None	Neither IN nor OUT alarm

8.3.1.4 EventTimeFilter

creationTime must have one of the following values:

creationTime	Description
0	All events (default, can be omitted)
1	Events generated in the last 15 minutes
2	Events generated in the last 30 minutes
3	Events generated in the last hour
4	Events generated last night
5	Events generated yesterday
6	Events generated today

8.3.1.5 EventSorting

sorting must have one of the following values.

sorting	Description
0	Sort by StateId, CategoryId, and descending CreationTime
1	Sort by StateId, CategoryId, DirectionId, and ascending CreationTime

8.3.1.6 EventFilter

Property	Type	Description
ids	Array of strings	Event identifier
caseSensitive	bool	
categories	Array of int	
cause	string	
creationTime	int	See EventTimeFilter [→206]
creationTimeFrom	DateTime	
creationTimeTo	DateTime	
disciplines	Array of int	
systemId	UInt	System identifier
states	Array of int	

8.3.1.7 EventSubscriptionStatus

Property	Type	Description
----------	------	-------------

ErrorCode	int	0: success 1: Failure
Sorting	int	See Event Sorting [→ 206]
EventsFilter	EventFilter object	See EventFilter [→206]
RequestId	string	Channelized subscription request identifier Note: Only available for channelize requests
RequestFor	string	Notification function name where client will receive further updates. Note: Only available for channelize requests

8.3.1.8 Maintenance State

“Maintenance” must have one of the following values.

SrcState	Description
All	Maintenance is All
MaintenanceOnly	MaintenanceState is Maintenance
GenuineOnly	MaintenanceState is Genuine

8.3.1.9 SuggestedAction

“SuggestedAction” must have one of the following values.

SuggestedAction	Description
Acknowledge	
Silence	
Suspend	
Reset	
Close	
CompleteOP	Complete operation
WaitforCondition	
None	

8.3.1.10 Next Command

“NextCommand” must have one of the following values.

NextCommand	Description
None	
Acknowledge	
Reset	
NoActionPossible	
StepMandatory	
Silence	
Close	

8.3.1.11 Automatic Treatment data

Property	Type	Description
HasFilter	boolean	
Stations	string[]	
Timeout	uint	
Users	uint[]	
CloseTreatmentWhen	string	Configuration value specifying condition for closing the treatment. Valid values: 1.SourceToNormal 2.UponUserRequest 3.EventAcknowledged 4.EventReset 5.EventClosed 6.Timeout
OnNewEvent	string	Configuration value specifying action on new event. Valid values: 1.DoNothing 2.OpenEventList 3.StartFastTreatment 4.StartInvestigativeTreatment 5.StartAssistedTreatment
OnNewHigherPriorityEvent	string	Configuration value specifying action on new higher priority event. Valid values: 1. DoNothing 2. SwitchToNew 3. AskUser

8.3.2 EventCommand

Property	Type	Description
_links	Array of Link	See Link [→ 199]
EventId	string	Unique identifier of an event
Id	EventCommandId	See Event Command ID [→ 51]
Configuration	Int	ValidationId of a propertyid If Configuration :0 ->None is required If Configuration :1 -> Only comment is required If Configuration : 2->Both Comment and Password(to re-authenticate) are required. if Configuration =3 : Comment , Current logged in user's credentials, Supervisor's Username and Supervisor's Password is expected if Configuration =4 : Supervisor's Username and Supervisor's Password is expected

8.3.3 ValidationData

Property	Type	Description
Password	String	Password required to re-authenticate event-commands
LangTextContent	LangTextContent	Input to provide comments Refer LangTextContent

SuperName	String	Supervisor's username required to authenticate event-commands
SuperPassword	String	Supervisor's password required to authenticate event-commands

8.3.4 LangTextContent

Property	Type	Description
CommonText	string	This property takes priority over MultiLangText. If value is assigned to CommonText then property MultiLangText will not be considered. Content of CommonText will be applied to all languages. Whitespace is treated as valid input.
LangTextContent	string[]	This property will accept input text for each language in system. Error is returned if number of entries does not match the number of languages in system. Whitespace is treated as valid input.

8.3.5 BatchEventsCommandExecution

Property	Type	Description
ValidationInput	ValidationData	Validation information of command execution See Link [→ 208]
EventIds	String[]	List of Event Id's for execution

8.3.6 EventDetailsList

Property	Type	Description
ViewId	uint	Unique identifier of a view
Descriptor	string	View details (Tree node or node details)

8.4 Event Counter Service

8.4.1 EventCounterList

Property	Type	Description
EventCategoryCounters	Array of EventCounter	Categorized list of event counters
TotalCounters	int	Total counters in the system for logged-in user
TotalUnprocessedCounters	int	Total unprocessed counters in the system for logged-in user

8.4.2 EventCounter

Property	Type	Description
_links	Array of Link	See Link [→ 199]
CategoryId	int	Category key

CategoryDescriptor	string	Localized category name
TotalCount	int	Total events of this category
UnprocessedCount	int	Total unprocessed events of this category
TotalSubsequentGrouping	int	Total number of grouped for subsequent Events for a category
UnprocessedSubsequentGrouping	int	Total number of unprocessed subsequent Events for a category

8.4.1 Category color details

EColorPalette Type(int)	EColorPalette description
1	TextButtonNormal
2	TextButtonPressed
3	TextEventSelected
4	TextEventNormal
5	TextEventHover
6	ButtonGradientBright
7	ButtonGradientDark
8	ButtonPressedGradientBright
9	ButtonPressedGradientDark
10	ButtonBlinkingBright
11	ButtonBlinkingDark
12	EventDescriptorSelected
13	EventDescriptorNormal

8.4.2 Subscription Status

Key	Type	description
ErrorCode	int	0: Success 1: Failure
RequestId	string	Channelized subscription request identifier Note: Only available for channelize requests
RequestFor	string	Notification function name where client will receive further updates Note: Only available for channelize requests

8.5 System Browser Service

8.5.1 View

Property	Type	Description
_links	Array of Link	See Link [→ 50]
Designation	string	Full view name
Descriptor	string	Localized view name
Name	string	View name
SystemId	uint	Unique ID of a system
SystemName	String	Name of system
ViewId	uint	Unique ID of a view within a system
ViewType	int	The type of view in Desigo CC: -3: Reserved (used internally) -2: Isolated (used for isolated nodes) -1: Undefined 0: Management 1: Application 2: Logical 3: Physical 4: UserDefined

8.5.2 BrowserNode

Property	Type	Description
_links	Array of Link	See Link [→ 50]
Attributes	Attributes	See Attributes [→ 51]
Descriptor	string	Localized browser object name
Designation	string	Full view name
HasChild	bool	Flag whether child nodes are available Note: Property omitted for searches and in notifications
Name	string	View name
Location	string	Localized full browser object name
ObjectId	string	Node address
SystemId	uint	Unique ID of a system
ViewId	uint	Unique ID of a view within a system

8.5.3 SystemBrowserSearchList

Property	Type	Description
ErrorCode	int	If value is >0, an error occurred
ObjectId	string	Searched Object Id
Nodes	Array of BrowserNode	Array of Browser nodes returned as search result for ObjectId see [→ 77]

8.5.4 SystemBrowserSubscription

Property	Type	Description
Action	ObservationAction	Enumeration to convey additional information about the action. See Enum for Action [→ 78]
Change	ObservationChange	Enumeration to convey the possible reasons for the event to have occurred. See Enum for Change [→ 78]
Node	BrowserObject	See BrowserObject [→ 77]
View	View	See View [→ 77]
SubscriptionKey	int	Subscription Key of View
RequestId	string	Channelized subscription request identifier Note: Only available for channelize requests

8.5.4.1 ObservationChange

Change type	Description
0	Reserved for 'no reason known'
1	System name changed
2	View separator changed
3	Change in the Nodes display name OR Views display name
4	Tree data changed
5	Structural change in the specified View, Tree OR Node. This also handles creation/deletion of System Browser-Views, a System Browser-Node OR System Browser-Subtree
100	Provider Reconnect

8.5.4.2 ObservationAction

Action type	Description
0	Reserver for 'no action known'
1	Changes in the System name
2	Creation of System Browser-View
3	Deletion of System Browser-View
5	Editing of System Browser-View
6	Editing of System Browser-View separator
7	Creation of System Browser-Tree
8	Deletion of System Browser-Tree
9	Editing of System Browser-Tree
10	Adding of a System Browser-Tree i.e. a new node somewhere in the Tree
11	Changes in the Node name
12	Changes in the Node data
100	All nodes in the client-hosted tree have been collapsed.

8.5.5 SystemBrowserSubscriptionKey

Property	Type	Description
_links	Array of Link	See Link [→ 199]
Key	int	Subscription key is specific for the requested ID
ErrorCode	int	If value is >0, an error occurred.
Designations	String[]	String array of designations for which client requested subscription
RequestId	string	Channelized subscription request identifier Note: Only available for channelize requests
RequestFor	string	Notification function name where client will receive further updates Note: Only available for channelize requests

8.6 Value Service

8.6.1 ValueDetails

Property	Type	Description
_links	Array of Link	See [→ 42]
AttributeId	string	Address of attribute to which value belongs to Note: In channelized subscription response data, this property is not set
DataType	ApiDataType	
ErrorCode	int	If value is >0, an error occurred
ObjectId	string	Address of object to which value belongs to Note: In channelized subscription response data, this property is not set
OriginalObjectOrPropertyId	string	Address of requested object or property Note: In channelized subscription response data, this property is not set
PropertyName	string	Name of property

		Note: In channelized subscription response data, this property is not set
Value	Value	See Value [→ 200]
IsArray	bool	Flag whether value is array or not
BackgroundColor	ColorRepresentation[See→ 201]	Background color of the icon, which denotes alarm state.
SubscriptionKey	Int	Subscription Key of OriginalObjectOrPropertyId

8.7 PropertyValue Service

8.7.1 Object<Type>

Property	Type	Description
_links	Array of Link	See Link [→ 199]
Attributes	Attributes	See Attributes [→ 201]
ErrorCode	int	If value is >0, an error occurred.
ObjectId	string	Address of object to which value belongs to
Properties	Array of <Type>	
FunctionProperties	Array of <Type>	

8.7.2 PropertyDetails

Property	Type	Description
_links	Array of Link	See Link [→ 199]
AllowWildcards	bool	Boolean flag for BACnet date/time values; indicates if the user can provide wildcards for any of the fields.
AllowDayOfWeek	bool	Boolean flag for BACnet date/time values; indicates if the day-of-week field is supported.
BackgroundColor	Color	
BACnetDateTimeDetail	uint	Indicates which parts of the BACnet date/time value to use: 1: date 2: time 3: date and time
BACnetDateTimeResolution	uint	When the time part of the BACnet date/time value is to be used, what is the resolution of the seconds? 0: seconds 1: tenths of seconds 2: hundredths of seconds
Descriptor	string	
DisplayOffNormalOnly	Boolean	When true, the user interface should only display the property when the value is off normal.
DisplayType	integer	Provides a 'hint' as to how the property value should be displayed. See DisplayType [→ 215]
DurationDisplayFormat	unsigned integer	If the property is of type "duration" this field will indicate how the duration is to be displayed.
DurationValueUnits	unsigned integer	If the property is of type "duration" this field will indicate the units of the raw property value (the raw value is an unsigned integer that can represent anything from milliseconds to days).

Property	Type	Description
ElementLabels	Array of strings	[Obsolete] Will be non-null (and contain string(s)) when a secondary text group has been assigned to the property. Normally used when the property value contains either an array of values or the value is a bit string. Strings will be in sorted order of Array indexes.
Elements	Array of Elements	Will be non-null when a secondary text group has been assigned to the property. Normally used when the property value contains either an array of values or the value is a bit string. Elements will be in sorted order of indexes. [→ 215]
MappedPropertId	string	Identifier of property to which a function property maps to.
Max	string	Allowed maximum value; empty if maximum value of datatype
Min	string	Allowed minimum value; empty if minimum value of data type
Normal	String	The „normal“ value for the property. Used primarily to determine when the property is in the „off-normal“ state.
Order	int	Order of the properties
PropertyAbsent	Boolean	True if the property is defined in the model but does not exist in the actual object.
PropertyName	string	The property's name. Name is invariant and serves as property identifier.
Resolution	int	If the property is floating point, this will be the desired resolution for display of values.
TextTable	string	If the property is of type enumeration, boolean, or bit string: this field will contain the name of the text table used to represent the property value.
Type	ApiDataType	
UnitDescriptor	string	Localized representation of unit
UnitId	uint	Unit of value
Usage	uint	
Value	Value	
ScaledInformation	ScaledFactor Representation	Scaled details of the property if applied. [→ 215]

8.7.3 Elements

Property	Type	Description
Id	uint	Id assigned to the
Descriptor	string	raw unit (index) of a property if configured

8.7.4 Scaled Property Details

Property	Description
Factor	scaling factor of a property if configured
Index	raw unit (index) of a property if configured
Offset	scaling offset of a property if configured

8.7.5 DisplayType

Change type	Description
0	Reserved for 'use default display'
1	Property is of type "command priority array" (normally 16 values, where any or all of those values can be "null")
2	Property is of type "event timestamps" (an array of three timestamps)

8.8 Properties Service

8.8.1 PropertyNames

Property	Type	Description
ObjectId	string	Address of object to which value belongs to
Properties	Array of string	Array of property names corresponding to the ObjectId
FunctionProperties	Array of string	Array of function property names corresponding to the ObjectId

8.9 Command Service

8.9.1 PropertyCommand

Property	Type	Description
Commands	Array of Command	See Command [→ 216]
PropertyId	int	
ErrorCode	int	-ErrorCode >0, an error occurred. -ErrorCode = 0 , Success
SubscriptionKey	int	-subscription key of propertyId

8.9.2 Command

Property	Type	Description
_links	Array of Link	See Link [→ 199]
Descriptor	string	Localized command name
GroupNumber	int	If provided, commands with same GroupNumber belong together
Parameters	Array of CommandParameter	See CommandParameter [→ 217]s
PropertyId	string	Address of property to which command belongs to
Id	string	Unique identifier of command
IsDefault	bool	True if this is the default command for the property.
Configuration	int	ValidationId of a propertyId If Configuration :0 ->None is required If Configuration :1 -> Only comment is required If Configuration : 2-> Comment and Password(to re-authenticate) are required.

8.9.3 CommandParameters

Property	Type	Description
AllowWildcards	bool	Boolean flag for BACnet date/time command parameters; indicates if the user can provide wildcards for any of the fields.
AllowDayOfWeek	bool	Boolean flag for BACnet date/time command parameters; indicates if the day-of-week field is supported.
Application	int	Integer indicating the application use of the parameter Application: 0 -> Normal Application: 1 -> The parameter is an index
BACnetDateTimeDetail	uint	Indicates which parts of the BACnet date/time value to use: 1: date 2: time 3: date and time
BACnetDateTimeResolution	uint	When the time part of the BACnet date/time value is to be used, what is the resolution of the seconds? 0: seconds 1: tenths of seconds 2: hundredths of seconds
DataType	ApiDataType	
DefaultValue	string	Default value of parameter
Descriptor	string	Localized command parameter name
DurationDisplayFormat	uint	How the duration is to be displayed. The following abbreviations are used: d: days, h: hours, m: minutes, s: seconds, ms: milliseconds 0: display raw value as-is 1: display is to be in units of d 2: display is to be in d + h 3: display is to be in d + h + m 4: display is to be in d + h + m + s 5: display is to be in d + h + m + s + ms 6: display is to be in h 7: display is to be in h + m 8: display is to be in h + m + s 9: display is to be in h + m + s + ms 10: display is to be in m 11: display is to be in m + s 12: display is to be in m + s + ms 13: display is to be in s 14: display is to be in s + ms
DurationValueUnits	uint	The content of the unsigned integer representing a duration value: 1: number of days 2: number of hours 3: number of minutes 4: number of seconds 5: number of deciseconds 6: number of centiseconds 7: number of milliseconds
EnumerationTexts	Array of EnumItem	See EnumItem [→ 218]
IsPassword	Bool	Boolean indicating if the parameter represents a password. This flag will only exist when the parameter is of type "BasicString"

Max	string	Maximum value if applicable
Min	string	Minimum value if applicable
Name	string	Unique identifier of command parameter within command
Order	int	Order of parameters within command
Resolution	int	For floating point parameters, the resolution
UnitDescriptor	string	For parameters of type real / integer / unsigned integer, if an engineering unit has been assigned (and can be found) this field will contain the localized text.

8.9.4 EnumItem

Property	Type	Description
Descriptor	string	Localized value
Value	int	Numeric value (unique within enumeration)

8.9.5 BatchExecution

Property	Type	Description
PropertyId	string	Address of property to which command belongs to
ErrorCode	int	-ErrorCode >0, an error occurred. -ErrorCode = 0 , Success
ErrorMessage	string	Error message containing reason for failure – comments not passed or missing validation information in request.

8.9.6 CommandInputForExecutionValidateParam

Inherits ValidationData See[[↗ 208](#)]

Property	Type	Description
Name	string	Parameter name
Value	int	Value of the parameter
DataType	ApiDataType	
MultiLangComments	string[]	Validation Profile : String array of comments for validation profile. Number of items in array should match the number of languages installed in project.
Password	string	Validation Profile : Logged in user's password for reauthentication
SuperName	string	Validation Profile : Supervisor's username
SuperPassword	string	Validation Profile : Supervisor's password

8.9.7 BatchCommandExecution

Property	Type	Description
CommandInputForExecution	CommandInputForExecutionValidateParam[]	Command parameter required for execution of commands
PropertyIds	string[]	String array of propertyids required for batch commanding

8.9.8 CommandInputForExecution

Property	Type	Description
Name	string	Parameter name
Value	int	Value of the parameter
DataType	ApiDataType	

8.9.9 BatchCommandExecutionValidateProperty

Inherits ValidationData See[→ 162]

Property	Type	Description
CommandInputForExecution	CommandInputForExecution[]	Command parameter required for execution of commands
PropertyIds	string[]	String array of propertyids required for batch commanding
LangTextContent	LangTextContent	Comments for logging
Password	String	Validation Profile : Logged in user's password for reauthentication
SuperName	String	Validation Profile : Supervisor's username
SuperPassword	String	Validation Profile : Supervisor's password

8.9.10 PropertyCommandInput

Property	Type	Description
CommandInputForExecution	CommandInputForExecutionValidateParam[]	Command parameter required for execution of commands
PropertyId	String	Property Id for execution
CommandId	string	Command Id for execution

8.9.1 PropertyCommandInputValidateProperty

Inherits ValidationData See[→ 162]

Property	Type	Description
CommandInputForExecution	CommandInputForExecution[]	Command parameter required for execution of commands
PropertyId	string	Property Id for execution
CommandId	string	Command Id for execution
LangTextContent	LangTextContent	Comments for logging
Password	string	Validation Profile : Logged in user's password for reauthentication
SuperName	string	Validation Profile : Supervisor's username
SuperPassword	string	Validation Profile : Supervisor's password

8.10 Trend Service

8.10.1 TrendCollector

Property	Type	Description
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CollectorObjectOrPropertyId	string	Address of collector object
PropertyIndex	uint	Index of Trended Property in the Trended Object. Start index: 1
ObjectId	string	Object ID of trended object
PropertyName	string	Trended property name
TrendseriesId	string	Identifier of corresponding trend series

8.10.2 TrendBorder

Property	Type	Description
From	DateTime	Minimum date/time from available values for requested trend series ID
To	DateTime	Maximum date/time from available values for requested trend series ID

8.10.3 TrendSeries

Property	Type	Description
Id	string	ID of trend series
Series	Array of Value	See Value [→ 200]
SeriesPropertyId	string	Trended object

8.10.4 TrendAggregateSeries

Property	Type	Description
Id	string	ID of trend series
Series	Array of TrendAggregate	See Value [→ 220]
SeriesPropertyId	string	Trend Aggregate object

8.10.5 TrendAggregate

Statistical analysis will be done for the samples of Values. All details are below are in reference of a Sample of Values

Property	Type	Description
AvgX	double	mean $x = (\sum x_i) / n$
AvgY	double	mean $y = (\sum y_i) / n$
Count	double	Total numbers of values taken for calculation
FromValue	double	First value in the Sample
FromValueDescriptor	string	Localized value to display FromValue
FromTime	DateTime	Time of the first value in the sample
Min	double	Minimum value in the Sample
Max	double	Maximum Value in the Sample
StDevX	double	Standard Deviation X
StDevY	double	Standard Deviation Y
SumX	double	Sum of all X Values
SumXX	double	Sum of all X^2

SumXY	double	Sum of all X*Y
SumY	double	Sum of all Y
SumYY	double	Sum of all Y ²
ToTime	DateTime	Last value in the Sample
ToValueDescriptor	double	Localized value to display ToValue
ToValue	double	Time of the last value in the sample
VarX	double	Variance of X
VarY	double	Variance of Y

8.10.6 TrendViewDefinitionRepresentation

Property	Type	Description
AxisX	AxisDefinitionRepresentation [→ 221]	
AxisYLeft	AxisDefinitionRepresentation [→ 221]	
AxisYRight	AxisDefinitionRepresentation [→ 221]	
DescriptionTvd	string	
ObjectIdOfTvd	string	
IdTvd	string	
NameTvd	string	
NumberDisplayedSamplesPerTrendSeries	integer	
RemoveOnlineTrendLogOfDeletedTrendSeries	boolean	
TimeRange	TimeRange [→ 221]	
TitleLeft	string	
TitleRight	string	
TitleTop	string	
TsdCollectionInfo	TrendSeriesDefinitionCollectionRepresentation [→ 223]	
TvdType	integer	
TvCovType	string	
_links	Array of Link [→ 199]	

8.10.7 AxisDefinitionRepresentation

Property	Type	Description
AutoScale	boolean	
AxisType	integer	
ScaleMax	number	
ScaleMin	number	
ShowTitle	boolean	
ShowZeroLine	boolean	
Title	string	

8.10.8 TimeRange

Property	Type	Description
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ValidTimeRange	integer	
AbsoluteTimeRange	AbsoluteTimeRange [→222]	
RelativeTimeRange	RelativeTimeRange [→222]	

8.10.9 TrendSeriesDefinitionRepresentation

Property	Type	Description
ReservedDateTime	string	
AccessDenied	boolean	
Alias	string	
ArchiveOn	boolean	
AxisAttachment	string	
ChartLineStyle	string	
ChartLineType	string	
Color	string	
DescriptionTIField	string	
DescriptionTIGms	string	
DescriptionTo	string	
DescriptionToProperty	string	
ObjectIdOfTrendLog	string	
ObjectIdOfTrendLogOnline	string	
ObjectIdOfTrendedObject	string	
Index	integer	
IsLineVisible	boolean	
LineWidth	integer	
ManagedType	string	
MaxTimeStamp	string	
MinTimeStamp	string	
ShowDataLabels	boolean	
ShowMarkers	boolean	
ShowQualityIcons	boolean	
TimeZoneServer	string	
Type	string	
UnitText	string	
ValueType	string	

8.10.10 RelativeTimeRange

Property	Type	Description
AnchorMode	integer	
AnchorTime	string	
NumberOfTimeUnits	integer	
TimeUnit	integer	

8.10.11 AbsoluteTimeRange

Property	Type	Description
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From	string	
To	string	

8.10.12 TrendSeriesDefinitionCollectionRepresentation

Property	Type	Description
TrendSeriesDefinitions	Array of TrendSeriesDefinitionRepresentation [→ 199]	

8.10.13 TrendViewDefinitionUpdateRepresentation

Property	Type	Description
AxisX	AxisDefinitionUpdateRepresentation [→ 221]	
AxisYLeft	AxisDefinitionUpdateRepresentation [→ 221]	
AxisYRight	AxisDefinitionUpdateRepresentation [→ 221]	
CNSDescription	string	
Location	string	
TvdObjectId	string	
NumberDisplayedSamplesPerTrendSerie	integer	
RemoveOnlineTrendLogOfDeletedTrendSerie	boolean	
TimeRange	TimeRange [→ 221]	
TitleLeft	string	
TitleRight	string	
TitleTop	string	
TsdCollectionInfo	TrendSeriesDefinitionUpdateCollectionRepresentation [→ 223]	
TvCovType	string	

8.10.14 TrendSeriesDefinitionUpdateRepresentation

Property	Type	Description
ArchiveOn	boolean	
AxisAttachment	string	
ChartLineStyle	string	
ChartLineType	string	
Color	string	
TrendLogObjectId	string	
TrendLogOnlineObjectId	string	
ObjectPropertyId	string	
TrendedObjectId	string	
IsLineVisible	boolean	
LineWidth	integer	
ShowDataLabels	boolean	

ShowMarkers	boolean	
ShowQualityIcons	boolean	
Type	string	
Smoothing	boolean	
CustomDescription	string	

8.10.15 TrendSeriesDefinitionUpdateCollectionRepresentation

Property	Type	Description
TrendSeriesDefinitions	Array of TrendSeriesDefinitionUpdateRepresentation	

8.10.16 AxisDefinitionUpdateRepresentation

Property	Type	Description
AutoScale	boolean	
ScaleMax	number	
ScaleMin	number	
ShowTitle	boolean	
ShowZeroLine	boolean	
Title	string	

8.11 History Logs Service

8.11.1 LogTable

Property	Type	Description
Name	string	Name of table which is unique and will be used for further operations in subsequent requests
Descriptor	string	Display name of table
Description	string	Description of table
HasChildColumns	bool	Flag whether table supports child columns or not
HasConditionFilter	bool	Flag whether table supports Condition Filter or not
HasNameFilter	bool	Flag whether table supports Name Filter or not
HasTimeFilter	bool	Flag whether table supports Time Filter or not
HasTimeFilterDefinedBySource	bool	Flag whether table supports Time Filter defined by source or not. Used in case of trend data retrieval. If set to TRUE then Time Filter provided in the request will be ignored and instead Time Filter defined by source will be considered to filter data
HasAndOperatorOnly	bool	Flag whether table only supports And-operator or not
HasMaxAge	bool	Flag whether table supports Max Age or not
HasSingleNameFilter	bool	Flag whether table supports Single Name Filter or not
SupportedTypes	Array of string	This is array of Managed Types which are supported by the table

8.11.2 LogColumn

Property	Type	Description
Name	string	Name of column which is unique and will be used for further operations in subsequent requests
Descriptor	string	Display name of column
DataType	string	Data type of the data which column supports
ErrorSupport	bool	Flag whether column supports display of error data (e.g., #COM, #ERR) or not
IsArray	bool	Flag whether column contains a data array or not
IsDefault	bool	Flag whether column is displayed by default or not. This will be useful while developing a user interface to create tables.
IsHidden	bool	Flag whether column is hidden or not
IsSortable	bool	Flag whether column is sortable or not
IsEnum	bool	Flag whether column contains enumerated data or not
IsFilterable	bool	Flag whether data from the column can be filtered using condition filter or not

8.11.3 LogConditionFilter

Property	Type	Description
Operators	Array of strings	Supported condition filter operators for column
ErrorInfo	Array of strings	Localized list of error information. It describes specific error cases.

8.11.4 LogEnumValues

Property	Type	Description
EnumValues	Array of strings	Supported enum values for column
ErrorInfo	Array of strings	Localized list of error information. It describes specific error cases.

8.11.5 LogSortInfo

Property	Type	Description
Name	string	Name of the Column which is to be sorted Required
SortType	LogSortType	See LogSortType [→ 225] Required

8.11.5.1 LogSortType

LogSortType must have one of the following values.

LogSortType	Description
Ascending	Sorts data in ascending order
Descending	Sorts data in descending order

8.11.6 LogColumnList

Property	Type	Description
Parent	Array of string	Column list for parent table
Child	Array of string	Column list for child table

8.11.7 LogTableData

Property	Type	Description
TableName	string	Name of the table
Page	int	Current page number
Size	int	Number of records per page
Result	Array of objects	At runtime each object is composed using selected columns to retrieve data and there value.
SnapshotId	string	Snapshot id of the snapshot created
Total	long	Total number of records found for the requested configuration
ErrorInfo	Array of strings	Localized list of error information. It describes specific error cases.
_links	Array of Link	See [→ 199]

8.11.8 LogDiscardedSnapshot

Property	Type	Description
TableName	string	Name of the table
SnapshotId	string	Snapshot Id which is discarded
IsSnapshotDiscarded	bool	Status of the discard request. If TRUE then discard is successful.

8.11.9 LogConditionFilterValidation

Property	Type	Description
ErrorPosition	int	In case of invalid string, Error position in the invalid string
ErrorMessageDescriptor	string	In case of invalid string, Localized Error message if string is not valid as per the standards
ToDoMessageDescriptor	string	In case of invalid string, Localized to-do message using which client can correct the Condition filter string if it is incorrect
IsValidConditionFilter	bool	Status of Condition filter string validation. If TRUE then string as per the standards.

8.11.10 LogChildData

Property	Type	Description
Page	int	Current page number
Size	int	Number of records per page
Total	long	Total number of records found for the requested configuration
Result	Array of objects	At runtime each object is composed using selected columns to retrieve data and there value.

ErrorInfo	Array of strings	Localized list of error information. It describes specific error cases.
_links	Array of Link	See [→ 199]

8.12 Language Service

8.12.1 Language

Property	Type	Description
Descriptor	string	Full localized language name
Code	string	IETF language tag according to RFC5646 [5] [→ 19]

8.13 Tables Service

8.13.1 SubDisciplines

Property	Type	Description
DisciplineId	Uint	Id of the Discipline
DisciplineDescriptor	string	Descriptor of the Discipline
SubDisciplines	Array of SubGroups	Array of SubDisciplines in SubGroups form for which filter is applied. See [→ 55]

8.13.2 SubObjectTypes

Property	Type	Description
ObjectTypeId	Uint	Id of the ObjectType
ObjectTypeDescriptor	string	Descriptor of the ObjectType
SubObjectTypes	Array of SubGroups	Array of SubObjectTypes in SubGroups form for which filter is applied. See [→ 55]

8.13.3 TextTableEntry

Property	Type	Description
Value	uint	Numeric value of the entry
Text	string	Text of the entry.
Color	uint?	The color assigned to the entry; this field is optional and if it does not exist, the property will not be included in the JSON representation.
Subtext	Array of TextTableEntry	Can be nullCollection of subtext.

8.13.4 LangTextTableEntry

Property	Type	Description
Value	uint	Numeric value of the entry
Color	uint?	The color assigned to the entry; this field is optional and if it does not exist, the property will not be included in the JSON representation.

LangText	String[]	Array of strings, one for each installed language.
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8.13.5 TextTableEntry Color Calculation

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. If a color property is provided, it is in the form of an unsigned 4 byte integer containing the following information:

Channel	Calculation
Alpha	(Color >> 24) & 0xFF
Red	(Color >> 16) & 0xFF
Green	(Color >> 8) & 0xFF
Blue	Color & 0xFF

8.14 Systems Service

8.14.1 System Languages

Property	Type	Description
Descriptor	string	Full localized language name
Code	string	IETF language tag according to RFC5646 [5] [→ 19]
ArrayIndex	Int	Index of the installed language

8.14.2 Systems

Property	Type	Description
Systems	SystemsInformation	Array of systems information. See[→228]
Languages	SystemsLanguages	Array of systems languages See[→228]
IsDistributed	bool	System is in distribution environment or not
IdLocal	uint	Id of system that is currently serving requests

8.14.3 SystemsInformation

Property	Type	Description
Name	string	System name
Id	uint	System id
IsOnline	bool	Obsolete: If system is available or not. MP4.0 onward: This field is not available in notification

8.14.4 ServicesSubscriptionsRepresentation

Property	Type	Description
SystemId	int	System id
ServiceId	int	Service id of the respective system id
ErrorCode	int	If that particular subscription for the respective system AND service is successful or not. 1 for error, 0 for

		success.
--	--	----------

8.14.5 ServicesRequestSubscriptionsRepresentation

Property	Type	Description
SystemId	int	System id
Ids	List<int>	List of Service ids of the respective system which are to be subscribed

8.14.6 ServicesNotificationRepresentation

Property	Type	Description
SystemId	uint	System id
ServiceId	int	Service id of the respective system id
Description	string	Name/description of the particular service for which notification is being received. NOTE: this will only be non-null when the service is connected.
IsConnected	bool	The current status of the service

8.15 Products Service

8.15.1 ProductInfo

Property	Type	Description
ProductNameShort	string	The name of the product; not localizable e.g. "Desigo CC" or "Cerberus DMS"
ProductName	string	The long name of the product; not localizable e.g. "Desigo CC ™"
VendorName	string	The name of the vendor of the product; not localizable e.g. "Siemens Switzerland Ltd." or "Daikin Ltd."
ProductCopyRights	string	Any statement; not localizable: © Siemens Switzerland Ltd, 2016
ProductVersion	string	Desigo CC project version
SupportLink	String	Technical support URL
Links	Array of Links	Direct URLs to access - Product picture wide - Product picture tall

8.16 Files Service

8.16.1 BasicFiles (Enum type)

Contain different file types.

Property	Type	Description
ProductPictures	int	a file type

8.16.2 File Details.

Property	Type	Description
FileName	string	contains file name
_links	Link	See Link [→ 199]

8.17 Graphics Service

8.17.1 GraphicalItem

Objects that are graphical in nature will be represented by this object type.

Property	Type	Description
Context	String	The designation of the object that 'owns' the graphical item.
ObjectId	String	The id of the object within DCC. Note that this id must include the system information to distinguish the object in a multi-system configuration.
ManagedType	String	The DCC managed type of the object.
DisplayName	String	A string that can be displayed to the user.

8.18 Scheduler Service

8.18.1 Calendar

Property	Type	Description
ScheduleCalendarType	integer	
CalendarEntries	Array of CalendarEntry[->230]	
AppliedSchedules	Array of string	
ContainerObjectId	string	
ObjectId	string	
Device	integer	

8.18.2 CalendarEntry

Property	Type	Description
StartDate	CalendarDate [->230]	
EndDate	CalendarDate [->230]	
Restrictions	CalendarDate [->230]	
Mode	integer	

8.18.3 CalendarDate

Property	Type	Description
Date	string	
IsAny	boolean	
Days	Array of integer	
Months	Array of integer	
DaysInMonth	Array of integer	
LastDayInMonth	boolean	

Years	Array of integer	
Weeks	Array of integer	
LastWeekInMonth	boolean	

8.18.4 WeeklySchedule

Property	Type	Description
DeviceId	string	
Name	string	
Setup	ScheduleSetup [->230]	
Outputs	Array of string	
ScheduleType	string	
Schedules	Array of Schedule [->230]	
ReferencedOutputDetails	Array of ReferencedOutputDetails [->231]	
Exceptions	Array of ScheduleException [->231]	

8.18.5 ScheduleSetup

Property	Type	Description
DefaultValue	string	
StartDate	CalendarDate [->230]	
EndDate	CalendarDate [->230]	
Priority	integer	
DataType	string	
ScheduleMin	number	
ScheduleMax	number	
TextGroupId	string	
Unit	integer	

8.18.6 Schedule

Property	Type	Description
Day	String	
Entries	Array of TimeTrigger [->232]	
OutputValuesActive	Array of string	
OutputValuesInactive	Array of string	

8.18.7 ReferencedOutputDetails

Property	Type	Description
Output	string	
TextGroupId	string	
Unit	integer	

8.18.8 ScheduleException

Property	Type	Description
----------	------	-------------

Priority	integer	
Mode	integer	
CalendarException	CalendarException [->232]	
CalendarEntry	CalendarEntry[->230]	
ExceptionTimes	Array of TimeTrigger [->232]	
ExceptionValuesActive	Array of string	
ExceptionValuesInactive	Array of string	

8.18.9 TimeTrigger

Property	Type	Description
Time	string	
Value	number	

8.18.10 CalendarException

Property	Type	Description
CalendarObjectId	string	
CalendarEntries	Array of CalendarEntry [->230]	
CalendarName	string	

8.19 Objects Service

8.19.1 Object

Objects will be represented by this type.

Property	Type	Description
Designation	string	Full node name under which child is supposed to be added
NameChildNode	string	Name of the newly created system browser node.
ObjectType	string	The type of the object to create.
Descriptor	Descriptor	Display name of the browser node. Multilingual text

8.19.2 Descriptor

Priority based text value definition.

Property	Type	Description
CommonText	string	Text to be initiated for all the languages Prio 1: If value is assigned to CommonText then attribute MultiLangText will not be considered. To assign value in MultiLangText set Descriptor to null. To be created node display name. Common descriptor will be applied in all languages.
MultiLangText	string[]	Language specific text for object Prio 2: MultiLangText has lower priority to Descriptor. It will not be considered if any value is assigned to attribute Descriptor Set Descriptor to null to assign MultiLangText. Language specific text will be updated for the node.

8.19.3 ObjectModel

Objects type will be represented by this type.

Property	Type	Description
Designation	string	Full node name
Description	string	Description of the object type.
ObjectType	string	The type of the object to create.
IsGenericDelete Allowed	bool	Generic delete configuration for the object
ChildObjects	ChildObjectModel Representation[]	List of available child types.

8.19.1 ChildObjectModel

Objects type will be represented by this type.

Property	Type	Description
Description	string	Description of the object type.
ObjectType	string	The type of the object to create.
IsGenericDelete Allowed	bool	Generic delete configuration for the object
IsGenericCreate Allowed	bool	Generic create configuration for the object
MaxChild	uint	Max number of Childs allowed for particular type
ExistingChildCount	int	Count of the Childs already added under the parent for current object types. Difference of ExistingChildCount and MaxChild can be used to identify that how many more childs can be added.

8.19.2 ServiceTextRepresentation

Information Text and Memo information will be represented by this type.

Property	Type	Description
InformationText	string	Information Text saved for an object in loggedin user's language.
Memo	string	Memo content for an object.

8.19.3 ObjectInfo

Response type for Information Text and Memo content.

Property	Type	Description
ObjectId	string	Object Id.
ServiceText	ServiceTextRepresentation [->233]	Information text and Memo details for an object. In case of failed request, this field is missing in Json response.

8.19.4 ServiceTextMemoRepresentation

Request type for accepting Memo text.

Property	Type	Description
Memo	string	Memo content for an object

8.20 Related Items Service

8.20.1 RelatedItemResult

This is the top level object returned from the “get related items” API call. It contains a single property at this time.

Property	Type	Description
RelatedResults	ObjectRelatedItem[]	An array: each element corresponds to an object id passed into the API call, where each element contains information about the related items for that object.

8.20.2 ObjectRelatedItem

This object contains the related item information for each object id passed to the “get related items” API.

Property	Type	Description
ErrorCode	integer	0: No error 1: Error
ObjectId	string	The object id passed into the API.
RelatedItems	RelatedItem[]	A collection of the related items for the object; this collection can be empty.

8.20.3 RelatedItem

As described elsewhere, two objects can be associated with each other – for example, an object in a report is ‘related to’ that report. This table defines the information that returned for each item that the object is related to.

Property	Type	Description
GroupDescriptor	string	If provided and non-blank, can be used to group items.
GroupOrder	ushort?	If a group descriptor is provided, this property can be used to sort the items within a group
ItemDescriptor	string	If provided and non-blank, should be used for the item's display name
Mode	ushort	The Mode in which the item should be visible: 0: All 1: Operation 2: Engineering
Nodes	BrowserNode[]	The array will contain one BrowserNode [→ 77] for each reference in a hierarchy to the object.
Parameter	string	An optional parameter added by a dynamic link provider.
Reference	string	If the related item is an external entity, this property will contain a URL – where the URL is of the form {key-word}{reference}. The list of valid keywords include: - file:// (reference is to a file) - file:/// (alternate form of file://) - http:// (HTTP location) - https:// (Secured HTTP location) - ftp:// (file transfer location)
SourceType	ushort	The source of the link: 0: NotSpecified 1: System 2: User 3: Dynamic

8.21 Access rights Service

8.21.1 AccessRightsRepresentation

Property	Type	Description
Application Rights	Array of ApplicationRightRepresentation [-> Error! Bookmark not defined.]	Collection of application rights assigned to logged in user

8.21.2 ApplicationRightRepresentation

Property	Type	Description
Name	string	Name of AccessRight in loggedin user's current language
Id	uint	Unique identifier assigned to access right
Operations	Array of OperationsRepresentation [->235]	Operations applicable within current access right

8.21.3 UserRoleRepresentation

Property	Type	Description
RolesMapping	List of type RoleInfo	Collection of roles assigned to logged in user

8.21.4 RoleInfo

Property	Type	Description
RoleName	string	Name of Role in loggedin user's current language
RoleId	uint	Unique identifier assigned to role
RoleDescription	string	Description of role functionality
RoleState	bool	Active or not.

8.21.5 OperationsRepresentation

Property	Type	Description
Name	string	Name of AccessRight in loggedin user's current language
Id	uint	Unique identifier assigned to operation within access right

8.22 Operating Procedures Service

8.22.1 Operating Procedure

Property	Type	Description
AlertCount	uint	
AlertSource	string	
AlertTime	datetime	
Id	string	Id of Operating procedure

InTreatmentBy	string	User performing the treatment
IsClosed	boolean	
ResetSteps	uint	
Sequential	boolean	Steps need to be sequential
Steps	Array of Operating Procedure Step [->236]	Steps available in Operating procedure
Subsequent	uint	
TemplateName	string	Name of Operating procedure template

8.22.2 Operating Procedure Step

Property	Type	Description
Attachments	string	[Property can be updated]
Attributes	string	
AutomaticDPE	string	
Configuration	string	
ErrorText	string	[Property can be updated]
FixedLink	string	
HasConfirmedExecution	bool	[Property can be updated]
IsCompleted	bool	[Property can be updated]
ManagedType	string	Internal managed type of Operating procedure step
Notes	string	Notes configured during Operating procedure step creation
Operator	string	[Property can be updated] Operator name
RuntimeStatus	string	[Property can be updated]
Status	string	[Property can be updated] Status of Operating procedure step Valid values are: Unknown, PartiallyFailed, Failed, Succesfull
StepId	string	[Required] Id of Operating procedure step
StepName	string	Name of Operating procedure step

8.22.3 Operating Procedure Subscription Representation

Property	Type	Description
AlertCount	uint	
AlertSource	string	
AlertTime	datetime	
Id	string	Id of Operating procedure
InTreatmentBy	string	User performing the treatment
IsClosed	boolean	

ResetSteps	uint	
Sequential	boolean	Steps need to be sequential
Steps	Array of Operating Procedure Step [->236]	Steps available in Operating procedure
Subsequent	uint	
TemplateName	string	Name of Operating procedure template
SubscriptionKey	string	Subscription key for which Notification is recieved

8.22.4 Operating Procedure Subscription Key Representation

Property	Type	Description
Id	string	Id of Operating procedure
Key	string	Subscription Key
ErrorCode	Int	Success = 0, Failure= -1
RequestId	string	Unique Id passed by client during subscription request
RequestFor	string	Hub method where notification will arrive

8.22.5 UnsubscriptionStatus

Property	Type	Description
Key	string	Subscription Key
ErrorCode	Int	Success = 0, Failure= -1

8.23 Diagnostics Service

8.23.1 DiagnosticsRepresentation

Property	Type	Description
ServerDateTime	DateTime	Server's UTC Datetime
ClientDateTime	DateTime	[optional] Client's UTC Datetime passed as input
NTPt1t0	string	[optional] Total milliseconds difference between server time and client time. Server's datetime – Client's datetime NOTE: This value would be double value and could be negative .

8.24 Licenses Service

8.24.1 LicenseMode

Property	Type	Description
LicenseModeName	string	The mode name of License Service

LicenseModeValue	uint	License mode value
ExpirationTime	uint	ExpirationTime of relative mode expression in second

8.24.2 LicenseOptions

Property	Type	Description
Id	string	The License option Id
Required	Int	Required property as int
Remaining	Int	Remaining property as int
Description	string	Description of licensing option
Available	int	Available property as int

8.25 Multi Monitors Service

8.25.1 MultiMonitorRepresentation

Property	Type	Description
Configuration	string	Json object for configuration of MultiMonitor Settings
OverruleSettings	bool	Any user can overrule the default multi monitor configuration with its own setting. This setting is used by the client to enable/disable the respective UI controls which support the multi monitor configuration. I.e. if it is set to false, the user (except the user which has the application right 'Configure multi monitor') cannot : • Open an additional system manager. • Undock a dedicated event manager. • Close an additional system or event manger. Move windows.
HostName	string	HostName of the client where multimonitor configuration being used.

8.25.2 MultiMonitorStationStatus

Property	Type	Description
stationIdentifier	string	• Identifier for the particular station preferably hostname of any particular machine • or it can unique id. If valid Client Certificate is present then it is given higher precedence than

		stationIdentifier. • Optional
DistinguishedSubjectName		• Subject name for client certificate
ErrorCode		• Status of response • 0 for success • 1 for failure

8.25.3 MultiMonitorUserStatus

Property	Type	Description
UserName	string	• LoggedIn UserId • an identification used by a person with access to a computer, network, or online service.

8.26 Validation Service

8.26.1 ValidationCommandOpRepresentation

Property	Type	Description
PropertyIds	String[]	The properties that are to be commanded.
GroupNumber	uint	Group Number of Command to execute.

8.26.2 ValidationEditOpRepresentation

Property	Type	Description
ObjectIds	String[]	Objects that are being edited.

8.26.3 ValidationCredentialRepresentation

Property	Type	Description
ObjectIds	String[]	The objects the supervisor should have access to.
CheckCredentials	int	0 validate both user and supervisor 1 validate user only, 2 validate supervisor only.
UserPassword	string	Password of user.
SupervisorName	string	Name of supervisor.
SupervisorPassword	string	Password of supervisor.

8.26.4 ValidateOpResponse

Property	Type	Description
IncrVersion	boolean	Should the Version number be incremented?
Log	boolean	Should a log entry be created?
Predefined Comments	TextGroupEntry	Pre-defined comments assigned to the object(s). If there are no pre-defined comments this will be an empty array.
CommentRule	string	Is a comment required? The value is an enumeration with values of "Never", "Optional", and "Mandatory". In practice the value is never or mandatory.
ReAuthentication	string	What kind of re-authentication is required? The value is an enumeration with values of "NoNeed" (no re-authentication is required) and "ReenterPw" (the user must re-enter their password).
IsFourEyesEnabled	boolean	Are a second user's credentials required?

8.27 Tags Service

8.27.1 TagsResponseRepresentation

Property	Type	Description
ObjectTagInfos	Array[TagsObjectRepresentation]	Array object of 8.27.2
GroupNumber	uint	Group Number of Command to execute.

8.27.2 TagsObjectRepresentation

Property	Type	Description
ErrorCode	integer	Error code specifying success or failure
ErrorMessage	String	Details specifying reason for failure.
ObjectId	String	Resolved object id
OriginalObjectId	String	Original object id passed in the request
KeyValueInfo	Dictionary<string, string>	Collection of tags in form of key value pair

8.27.3 TagsTraversalResponseRepresentation

Property	Type
TagsTraversalInfos	Associated Array of 8.27.5

8.27.4 TagsTraversalInfoRepresentation

Property	Type	Description
Node	IsolatedSystemBrowserNodeRepresentation	Object of 8.27.5
ErrorCode	integer	Error code specifying success or failure
ErrorMessage	String	Details specifying reason for failure
ObjectId	String	Resolved object id
OriginalObjectId	String	Original object id passed in the request
KeyValueInfo	Dictionary<string, string>	Collection of tags in form of key value pair

8.27.5 IsolatedSystemBrowserNodeRepresentation

Property	Type	Description
SystemId	Integer	Object of
Name	String	Error code specifying success or failure
Descriptor	String	Details specifying reason for failure
Designation	String	Resolved object id
ObjectId	String	Original object id passed in the request
_links	Array[HalLink]	Array of 8.27.6

8.27.6 HalLink

Property	Type
Rel	String
Href	String
IsTemplated	String

8.28 Operator Task

8.28.3 Operator Task Subscription

After subscribing for operator tasks, the current operator tasks (based on task filter body which we passing while subscribing operator tasks) will be sent in an initial notification. Afterwards, any change will be notified(based on task filter body).

Type	Name (case sensitive)		
Hub	3. norisHub : recommended		
Function	notifyChangeOfTasks (data)		
	Property	Type	Description
	data	OperatorTasksList	Representation of OperatorTasksList

8.28.4 Creating Channelized Operator Taks Subscription

If subscription status will be pushed through notifySubscriptionStatus function available at client side. Operator Tasks changes will be notified over notifyChangeOfTasks for any change in operator tasks, notifyOnModifyFilter for any change in subscription filter and notifyOnRealign for on Realign tasks , .

Resource Path	/sr/eventcounterssubscriptions/channelize/:requestId/:connectionId		
Category	Public		
HTTP Method	POST		
Parameters	Property	Type	Description
	requestId	String	Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client. It is expected from client that client should use unique id for for each request. Recommendation: Use GUID Required
	connectionId	String	Identifier of subscription hub

			connection Required
	taskFilterBody	TaskFilterBody Example: { "IsEnabled": true, "SystemId": 0, "TaskId": ["string"], "TaskStatus": [10] }	To filter the tasks , So based on filter user will get notified on any change. If IsEnabled is false then subscription will get happened for all tasks.
	isSuspended	boolean	Default parameter with false value
Response	HTTP 200 OK		
Examples	Request: POST api/sr/operatortaskssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/448b3117-44ef-4d8f-baa2-7274aa3050af Response: HTTP/1.1 200 OK 1. notifySubscriptionStatus: { "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifyChangeOfTasks " } 2. notifySubscriptionStatus: { "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifyOnRealign " } 3. notifySubscriptionStatus: { "ErrorCode":0, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor":"notifyOnModifyFilter " } }		

8.28.5 Deleting Operator Tasks Subscription

Deleting Operator Tasks Subscription

Resource Path	api/sr/operatortaskssubscriptions/ connectionId		
Categeory	Public		
Method	DELETE		
Parameters	Property	Type	Description

	connectionid	string	identifier of sub- scription Re- quired
Response	200 Ok		

8.29 Tags Service

8.29.1 Naming

8.30 Session Service

8.30.1 Subscription Status with Subscription Key

Key	Type	description
ErrorCode	int	0: Success 1: Failure
Key	string	Subscription Key
RequestId	string	Channelized subscription request identifier Note: Only available for channelize requests
RequestFor	string	Notification function name where client will receive further updates. Note: Only available for channelize requests

8.30.2 SessionRepresentation

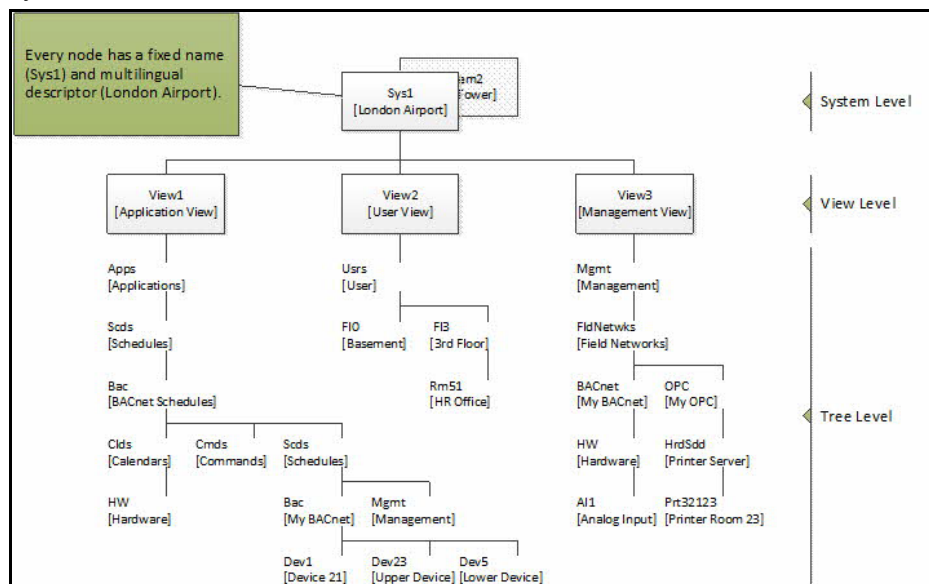
Property	Type	description
SessionId	String	Session identifier
UserId	int	User id
UserName	string	Username
HostName	string	Host Name / Station Name
SessionType	EnumKeyVal ueRepresent ation	Key Value representation to get key and value of session type e.g {Key: 3, Value: "Wsi"} All possible key and values of Session Types Desktop = 0, Web = 1, Sso = 2,

		Wsi = 3,
LogonTime	DateTime	
ClientProfile	string	Profile Name
SessionState	EnumKeyVal ueRepresent ation	Key Value representation to get key and value of session state e.g {Key: 0, Value: "Active"} All possible keys and values of Session State Active = 0, TerminationOnRequest = 1, Deactivated = 2
AccessPoint	string	Instance Point Name from where user is logged in

9 Concepts

9.1 Naming

The following is an overview about data hierarchies and naming conventions in the system.



Name: The name is a fixed (untranslatable) literal of a node. In order to uniquely identify nodes deeper in the hierarchy, nodes can be concatenated. Further, to ensure uniqueness sibling nodes can never have the same name. Concatenated Names are called **Designation**.

`<Designation> ::= <SystemName>.<ViewName>:{<TreeNodeName>}`

Example of Designation (node **HW [Hardware]**):

`Sys1.View1:Apps.Scds.Bac.Clds.HW`

Descriptor: The descriptor is a multilingual text of a node. Although nodes deeper in the hierarchy can be concatenated, they DO NOT have to be unique! Concatenated Descriptors are called **Location**.

`<Location> ::=`

`<SystemDescriptor>.<ViewDescriptor>:{<TreeNodeDescriptor>}`

Example of Location (node **HW [Hardware]**):

```
London Airport.Application View:Applications.Schedules.BACnet
Schedules.Calendar.Hardware
```

9.2 Extensibility

In order to deploy its own Web Service Interface extension one can create and deploy its own DLL. The DLL has to meet the following prerequisites:

- DLL needs to have literal "WsiExtension" as part of its name
- DLL needs to be deployed into the GMSMainProject\bin folder
- Class Attribute `[Authorize]`: It is an optional attribute and that is required if GMS user authorization is required.
- Class Attribute `[ControllerLicense(<Put License Name>)]`: It is a mandatory attribute. Provide here a valid GMS License information against which extension module feature should be available.
- For 3rd party license key "sbt_gms_opt_ws_3rd" can be used.
- If WSI extension wants license verification feature to be implemented then extension team should inform WSI team to incorporate License key along with it's Service ApiController name.

Each service comes as a class named with the service-name and a "Controller"-suffix. It derives from the ApiController base class. The following service could be reached at `http://[Your Host]/api/extensions/` :

Example with controller mapping:

```
namespace Siemens.Gms.WsiExtensionTest.Controller
{
    [Authorize]
    [ControllerLicense(ControllerNames.ServiceApiController, Licensin-
gOptions.sbt_gms_opt_ws_base)]
    [RoutePrefix("api/MyService")]
    public class ExtensionsApiController : ControllerBase
    {
        private const string TraceChannel = "MyService";
        public MyServiceController()
            : base(TraceChannel){}

        public MyServiceController(IStaticManager staticManager)
            : base(staticManager, TraceChannel){ }

        public HttpResponseMessage Get()
        {
            return Request.CreateResponse(HttpStatusCode.OK);
        }
    }
}
```

Example without controller mapping:

```
namespace Siemens.Gms.WsiExtensionTest.Controller
{
    [Authorize]
    [ControllerLicense(<Put License Name>)]
    [RoutePrefix("api/MyService")]
    public class MyServiceController : ControllerBase
    {

```

```
private const string TraceChannel = "MyService";
public MyServiceController()
    : base(TraceChannel){}

public MyServiceController(IStaticManager staticManager)
    : base(staticManager, TraceChannel){ }
}

public HttpResponseMessage Get()
{
    return Request.CreateResponse(HttpStatusCode.OK);
}
}
```

For using push-notifications one can take advantage of the existing push infrastructure.

```
[System.Web.Http.Authorize]
[ControllerLicense("sbt_gms_opt_ws_base")]
[RoutePrefix("api/sr/SubExtensions")]
public class SubExtensionsController : ControllerBase
{
    #region SignalR Hub
    private const string TraceChannel = "SubExtensions";
    readonly Lazy<IHubContext> _hub = new Lazy<IHubContext>(() =>
        GlobalHost.ConnectionManager.GetHubContext<ExtensionsHub>());

    protected IHubContext Hub { get { return _hub.Value; } }

    #endregion

    public SubExtensionsController(): base(TraceChannel){}

    public SubExtensionsController(IStaticManager staticManager)
        : base(staticManager, TraceChannel)

    public HttpResponseMessage Post(string id)
    {
        //send message to all clients
        Hub.Clients.All.notifyMessage(id);
        return Request.CreateResponse(HttpStatusCode.OK);
    }
}
```

For details on how push notifications work check out SignalR documentation [20].

9.3 Support for Web Service Interface extensions

9.3.1 Initialization support

- WSI now provides support for an extension module to initialize.

```
namespace Siemens.Gms.WsiExtensionTest.Init
{
    public class ServiceInitialization : ExtensionBase
    {
        public IDpProxy DpProxy { get; private set; }
        public override void Cleanup()
```

```

    {
    }
    public override void Initialize()
    {
        WsiTracingManager.Instance.TraceDebug("Client1");
        DpProxy = new DpProxy(StaticManager.OperationContext,
GlobalDefinitions.ClientName, null, GlobalDefinitions.ClientName);
        var dpConn = DpProxy.Connect(null, null);
        dpConn.WaitCompleted();
    }
}
}

```

9.3.2 Caching support

WSI now provides support for an extension module to cache.

```

namespace Siemens.Gms.WsiExtensionTest.Init
{
    public class CacheInitialization : InitializeCacheExtensionBase
    {
        public override void Initialize()
        {
        }
    }
}

```

9.4 Support for Browser Caching

WSI now support browser caching support via Cache-Control and Etag headers.

Browser handles caching based on response headers thus client application does not need to make any explicit change to take advantage of caching.

Following services now returns Cache-Control headers in response.

- Files service [→149]
- Graphics service [→150]
- Image service [→ 129]
- Properties service [→86]
- Tables service [→130]

Following services returns Etag headers in response.

- Graphics service [→150]

WSI also returns Access-Control-Max-Age for preflight requests allowing browsers to cache preflight response.

In default configuration, Caching max time is set to 30 minutes and Vary header is set with value *Authorization* to refresh cache based on user login session.

Caching configuration can be modified in config file [→Error! Bookmark not defined.] .

Developer Resources:

- **Cache-Control** : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Cache-Control>

- **Access-Control-Max-Age** : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Access-Control-Max-Age>
- **Etag** : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/ETag>
- **Vary** : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Vary>



NOTE:

Browser supports only GET response to be cached. POST ,PUT and DELETE response is not cached by browsers.

In case of HTTPS connection , Chrome supports caching only on valid certificate.

9.5 Support for CORS

WSI provides support to apply CORS policy for list of known domains.

In default configuration , CORS policy allows all domains.

White list of domains can be provided in config file to modify default configuration and apply CORS policy for specific domains. See [→**Error! Bookmark not defined.**].

10 Appendix

10.1 Error Response

To identify errors in the responses WSI follows standards so that error location and service can be easily identified. Any error response shall contain below information:

Property	Type	Description
Id	Int	Error codes. It will be return code from service specific range mentioned in Error Codes [250]
Error	String	Hint text to identify error and service. For details refer Errors [251]
Details	String	One liner information for short description of error

10.1.1 Errors Codes

S.No.	Service	Max	Min
1	Authentication/ Token	2400000	2400099
2	Licensing	2400100	2400199
3	General	2400200	2400299
4	Command	2400300	2400399
5	commandSubscriptions	2400400	2400499
6	EventCounters	2400500	2400599
7	EventCountersSubscriptions	2400600	2400699
8	EventsCommands	2400700	2400799
9	Events	2400800	2400899
10	EventsSubscriptions	2400900	2400999
11	Images	2401000	2401099
12	Properties	2401100	2401199
13	PropertyValues	2401200	2401299
14	SystemBrowser	2401300	2401399
15	SystemBrowserSubscriptions	2401400	2401499
16	Tables	2401500	2401599
17	TrendSeries	2401600	2401699
18	TrendSeriesInfo	2401700	2401799
19	Values	2401800	2401899
20	ValuesSubscriptions	2401900	2401999
21	HistoryLog	2402000	2402099
22	Files	2402100	2402199
23	Graphics	2402200	2402300
24	EventCategorySounds	2402400	2402499
25	EventCategorySoundsSubscriptions	2402301	2402399
26	SystemsSubscriptions	2402500	2402599
27	Schedule	2402600	2402699
28	Licenses	2403200	2403299
29	Validation	2404000	2404099
30	MultiMonitors	2403800	24003899

10.1.2 Errors

S.No.	Resource Path	Error	Id	Details	Remarks
1	POST /token	invalid_grant	2400000	The user name or password is incorrect.	Details not displayed because of some design constraint of OAuthGrantResourceOwnerCredentialsContext
2		invalid_token_grant	2400001	The SSO Token is incorrect or Expired	Provided SSO Token is either expired or invalid, continue with new SSO Token.
3		password_expired	2400003	Credentials of account are expired and need to get changed first.	Credentials of account are expired and should be changed from DesigoCC first.
4		license_not_available	2400100	A needed license is missing or expired	Required license is consumed or not available
5		license_option_not_available	2400102	A required license option is missing	This resource requires a license option which is not available.
6	PUT /users/password	invalid_details	2400002	The user name or password is incorrect or missing.	Provided user details are invalid or missing.
		server_error	2400008	Server is not able to process the request	Problem with reading from the server.
		password_not_compatible	2400004	The password does not meet policy requirements.	The password does not meet policy requirements.
7	GET /commands/:propertyId	cmd_missing_input	2400300	propertyIds are missing	propertyIds is blank or Null
8		cmd_invalid_input	2400301	Unknown clientType	Invalid Client Type
9		cmd_failure	2400399	Couldn't retrieve a list of commands	Exception occurred
10	GET /commands	cmd_missing_input	2400300	propertyIds are missing	propertyIds is blank or Null
11		cmd_invalid_input	2400301	Unknown clientType	Invalid Client Type
12		cmd_failure	2400399	Couldn't retrieve a list of commands	Exception occurred
13	POST /commands/:propertyId:command	cmd_missing_input	2400300	propertyIds are missing	propertyIds is blank or Null

S.No.	Resource Path	Error	Id	Details	Remarks
14	dld	cmd_invalid_input	2400301	propertyId is invalid	Invalid Property Id - DplIdentifier instantiation
15		cmd_execution_failed	2400398	command execution failed!	Command execution could not succeed
16		cmd_failure	2400399	Command execution failed	Command execution could not succeed
17	POST /commands/:commandId	cmd_missing_input	2400300	propertyIds are missing	propertyIds is blank or Null
18		cmd_execution_failed	2400398	command execution failed!	Command execution could not succeed
19		cmd_failure	2400399	Command could not execute	Exception occurred
20	POST /sr/commandssubscriptions/:connectionId or POST /sr/commandssubscriptions /channelize/:requestId/:connectionId	cmd-sub_invalid_input	2400400	Specify a valid propertyIdList	
21		cmd-sub_missing_input	2400401	connectionId is missing	
22		cmd-sub_subscription_failed	2400402	Error occurred while creating command subscription --> Command subscription failed	
23		cmdsub_failure	2400499	Could not retrieve command(s)	Exception occurred
24	DELETE /sr/commandssubscriptions/:connectionId	cmd-sub_missing_input	2400399	connectionId or subscriptionkey not available	subscriptionkey or connectionId null or whitespace
25		cmd-sub_unsubscribe_failed	2400498	Command unsubscription failed for commands with subscriptionkey	
26		cmdsub_failure	2400499	Command unsubscription failed for commands with connectionId	Exception occurred
27	GET /eventcounters/:categoryId	evtcnt_failure	2400599	Could not retrieve Event Counters	
28	POST /sr/eventcounterssubscriptions/:connectionId or POST /sr/eventcounterssubscriptions /channelize/:requestId/:connectionId	evtcntsub_missing_input	2400600	connectionId is missing	
29		evtcntsub_failure	2400699	Event Counter subscription unsuccessful for connection-id	Exception occurred

S.No.	Resource Path	Error	Id	Details	Remarks
30	DELETE /sr/eventcounterssubscriptions/:connectionId	ev-tcntsub_missing_input	2400600	connectionId is missing	
31		evtcntsub_failure	2400699	Event-counters unsubscription unsuccessful for connection-id	Exception occurred
32	POST /eventcommands/:eventId/:commandId	ev-tcmd_unknown_command	2400700	Unknown command	
33		ev-tcmd_missing_supervisor	2400794	Missing supervisor credentials for validation input	
34		ev-tcmd_invalid_supervisor	2400795	Invalid supervisor's credential passed	
35		ev-tcmd_invalid_comment	2400796	Mandatory comments not passed in case validation profile is applicable	
36		ev-tcmd_invalid_password	2400797	Invalid user credentials are passed in case validation profile is applicable	
37		ev-tcmd_execution_failure	2400798	Command execution failed	
38		evtcmd_failed	2400799	Command execution failed	
39	GET /events	evt_failure	2400899	Could not retrieve Events List	Exception occurred
40	POST /sr/eventsubscriptions/:connectionId or POST /sr/eventsubscriptions/channelize/:requestId/:connectionId	ev-tsub_missing_input	2400900	connectionId is missing	
41		evtsub_failure	2400999	Event Subscription unsuccessful for connection-id	Exception occurred
42	PUT /sr/eventsubscriptions/:connectionId	ev-tsub_missing_input	2400900	connectionId is missing	
43		evtcntsub_failure	2400699	Error occurred while changing an event subscription	Exception occurred
44	DELETE /sr/eventsubscriptions/:connectionId	eventsub_missing_input	2400900	connectionId is missing	

S.No.	Resource Path	Error	Id	Details	Remarks
45	GET /images/:imageId	img_format_unknown	2401000	Image format unknown!, format	
46		img_invalid_path	2401001	Image path not allowed!	
47		img_failure	2401099	Could not retrieve image	Exception occurred
48	GET /properties/:objectOrPropertyId	prop_missing_input	2401100	ObjectOrPropertyId is missing	
49		prop_invalid_input	2401101	Invalid Request Type requestType	
50		prop_invalid_input	2401101	Specify a valid Object OR Property Id	
51		prop_failure	2401199	Failed to read properties for ObjectOr-PropertyId	Exception occurred
52	POST /properties	prop_missing_input	2401100	Object OR Property Ids are missing!	
53		prop_invalid_input	2401101	Invalid Request Type	
54		prop_failure	2401199	Failed to read properties	Exception occurred
55	GET /propertyvalues/:objectOrPropertyId	propVal_missing_input	2401200	Object OR Property Ids are missing!	
56		propVal_invalid_input	2401201	Specify a valid Object OR Property Id	
57		propVal_failure	2401299	Failed to get values for properties	Exception occurred
58	GET /systembrowser	sysbrow_failure	2401399	Failed to retrieve Views	Exception occurred
59	GET /systembrowser/:systemId/:viewId/:node	sys-brow_missing_input	2401300	View OR Node details are missing while retrieving Browser Objects	
60		sysbrow_failure	2401399	Failed to retrieve Nodes	Exception occurred
61	POST /systembrowser/:systemId/:viewId	sys-brow_missing_input	2401300	Object Ids are missing	
62		sysbrow_failure	2401399	Failed to search multiple object	Exception occurred

S.No.	Resource Path	Error	Id	Details	Remarks
63	POST /sr/SystemBrowserSubscriptions/ connectionId	sys- browsub_mis sinput	2401400	connectionId is missing	
64	or POST /sr/SystemBrowserSubscriptions /channelize/:requestId/:connectio nId	sys- browsub_fa ilure	2401499	SystemBrowser Subscription failed	Exception occurred
65	DELETE /sr/SystemBrowserSubscriptions/ connectionId	sys- browsub_mis sinput	2401400	connectionId is missing	
66		sys- browsub_fa ilure	2401499	Failed to unsubscribe Cns	Exception occurred
67	GET /tables/:tableId	tbl_missing_inpu t	2401500	tableId is missing	
68		tbl_unknown_tab leId	2401500	Unknown table	
69		tbl_failure	2401599	Failed to Get Tables using TableId	Exception occurred
70	GET /tables/:tableId/:tableEntryId	tbl_missing_inpu t	2401500	tableId is missing	
71		tbl_failure	2401599	Failed to get table	Exception occurred
72		tbl_unknown_id	2401501	Unknown id	
73		tbl_unknown_tab leId	2401502	Unknown table	
74	GET /tables/:tableId/subtables/:subtabl eId	tbl_missing_inpu t	2401500	subtableId is missing!	
75		tbl_unknown_su btable	2401503	Unknown subtable	
76		tbl_failure	2401599	Failed to Get Sub Tables	Exception occurred
77	GET /tables/:tableId/subgroups	tbl_missing_inpu t	2401500	tableId is missing!	
78		tbl_unknown_su bgroup	2401504	Unknown subGroups	
79		tbl_failure	2401599	Failed to Get Sub Groups	Exception occurred

S.No.	Resource Path	Error	Id	Details	Remarks
80	GET /trendseries/:trendseriesId/borders	tnd_missing_input	2401600	TrendSeriesId is missing	
81		tnd_failure	2401699	Could not retrieve Borders	Exception occurred
82	GET /trendseries/:trendseriesId	tnd_missing_input	2401600	one or some of the input fields are missing	
83		tnd_notfound	2401601	Timeseries data not available	
84		tnd_failure	2401699	Could not retrieve Timeseries data	Exception occurred
85	GET /trendseriesinfo/:systemId	tndinfo_notfound	2401700	Get All TrendSeriesInfo Trendseries not available	
86	GET /trendseriesinfo/:objectOrPropertyId	tndinfo_notfound	2401702	Trendserie not available.	
87		tndinfo_failure	2401799	Could not retrieve Trendseries Information	Exception occurred
88	GET /trendseriesinfo/tvd/:tvdDplName	tnd_missing_input	2401600	one or some of the input fields are missing	
89		tndinfo_failure	2401799	Could not retrieve Trendseries Information	Not a valid objectid or Exception occurred
90	GET /values/:objectOrPropertyId	val_missing_input	2401800	Object OR Property Ids are missing!	
91		val_failure	2401899	Could not retrieve Value	Exception occurred
92	POST /values	val_missing_input	2401800	objectOrPropertyIds not provided	
93		val_failure	2401899	Could not retrieve Value	Exception occurred
94	POST /sr/valuessubscriptions/:connectionId or POST /sr/valuessubscriptions/channelize/:requestId/:connectionId	valsub_missing_input	2401900	connectionId or objectOrPropertyIds not available	
95		valsub_subscription_unsuccessful	2401901	Subscription unsuccessful	
96		valsub_failure	2401999	Could not subscribe	Exception occurred
97	DELETE /sr/valuessubscriptions/:connectionId	valsub_missing_input	2401900	connectionId or subscriptionkey not available	
98		valsub_notfound	2401902	subscription not available for provided subscription key	unsubscribe not done for any of the input
99		valsub_failure	2401999	Could not remove subscription	Exception occurred

S.No.	Resource Path	Error	Id	Details	Remarks
100	GET /historylogs/:systemId	htlog_failure	2402099	Could not retrieve list of available Tables.	Exception occurred
101	GET /historylogs/:systemId/:tableName	htlog_missing_in put	2402000	Table name is missing while retrieving column names for the table.	tableName is blank or Null
102		htlog_failure	2402099	Could not retrieve list of column details for Table	Exception occurred
103	GET /historylogs/:systemId/:tableName /operators/:columnName	htlog_missing_in put	2402000	Table name is missing while retrieving condition filter operators for the column.	tableName is blank or Null
104		htlog_missing_in put	2402000	Column name is missing while retrieving condition filter operators for the Table	columnName is blank or Null
105		htlog_failure	2402099	Could not retrieve Condition filter operators for Column	Exception occurred
106	GET /historylogs/:systemId/:tableName /enumvalues/:columnName	htlog_missing_in put	2402000	Table name is missing while retrieving Enum values for the column.	tableName is blank or Null
107		htlog_missing_in put	2402000	Column name is missing while retrieving Enum values for Table	columnName is blank or Null
108		htlog_failure	2402099	Could not retrieve Enum values for Column	Exception occurred
109	GET /historylogs/:systemId/:tableName /:snapshotId	htlog_missing_in put	2402000	Table name is missing while retrieving Table Data.	tableName is blank or Null
110		htlog_missing_in put	2402000	Column names not available while retrieving Table Data from Table	No valid list of columns
111		htlog_failure	2402099	Could not retrieve page wise data for Table	Exception occurred
112	DELETE /historylogs/:systemId/:tableName /:snapshotId	htlog_missing_in put	2402000	Table name is missing in the request to discard the snapshot.	tableName is blank or Null
113		htlog_missing_in put	2402000	Snapshot id is missing in the request to discard the snapshot for Table	snapshotId is blank or Null
114		htlog_failure	2402099	Could not discard snapshot with SnapshotId	Exception occurred

S.No.	Resource Path	Error	Id	Details	Remarks
115	GET /historylogs/:systemId/:tableName /validateconditionfilter/:conditionFilter	htlog_missing_in put	2402000	Table name is missing in the request to validate condition filter string.	tableName is blank or Null
116		htlog_missing_in put	2402000	Condition filter string is missing in the request to validate condition filter string for Table	conditionFilter is blank or Null
117		htlog_failure	2402099	Could not validate condition filter string	Exception occurred
118	GET /historylogs/:systemId/:tableName /:snapshotId/:parentId	htlog_missing_in put	2402000	Table name is missing in the request to retrieve child data.	tableName is blank or Null
119		htlog_missing_in put	2402000	Snapshot Id is missing in the request to retrieve child data for Table	snapshotId is blank or Null
120		htlog_missing_in put	2402000	Parent information is missing in the request to retrieve child data	parentId is blank or Null
121		htlog_missing_in put	2402000	Column names not available while retrieving child Data for Table	No valid list of columns
122		htlog_cancelled	2402098	Request preliminary canceled by the user	Report cancellation request aborted report generation
123		htlog_failure	2402099	Could not retrieve Child data for Table	Exception occurred
124	GET /files/:filetype	file_retrieval_fail ed	2402199		
125	GET /files/:systemId/:relativeFilePath	file_directorynotf ound	2402100		
126		file_filenotfound	2402101		
127		file_inputdatainv alid	2402102		
128		file_inputdatanot expected	2402103		

S.No.	Resource Path	Error	Id	Details	Remarks
129		file_pathinvalid	2402104		
130		file_pathnull	2402105		
131		file_pathtoolong	2402106		
132		file_unauthoirzed access	2402107		
133		file_retrieval_failed	2402199		
134	GET /files/documents/:designation	file_inputdatainvalid	2402102		
135		file_retrival_failed	2402199		
136	GET /graphics/:objectId	object_not_known	2402200	The object id is invalid.	
137		failure	2402201	Could not identify the object or could not determine if it is graphical.	
138	GET /graphics/itemIds/:objectId	object_not_known	2402200	The object id is invalid.	
139		failure	2402201	Could not identify the object or there was a problem finding graphical items related to the object.	
140	GET /graphics/items/:objectId	object_not_known	2402200	The object id is invalid.	
141		failure	2402201	Could not identify the object or there was a problem finding (or loading) the specified graphic.	The object id must specify a graphical item.
142	GET api/eventcategorysounds	event_sound_retrieval_failure	2402499		

S.No.	Resource Path	Error	Id	Details	Remarks
143	POST api/sr/eventcategorysoundssubscriptions/:connectionId or POST api/sr/eventcategorysoundssubscriptions/channelize/:requestId/:connectionId	evtsound-sub_missing_input	2402301	Connection id missing	
144	api/sr/eventcategorysoundssubscriptions/channelize/:requestId/:connectionId	evtsound-sub_client_already_subscribed	2402302		
145		evtsound-sub_failure	2402399		
146	DELETE api/sr/eventcategorysoundssubscriptions/:connectionId	evtsoundUn-sub_failure	240398		
147	POST /api/sr/systemssubscriptions/channelize/:requestId/:connectionId	systems-sub_missing_input	2402500		
148	DELETE /api/sr/systemssubscriptions/channelize/:connectionId	systems-sub_invalid_connectionId	2402501		
149	POST /api/sr/systemssubscriptions/channelize/:requestId/:connectionId	systemsun-sub_failure	2402598		
150	POST /api/sr/systemssubscriptions/channelize/:requestId/:connectionId	systems-sub_failure	2402599		
151	GET /api/scheduler/schedule/:objectId	sch_Invalid_schedule	2402600		
152		sch_unknown_schedule	2402603		
153		sch_failure	2402602		
154	POST /api/scheduler/schedule/save	sch_Invalid_schedule	2402600		
155		sch_unknown_schedule	2402603		

S.No.	Resource Path	Error	Id	Details	Remarks
156		sch_failure	2402602		
157	POST /api/scheduler/exception/save	sch_Invalid_schedule	2402600		
158		sch_unkown_schedule	2402603		
159		sch_failure	2402602		
160	GET /api/scheduler/schedule/saveDefaultValue	sch_Invalid_schedule	2402600		
161		sch_unkown_schedule	2402603		
162		sch_failure	2402602		
163	GET /api/scheduler/calendarList/:objectId	sch_Invalid_Calendar	2402601		
164		cal_unkown_calendar	2402606		
165		sch_failure	2402602		
166	GET /api/scheduler/calendar/:objectId	cal_failure	2402607		
167		sch_Invalid_Calendar	2402601		
168		cal_unkown_calendar	2402606		
169		sch_failure	2402602		

S.No.	Resource Path	Error	Id	Details	Remarks
170	GET /api/objects/:designation	obj_invalid_object	2402709		
171		obj_failure	2402700		
172	POST /api/objects	obj_object_validation_failed	2402708		
173		obj_object_creation_failed	2402703		
174		obj_new_object_not_allowed	2402704		
175		obj_object_already_exists	2402705		
176		obj_failure	2402700		
177	DELETE /api/objects/:designation	obj_deletion_failed	2402710		
178	GET /api/licenses	license_failure	2403299		
179	/POST/api/getComamndValidationOperation	validation_missing_input	2404000		Valid Property Ids are missing
180		validation_invalid_input	2404001		Some of the Property Ids are invalid
181		validation_execution_failure	2400409		Exception
182	/api/getEditValidationOperation	validation_missing_input	2404000		Valid Object Ids are missing
183		validation_invalid_input	2404001		Some of the Object Ids are invalid

S.No.	Resource Path	Error	Id	Details	Remarks
184		validation_execution_failure	24004099		Exception
185	/api/validateCredential	validation_execution_failure	24004099		Exception
186	api/multimonitors	multi-mon_failure	24003899		Exception
187	api/multimonitors	multi-mon_invalid_input	2403801		Some input Parameters are missing

Issued by
Siemens Switzerland Ltd
Smart Infrastructure
International Headquarters
Theilerstrasse 1a
CH-6301 Zug
Tel. +41 58-724 24 24
www.siemens.com/buildingtechnologies

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